COURSE SLIDES · WSQ

CompTIA Certified Network+ Training (N10-009)

WSQ Course Code: TGS-2025054472

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Networking, cybersecurity, cloud infrastructure, data science and AI — 20+ years of training and consulting.

DELIVERS
WSQ courses on CompTIA Network+, Security+, Linux+, cloud computing and applied AI.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your experience with networking, Linux or IT support (if any).
- What network you manage or plan to build after this course.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1

1 · Go to the LMS portal

Open https://lms-tms.tertiaryinfotech.com in your browser.

2

2 · Log in

Sign in with the account e-mail you used to register for this course.

3

3 · Open this course

Select 'CompTIA Certified Network+ Training' in My Courses.

4

4 · Download

Download the Learner Slides, Learner Guide and Lesson Plan (PDF).

5

5 · Get the labs

Open the lab repository on GitHub — the link is on the course page.

6

6 · Keep them open

You may use these materials during the open-book assessment.

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SCHEDULE · 9:30 AM – 6:30 PM DAILY

Lesson Plan — 5 Days · 8 Hours a Day

1

Day 1 · Networking Concepts

OSI model, appliances, cloud, ports & protocols, media, topologies, IPv4/IPv6 · Labs 1–7.

2

Day 2 · Network Implementation

Routing, NAT/PAT, VLANs & trunking, STP, link aggregation, wireless, installations · Labs 8–14.

3

Day 3 · Network Operations

Documentation, change & configuration management, monitoring, DR, network services · Labs 15–20.

4

Day 4 · Network Security

Encryption & PKI, IAM/AAA, segmentation, attacks and defence features · Labs 21–27.

5

Day 5 · Network Troubleshooting & Assessment

Seven-step methodology, layer-by-layer diagnosis, performance · Labs 28–36 · Final Assessment (WA + PP).

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Explain networking concepts

OSI model, appliances, cloud, ports & protocols, traffic types, media, topologies and IP addressing.

2

LO2 · Implement networks

Routing (BGP/EIGRP/OSPF), NAT/PAT, VLANs, STP, link aggregation, wireless and installations.

3

LO3 · Operate networks

Documentation, IPAM, change management, SNMP/flow/log monitoring, DR and network services.

4

LO4 · Secure networks

Encryption, PKI, IAM, AAA, segmentation, attack recognition and defence features.

5

LO5 · Troubleshoot networks

The CompTIA methodology applied with the right tool at every layer.

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FIVE DOMAINS · 36 HANDS-ON LABS

Course Outline — the Five N10-009 Domains

1

Domain 1.0 · Networking Concepts · 23%

OSI, appliances, cloud, ports & protocols, traffic, media, topologies, IPv4 · Labs 1–7.

2

Domain 2.0 · Network Implementation · 20%

Routing, switching, wireless and physical installations · Labs 8–14.

3

Domain 3.0 · Network Operations · 19%

Documentation, life-cycle, change, monitoring, DR, services, access · Labs 15–20.

4

Domain 4.0 · Network Security · 14%

Logical/physical security, attacks, segmentation and defence · Labs 21–27.

5

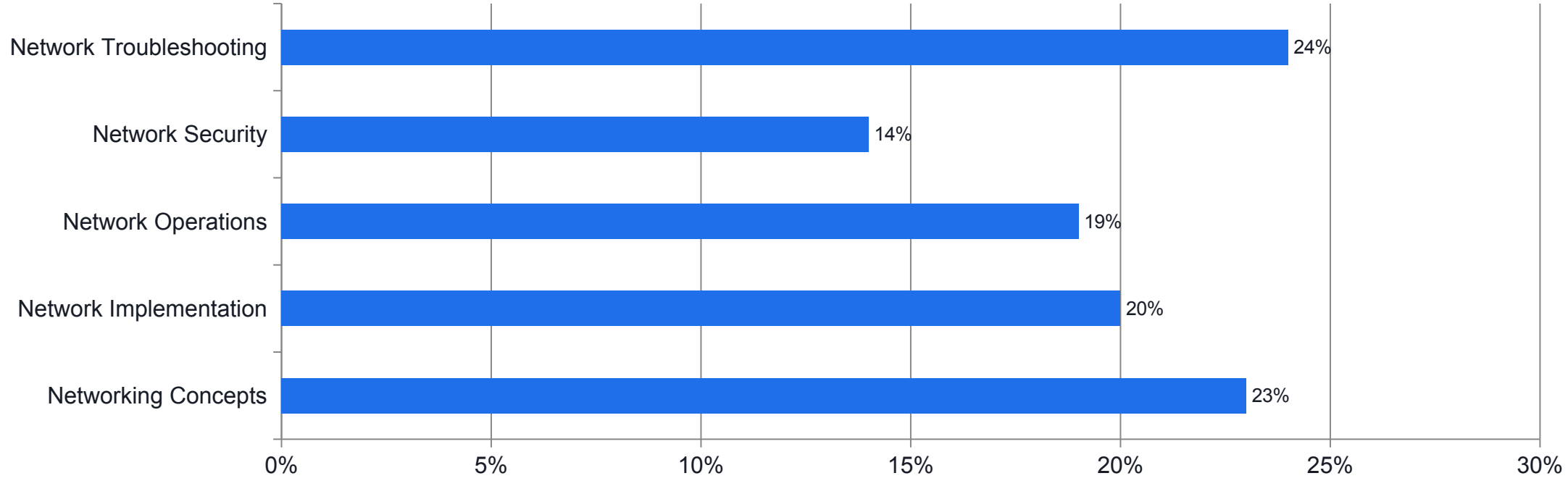
Domain 5.0 · Network Troubleshooting · 24%

Methodology, cabling, services, performance, tools · Labs 28–36.

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KNOW THE BLUEPRINT

N10-009 Exam Weighting by Domain



WHAT THE DATA SHOWS

Troubleshooting (24%) and Networking Concepts (23%) carry almost half the exam — the course spends Day 1 and Day 5 on them and every lab ends with a verify step.

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THE CERTIFICATION TARGET

About the CompTIA Network+ Exam (N10-009)

1

Format

Maximum of 90 questions — multiple-choice and performance-based — in 90 minutes.

2

Passing score

720 on a 100–900 scale. Book at Pearson VUE test centres or online proctoring.

3

No prerequisites

CompTIA recommends A+ and 9–12 months of networking experience, but neither is required.

4

Renewal

The certification is valid for 3 years and renews through CompTIA CE credits.

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FREE TOOLS USED THROUGHOUT

Your Lab Environment

1

Killercoda Ubuntu Playground

A free, throwaway Ubuntu VM in your browser — every Linux lab runs here. https://killercoda.com/playgrounds/scenario/ubuntu

2

TIS IP Calculator

Web subnetting tool for CIDR/VLSM planning and verification. https://alfredang.github.io/ipcalculator/

3

TIS PCAP Analyzer

Upload and dissect packet captures layer by layer in the browser. https://alfredang.github.io/pcapanalyzer/

4

TIS Cybersecurity Simulator

Practise recognising network attacks in a safe simulated environment. https://alfredang.github.io/cybersecuritysimulator/

5

TIS Regex Generator

Build the regular expressions used to search and filter log files. https://alfredang.github.io/regexgenerator/

6

No installs needed

Everything is browser-based; packages are pulled with apt inside the playground VM.

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EVERY LAB IS A FOLDER, NOT ONE FILE

What Ships With Every Lab

1

README.md

The lab itself — the concept, every command, the verify step and troubleshooting.

2

checklist.md

A tick-box list of what you must achieve — use it to self-check before moving on.

3

setup.sh / run.sh

The prep and the command sequence, so a long lab is not retyped by hand.

4

verify.sh / cleanup.sh

Prove the lab worked, then restore the box so the next lab starts clean.

5

Config files

Real service configs (dhcpd.conf, zone files, wg0.conf, rules) — copy them into place.

6

data/*.csv

Subnet plans, port lists, log samples and baseline sheets to work from and fill in.

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

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FINAL ASSESSMENT · DAY 5

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 1 hour · open book. Tests the underpinning networking knowledge.

2

Practical Performance (PP)

Hands-on networking tasks · 90 minutes · open book. Tests the ability you built in the 36 labs.

3

Open book

You may use the course slides, the Learner Guide and approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

You are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM survey

Scan the QR on the LMS

2

Digital attendance

Scan the SSG QR

3

Sit WA then PP

Open book · 60 + 90 min

4

Submit on the LMS

Upload your answers

5

Sign the record

Assessment Summary

REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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WSQ FUNDING

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

TRAQOM survey

Complete the mandatory TRAQOM course feedback survey on the LMS.

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1.0

DOMAIN 1.0 · 23% OF THE EXAM

Networking Concepts

OSI model · appliances · cloud · ports & protocols · traffic types · media · transceivers & connectors · topologies · IPv4 addressing

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WHAT YOU'LL DO

Hands-On Labs — Networking Concepts

Labs 1–3

- OSI Model Packet Inspection
- IPv4 Subnetting and CIDR
- IPv6 Addressing and Link-Local

Labs 4–6

- Common Ports and Protocols
- DNS Record Types
- DHCP Lease Capture (DORA)

Lab 7

- Simulating a Cloud VPC with Namespaces

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DOMAIN 1 · 23% OF THE EXAM

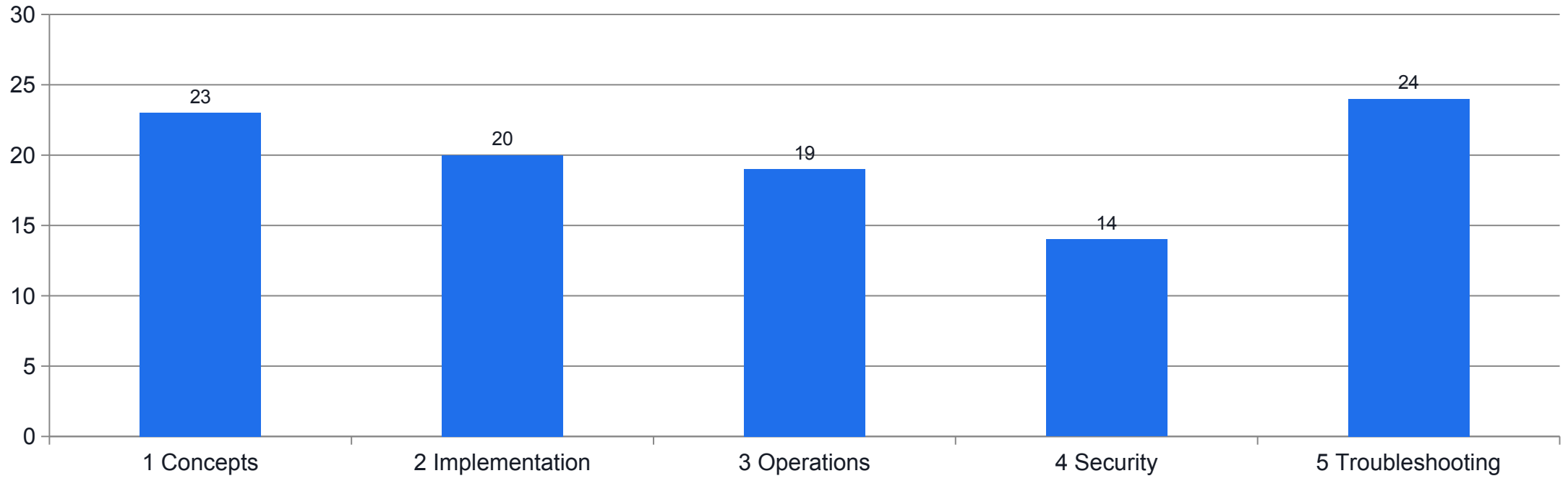
Domain 1 — Networking Concepts.

Seven layers, addressing, ports, media, topologies and cloud — 23% of the N10-009 exam and the foundation every other domain builds on.

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EXAM BLUEPRINT

N10-009 Exam Domain Weights



WHAT THE DATA SHOWS

Concepts and Troubleshooting together are nearly half the exam — and troubleshooting questions lean on the concepts you learn here.

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Why We Model Networks

1

Models simplify

A model is a basic representation that makes a complex system explainable one piece at a time.

2

Frames

Devices exchange data in discrete chunks called frames, built and destroyed inside the NIC.

3

1,500 bytes

The default Ethernet payload limit (MTU); jumbo frames stretch to 9,000 bytes.

4

OSI vs TCP/IP

OSI is the 7-layer teaching and troubleshooting model; TCP/IP is the 4-layer practical stack.

5

PDU

Protocol Data Unit — the name of the data chunk changes at each layer as headers are added.

6

Exam anchor

N10-009 expects you to place any protocol, device or symptom at its correct layer.

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The Seven Layers

1

L7 — Application

Protocols the software speaks: HTTP, FTP, SMTP, DNS. Where data is produced and consumed.

2

L6 — Presentation

Formats and translates data: encryption (TLS), compression, character encoding.

3

L5 — Session

Opens, manages and tears down dialogs between hosts; think RPC and API sessions.

4

L4 — Transport

End-to-end delivery: TCP (reliable) and UDP (best-effort), addressed by port numbers.

5

L3 — Network

Logical addressing and routing between networks: IP and ICMP. Routers live here.

6

L2 — Data Link

Frames and MAC addresses on the local link; switches forward here, FCS detects errors.

7

L1 — Physical

Bits on the wire or in the air: cables, connectors, transceivers, hubs, radio.

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PDUs and Devices by Layer

Layer(s)	PDU	Typical devices / protocols
L5-L7 upper layers	Data	HTTP, TLS, DNS, proxies
L4 Transport	Segment / datagram	TCP, UDP, load balancer
L3 Network	Packet	IP, ICMP, router, L3 switch
L2 Data Link	Frame	Ethernet, switch, NIC, AP
L1 Physical	Bits	Cables, hubs, transceivers

WHEN IT MATTERS Exam shortcut: name the PDU and you have named the layer.

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Encapsulation: Data Down the Stack



WHY IT MATTERS

Every capture you will read shows these headers nested in exactly this order — and decapsulation reverses the sequence at the receiver.

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OSI vs TCP/IP Model

OSI (7 layers)

- 7-6-5: Application, Presentation, Session
- 4: Transport — segments, ports
- 3: Network — packets, routing
- 2-1: Data Link and Physical

TCP/IP (4 layers)

- Application — HTTP, DNS, SMTP, TLS
- Transport — TCP and UDP
- Internet — IP, ICMP, IGMP
- Network Access — Ethernet, Wi-Fi, media

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Upper Layers in Practice

1

Session (L5)

Establishes and resumes conversations — RPC calls, NetBIOS sessions, SQL connections.

2

Presentation (L6)

TLS encryption, JPEG/PNG encoding, ASCII vs Unicode — data made readable and safe.

3

Application (L7)

The protocol carrying user intent: a GET request, a mail submission, a DNS query.

4

Why it blurs

Real stacks fold L5-L7 together — TLS spans layers — but the exam still tests the clean split.

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The MAC Address

1

48-bit physical address

Burned into every NIC: first 24 bits are the vendor OUI, last 24 the unique device ID.

2

Hex notation

Example 00:16:3e:4a:99:d1 — view yours with ip link (Linux) or ipconfig /all (Windows).

3

Destination + source

Every frame carries both; a NIC normally processes only frames addressed to it.

4

FF-FF-FF-FF-FF-FF

The broadcast MAC — every NIC in the broadcast domain must process the frame.

5

Duplicate MACs

Rare but real — cloned VMs are a classic source of duplicate-MAC connectivity faults.

6

Physical vs logical

MAC = physical address (L2); IP = logical address (L3). ARP glues them together.

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Anatomy of an Ethernet Frame

1

Preamble

7 bytes of alternating bits plus a start delimiter — lets receivers sync their clocks.

2

Destination + source MAC

6 bytes each — switches forward on the destination, learn from the source.

3

EtherType

Declares the payload protocol: 0x0800 IPv4, 0x86DD IPv6, 0x0806 ARP.

4

Payload

46-1,500 bytes of upper-layer data; padded if too short.

5

FCS

4-byte CRC checksum — corrupt frames are silently dropped, upper layers must recover.

6

Size range

64 to 1,522 bytes standard; jumbo frames up to 9,000 for storage and DC traffic.

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Traffic Types

1

Unicast

One sender to one receiver — the vast majority of traffic, e.g. your HTTP session.

2

Broadcast

One to all in the broadcast domain (FF-FF... / 255.255.255.255). IPv6 abolishes it.

3

Multicast

One to a subscribed group — IPv4 224.0.0.0/4, IPv6 ff00::/8; membership managed by IGMP.

4

Anycast

One address advertised from many places; routing delivers to the nearest — how 8.8.8.8 scales.

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Reading a Capture Layer by Layer

One tcpdump line contains headers from three layers — the payload adds the fourth.

```
# tcpdump -i eth0 -nn -v -c 5 'tcp port 80'
IP 10.244.0.5.55512 > 93.184.216.34.80:
   Flags [S], seq 1287312, win 64240

# add -e for the link-layer (L2) header:
00:16:3e:4a:99:d1 > 52:54:00:12:35:02

# add -A for the ASCII payload (L7):
GET / HTTP/1.1
Host: example.com
```

L2 Data Link

MAC-to-MAC delivery on the local link — visible only with the -e flag.

L3 Network

Source and destination IP addresses route the packet between networks.

L4 Transport

Ports 55512 > 80 plus the SYN flag — a TCP connection being opened.

L7 Application

The HTTP GET request itself — the data everything else exists to carry.

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From URL to Wire

1. Browser builds an HTTP GET (L7)
2. TCP adds ports + starts handshake (L4)
3. IP adds source/destination address (L3)
4. NIC frames it with MACs + FCS (L2)
5. Bits transmitted on the medium (L1)

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Troubleshooting by Layer

1

L1 first

Link lights, cables, transceivers, Wi-Fi signal — most 'network down' calls end here.

2

L2 next

Check MAC learning, VLAN membership, ARP entries; switches misbehave at this layer.

3

L3 routing

ping and traceroute test addressing and routes — wrong gateway or mask lives here.

4

L4 ports

Service unreachable but host pings? Check listening ports and firewall rules.

5

L7 last

DNS answers, certificates, application errors — the top of the bottom-up method.

6

Name the layer

Exam questions describe a symptom and expect the layer — practise the mapping now.

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DOMAIN 1.0 · HANDS-ON

OSI Model Packet Inspection

LAB 1

Capture live traffic and map each protocol header to its OSI layer.

Capture HTTP traffic with tcpdump on a Killercoda Ubuntu host, generate requests with curl, and match the MAC, IP, TCP and HTTP headers you see to OSI layers 2, 3, 4 and 7.

YOU'LL BUILD

A live capture and a saved .pcap where you can point at every header and name its OSI layer.

TOOLCHAIN

tcpdump, curl, Killercoda Ubuntu, TIS PCAP Analyzer (alfredang.github.io/pcapanalyzer)

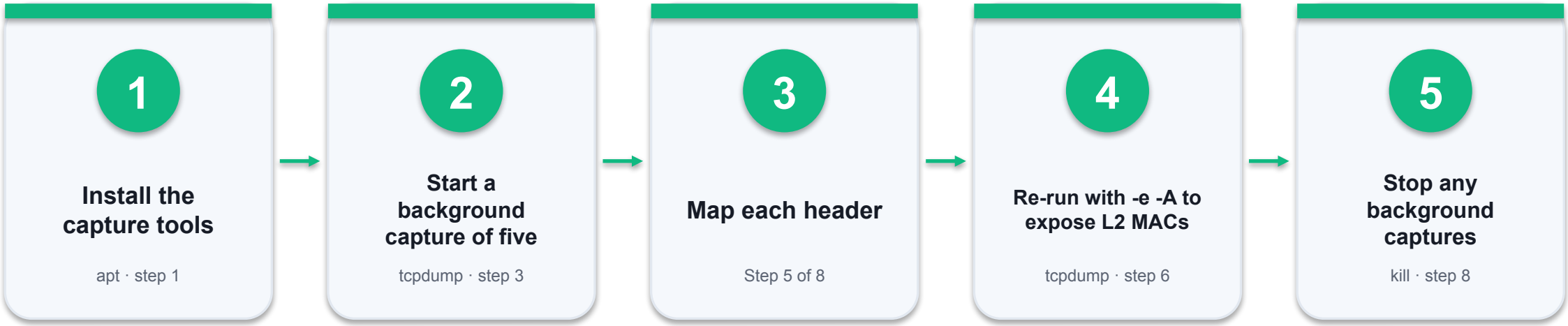
DONE WHEN

tcpdump shows SYN/ACK/GET packets and you can name the OSI layer of every header; the TIS PCAP Analyzer decodes the same layers from the downloaded osi-lab.pcap.

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LAB 1 · PROCESS MAP

How Lab 1 Runs



THE POINT

Capture live traffic and map each protocol header to its OSI layer.

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LAB 1 · VERIFY

OSI Model Packet Inspection

✅ Expected result

tcpdump shows SYN/ACK/GET packets and you can name the OSI layer of every header; the TIS PCAP Analyzer decodes the same layers from the downloaded osi-lab.pcap.

IF IT DOESN'T WORK

No packets captured

Check the interface with ip -brief addr — capture on the interface marked UP, not lo.

curl hangs or fails

The playground may block port 80 — try curl -s http://neverssl.com instead.

tcpdump: command not found

Re-run apt update && apt install -y tcpdump curl — the install step may have failed.

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Core Networking Appliances

1

Router

Connects networks and forwards packets by IP; every LAN's default gateway is a router interface.

2

Switch

Forwards frames by MAC using its learned MAC table; each port is its own collision domain.

3

Layer 3 switch

Switches inside VLANs and routes between them in hardware at wire speed.

4

Hub (legacy)

A multiport repeater — one shared collision domain using CSMA/CD; replaced by switches.

5

Access point

Bridges 802.11 wireless clients onto the wired LAN at Layer 2.

6

Wireless controller

Centrally manages fleets of APs: channels, power, roaming and policy.

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Which Layer Does Each Device Read?

Device	Layer	Forwards / filters on
Hub / repeater	L1	Electrical signal only
Switch, bridge, AP	L2	MAC addresses
Router, L3 switch	L3	IP addresses and routes
Load balancer	L4-L7	Ports, sessions, URLs
Firewall / IPS	L3-L7	Rules, state, signatures

WHEN IT MATTERS A device's layer tells you what information it can see — and what it can never see.

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Security and Application Appliances

1

Firewall

Filters traffic between zones by rule: addresses, ports, and connection state.

2

IDS

Passively inspects mirrored traffic and raises alerts on signatures or anomalies.

3

IPS

Sits inline in the traffic path and actively blocks malicious traffic in real time.

4

Load balancer

Distributes client sessions across a server pool and health-checks every member.

5

Forward proxy

Fetches web content on behalf of internal clients — filtering, caching, logging.

6

Reverse proxy

Fronts your servers: TLS offload, caching, and shielding the origin from the Internet.

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IDS or IPS?



YES — deploy an IPS

Place it inline so it can drop packets in real time — and accept that a false positive can block legitimate traffic.

NO — deploy an IDS

Feed it a mirror/SPAN copy. It only alerts, so it can never break the network — but a human must act on the alert.

Many NGFW platforms bundle both modes — the inline-vs-passive placement decision still applies.

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Forward vs Reverse Proxy

Forward proxy (protects clients)

- Sits between your users and the Internet
- Enforces acceptable-use and URL filtering
- Caches popular content to save bandwidth
- Hides internal client identities

Reverse proxy (protects servers)

- Sits between the Internet and your servers
- Terminates TLS and offloads encryption
- Load-balances and caches for the backend
- Hides and shields the origin servers

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NAS vs SAN

Attribute	NAS	SAN
Access level	File (SMB, NFS)	Block (raw volumes)
Network	Standard Ethernet LAN	Fibre Channel or iSCSI
Appears as	Shared folders	Local disks to the OS
Hardware	NAS appliance	HBAs, FC switches, arrays
Typical use	Team file shares	Databases, VM datastores

WHEN IT MATTERS FCoE carries Fibre Channel over Ethernet; multipathing keeps SAN storage reachable through failures.

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Physical vs Virtual Appliances

Physical appliance

- Dedicated hardware, predictable throughput
- ASICs accelerate switching and crypto
- Scaling means buying and racking boxes
- Failure domain is the chassis

Virtual appliance (NFV)

- Router, firewall, LB run as VMs/containers
- Deployed in minutes from an image
- Scales out on shared compute
- Portable across data center and cloud

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Copper: Coaxial and Twinaxial

1

Coaxial

A single center conductor inside a shield — highly resistant to EMI; used for cable broadband.

2

RG-6

75-ohm coax for cable TV and DOCSIS internet; terminated with threaded F-type connectors.

3

BNC

Bayonet-twist coax connector found on older gear and lab equipment.

4

Twinax / DAC

Twin conductors; direct-attach copper links switches over short 10-100G runs, cheaper than optics.

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Twisted Pair Essentials

1

Four twisted pairs

The twists cancel EMI and crosstalk between pairs — tighter twists, higher category.

2

UTP vs STP

Unshielded is the default; shielded adds foil/braid for electrically noisy environments.

3

100 m channel limit

The maximum twisted-pair run: 90 m horizontal plus 10 m of patch cords.

4

RJ45 (8P8C)

The standard data connector; RJ11 is the smaller 2-3 pair telephone jack.

5

T568A / T568B

The two pin-out standards — use one consistently; mixing them creates a crossover.

6

Solid vs stranded

Solid core for in-wall horizontal runs; stranded for flexible patch cables.

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Twisted-Pair Categories

Category	Max speed	Bandwidth	Notes
Cat 5e	1 Gbps	100 MHz	Minimum for gigabit
Cat 6	1 G (10 G to 55 m)	250 MHz	Internal separator
Cat 6a	10 Gbps to 100 m	500 MHz	New-build standard
Cat 7	10 Gbps	600 MHz	Shielded, GG45/TERA
Cat 8	25/40 Gbps to 30 m	2,000 MHz	Data center ToR runs

WHEN IT MATTERS Every link component — cable, jacks, patch panel — must meet the category or the link downgrades.

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Fiber: Single-Mode vs Multimode

Single-mode (SMF)

- Laser source, 8-10 micron core
- Runs of tens of kilometers
- Yellow jacket by convention
- WAN, metro and carrier links

Multimode (MMF)

- LED/VCSEL source, 50 or 62.5 micron core
- Hundreds of meters (OM3/OM4/OM5)
- Aqua or orange jacket
- Inside buildings and data centers

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Ethernet Standards to Recognise

Standard	Speed	Medium	Distance
100BASE-TX	100 Mbps	Cat 5 UTP	100 m
1000BASE-T	1 Gbps	Cat 5e UTP	100 m
10GBASE-T	10 Gbps	Cat 6a UTP	100 m
1000BASE-SX	1 Gbps	Multimode fiber	~550 m
1000BASE-LX	1 Gbps	Single-mode fiber	5-10 km
10GBASE-SR / LR	10 Gbps	MMF / SMF	400 m / 10 km

WHEN IT MATTERS Read the name: speed, BASE (baseband), then media hint — T twisted pair, S short fiber, L long fiber.

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Hubs, Switches and Domains

1

CSMA/CD

Hub-era rule: listen before sending, detect collisions, back off and retry.

2

Collision domain

Where frames can collide — an entire hub, but only a single switch port.

3

Broadcast domain

How far a broadcast reaches — the whole LAN/VLAN, bounded only by routers.

4

MAC address table

Switches learn source MACs per port and forward frames only where needed.

5

Full duplex

Switched ports send and receive simultaneously — collisions disappear.

6

STP

Spanning Tree Protocol blocks redundant links so switching loops cannot melt the LAN.

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Cable Ratings and Fire Safety

1

Plenum (CMP)

Fire-retardant, low-smoke jacket — required in air-handling spaces above ceilings.

2

Riser (CMR)

Rated for vertical runs between floors — flame-resistant but not plenum-grade.

3

PVC / non-plenum

General patching only — burns readily and emits toxic smoke.

4

Check the box

Fire rating is printed on the manufacturer's box and cable jacket — inspectors will.

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Structured Cabling Spaces

1

MDF

Main distribution frame — the primary equipment room where the backbone terminates.

2

IDF

Intermediate frames per floor/zone, uplinked back to the MDF.

3

Demarc

Where the provider's responsibility ends and yours begins.

4

Horizontal run

Solid-core cable from the IDF patch panel to each work-area wall outlet.

5

Patch panel

110-punchdown terminations for runs; patch cables jump to switch ports.

6

Racks and U

Equipment racks are 19 inches wide; height is measured in 1.75-inch units (U).

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Testing and Locating Cable

1. **Wiremap tester**
   Verifies each pin lands on the right pin — catches opens, shorts and split pairs.

2. **TDR / OTDR**
   Time-domain reflectometers measure length and locate mid-cable breaks (optical for fiber).

3. **Toner and probe**
   The 'fox and hound' — a tone generator plus probe finds one cable in a bundle.

4. **Loopback plug**
   Tests a NIC's ability to send and receive to itself.

5. **Crosstalk metrics**
   NEXT/FEXT measurements catch poor terminations that whisper between pairs.

6. **Attenuation**
   Signal loss over distance — why category ratings carry a length limit.

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Wireless Media

1

802.11 Wi-Fi

Unlicensed 2.4/5/6 GHz radio LANs, defined by the IEEE 802.11 family.

2

Cellular

Carrier-licensed 4G/5G — WAN links for mobile users, IoT and branch backup.

3

Satellite

Coverage anywhere with sky view; GEO adds ~600 ms latency, LEO constellations far less.

4

The trade

Wireless wins on mobility and reach — traded against interference and shared airtime.

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Transceiver Form Factors

Form factor	Speed	Notes
SFP	1 Gbps	Hot-swappable; fiber or RJ45
SFP+	10 Gbps	Same cage as SFP
QSFP	4 x 1 G / 40 G	Four lanes in one module
QSFP+ / QSFP28	40 G / 100 G	Spine and uplink ports

WHEN IT MATTERS Multi-source agreements (MSAs) keep modules interoperable across vendors — GBIC was the ancestor.

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Connectors to Recognise on Sight

1. **LC**
   Small square duplex fiber connector — the default on SFP modules.

2. **SC**
   Larger square push-pull fiber connector — 'stick and click'.

3. **ST**
   Round bayonet-twist fiber connector on legacy multimode plants.

4. **MPO / MTP**
   Multi-fiber ribbon connector feeding 40/100G parallel optics.

5. **RJ45 and RJ11**
   8-pin Ethernet jack vs the smaller 4/6-pin telephone jack.

6. **F-type and BNC**
   Threaded and bayonet coax connectors for broadband and lab gear.

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Network Topologies

1

Star / hub-and-spoke

All nodes home to a central device — today's standard LAN; the center is the failure point.

2

Full mesh

Every node links to every other: n(n-1)/2 links — maximum resilience, maximum cost.

3

Partial mesh / hybrid

Key sites meshed, others single-homed — the practical WAN compromise.

4

Point-to-point

One dedicated link between two endpoints — leased lines and wireless bridges.

5

Bus and ring (legacy)

Shared coax and token passing — obsolete, but the exam expects recognition.

6

Physical vs logical

The cabling layout and the data flow can differ — describe both when documenting.

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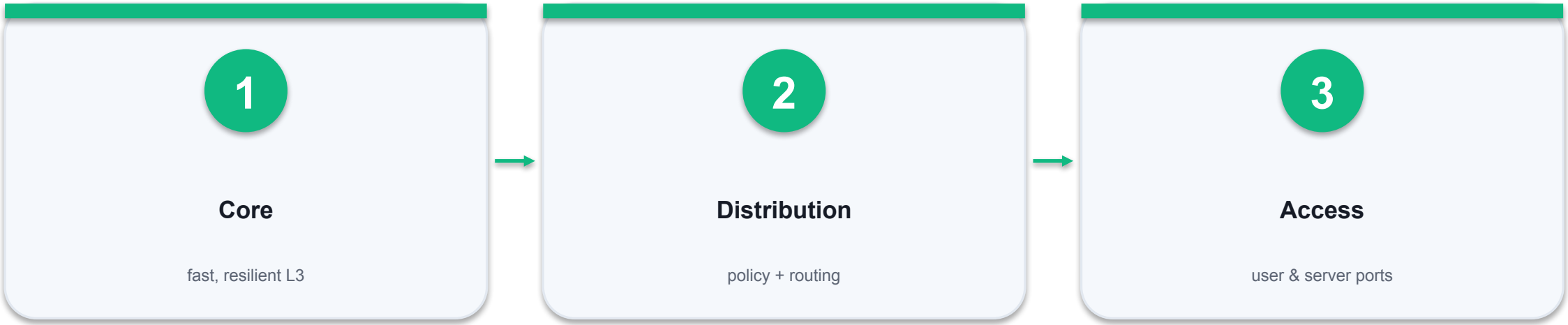
Choosing a Topology

Topology	Resilience	Cost	Best fit
Star	Low (center fails)	Low	Access LANs
Full mesh	Highest	Highest	Small WAN cores
Partial mesh	High	Medium	Regional WANs
Hub-and-spoke	Low	Low	Branch-to-HQ, SD-WAN
Point-to-point	Per-link	Low	Site interconnects

WHEN IT MATTERS Count the links before you promise resilience — every extra path is money and management.

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Three-Tier Architecture



WHY IT MATTERS

Each tier has one job: access connects hosts, distribution applies policy, core moves packets fast. Collapsed core merges the top two for smaller sites.

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Spine-and-Leaf vs Three-Tier

Three-tier (classic)

- Optimised for north-south client traffic
- Hosts can be several hops apart
- STP blocks redundant links
- Scale up: buy bigger core boxes

Spine-and-leaf (modern DC)

- Optimised for east-west server traffic
- Any leaf is two hops from any leaf
- All links active with ECMP routing
- Scale out: add spines and leaves

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Traffic Flows and Network Types

1

North-south

Traffic entering or leaving the data center — client to server.

2

East-west

Server-to-server traffic inside the DC — dominant since virtualization and microservices.

3

LAN / WLAN

A single site's wired or wireless network.

4

WAN

Links sites across geography — the Internet is the biggest WAN of all.

5

CAN / MAN

Campus and metropolitan scale sit between LAN and WAN.

6

PAN

Personal-area links: Bluetooth, USB tethering, wearables.

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Special-Purpose Networks

1

IoT devices

Sensors, cameras, controllers — isolate on their own SSID/VLAN with ACLs; patch firmware.

2

VoIP / UC

Voice, video and chat over IP: SIP signalling on 5060/5061, RTP media streams.

3

ICS / SCADA

Industrial control: PLCs, HMIs and RTUs supervised over wide areas — segment strictly from IT.

4

Storage networks

NAS/SAN traffic often rides its own network for performance and isolation.

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OBJ 1.7 · IPv4 ADDRESSING

IP addressing is the most-rewarded skill on the exam.

Every router, firewall rule and cloud VPC starts with correct CIDR math — this section builds it from binary upward.

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IPv4 Address Anatomy

1

32 bits, four octets

Dotted decimal such as 192.168.4.12 — each octet ranges 0-255.

2

Network + host portions

The subnet mask draws the boundary line between the two.

3

Binary underneath

192 is 11000000 — subnetting is just moving the 1s/0s boundary rightward.

4

Subnet mask

All 1s on the left, all 0s on the right — 255.255.255.0 is written /24 in CIDR.

5

Default gateway

The local router interface a host uses to reach remote networks.

6

Unique per network

Duplicate IPs break connectivity — usually a rogue static assignment.

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Decimal to Binary, the Exam Way

Each octet is 8 bits with place values 128 64 32 16 8 4 2 1.

```
Place: 128  64  32  16   8   4   2   1

171 -> 128? yes (43 left)   64? no
        32? yes (11 left)   16? no
         8? yes (3 left)     4? no
         2? yes              1? yes
171  =  1 0 1 0 1 0 1 1

224  =  11100000     95 = 01011111
255  =  11111111      0 = 00000000
```

Method

Subtract the largest place value that fits and write 1; write 0 where it does not fit.

Check

The place values you marked with 1 must sum back to the original number.

Why bother

Masks and CIDR boundaries only make sense at the bit level — this is the foundation.

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Classful Addressing (Historical)

Class	First octet	Default mask	Purpose
A	1-126	/8 (255.0.0.0)	Huge networks
B	128-191	/16 (255.255.0.0)	Medium networks
C	192-223	/24 (255.255.255.0)	Small networks
D	224-239	n/a	Multicast
E	240-255	n/a	Experimental

WHEN IT MATTERS 127.x.x.x is reserved for loopback. CIDR replaced classes in 1993, but the exam still tests them.

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RFC 1918 Private Ranges

CIDR block	Range	Addresses	Typical use
10.0.0.0/8	10.0.0.0 - 10.255.255.255	16.7 M	Enterprises, clouds
172.16.0.0/12	172.16.0.0 - 172.31.255.255	1 M	Mid-size nets, VPCs
192.168.0.0/16	192.168.0.0 - 192.168.255.255	65,536	Home and SOHO

WHEN IT MATTERS Private addresses never route on the Internet — NAT translates them at the edge.

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Special IPv4 Addresses

1

127.0.0.1

IPv4 loopback — the host talking to itself; IPv6 uses ::1.

2

169.254.x.x (APIPA)

Self-assigned when DHCP fails — a client stuck here cannot reach its DHCP server.

3

Network address

Host bits all 0 — names the subnet itself; never assignable to a host.

4

Broadcast address

Host bits all 1 — reaches every host on the subnet.

5

0.0.0.0

The 'any' address — unconfigured source, or the default route 0.0.0.0/0.

6

Public addresses

Everything routable outside private/reserved space — allocated through RIRs.

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How NAT Hides Private Addresses

1. Host 192.168.1.10 sends toward 8.8.8.8
2. Edge router swaps source to its public IP
3. Translation recorded in the NAT table
4. Reply arrives at the public address
5. Router maps it back to the private host

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ARP: Gluing Layer 3 to Layer 2

1

The problem

A host knows the destination IP, but the frame it builds needs a destination MAC.

2

ARP request

Broadcast: 'Who has 192.168.1.1?' — every host in the broadcast domain hears it.

3

ARP reply

The owner answers unicast with its MAC; the mapping is cached for reuse.

4

arp -a / ip neigh

Inspect the cache — stale or poisoned entries cause 'unreachable' faults.

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Local or Remote? The Mask Decides

1. AND your IP with the subnet mask
2. AND the destination with the same mask
3. Results equal? The destination is local
4. Local: ARP for the destination's MAC
5. Remote: ARP for the default gateway

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CIDR: Splitting a /24 into /26s

Borrow 2 host bits: 2^2 = 4 subnets, 2^6 - 2 = 62 hosts each.

```
192.168.10.0/24  ->  four /26 blocks of 64

Sub  Network         First  Last   Bcast
1    192.168.10.0    .1     .62    .63
2    192.168.10.64   .65    .126   .127
3    192.168.10.128  .129   .190   .191
4    192.168.10.192  .193   .254   .255

Mask /26 = 255.255.255.192  (...11000000)
```

Block size

256 - 192 = 64, so subnets start every 64: .0, .64, .128, .192.

Usable hosts

64 addresses minus network and broadcast leaves 62 per subnet.

Verify

The TIS IP Calculator confirms every range — exactly what you do in Lab 2.

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CIDR Host Counts to Memorise

Prefix	Mask	Subnets from a /24	Usable hosts
/25	255.255.255.128	2	126
/26	255.255.255.192	4	62
/27	255.255.255.224	8	30
/28	255.255.255.240	16	14
/29	255.255.255.248	32	6
/30	255.255.255.252	64	2 (P2P links)

WHEN IT MATTERS Pattern: each bit borrowed doubles the subnets and roughly halves the hosts.

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VLSM: Right-Sizing Subnets

1

The idea

Carve one block into different-sized subnets matched to each segment's host count.

2

Largest first

Allocate the biggest requirement first, then continue at the next free boundary.

3

Worked example

From a /24: a /25 for 100 users, a /26 for 50 servers, /30s for router links.

4

Why it matters

No wasted space — critical in cloud VPCs, where the CIDR is fixed at creation.

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IPv4 Addressing: Classic Fault Patterns

- Duplicate IP address — two hosts configured statically with the same address
- Duplicate MAC address — usually cloned virtual machines
- Wrong default gateway — local traffic works, remote traffic fails
- Wrong subnet mask — 'random' unreachable hosts on the same LAN
- APIPA 169.254.x.x — the DHCP server never answered
- Expired or exhausted lease — check scope size and lease times

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DOMAIN 1.0 · HANDS-ON

IPv4 Subnetting and CIDR

LAB 2

Split a /24 into four /26 subnets and verify the math with the TIS IP Calculator.

Use the TIS IP Calculator to break 192.168.10.0/24 into four /26 subnets, record each network, host range and broadcast, then assign an address from subnet 2 to a Linux interface.

YOU'LL BUILD

A completed four-subnet plan and a Linux interface carrying 192.168.10.65/26 as a labelled secondary address.

TOOLCHAIN

TIS IP Calculator (alfredang.github.io/ipcalculator), iproute2, Killercoda Ubuntu

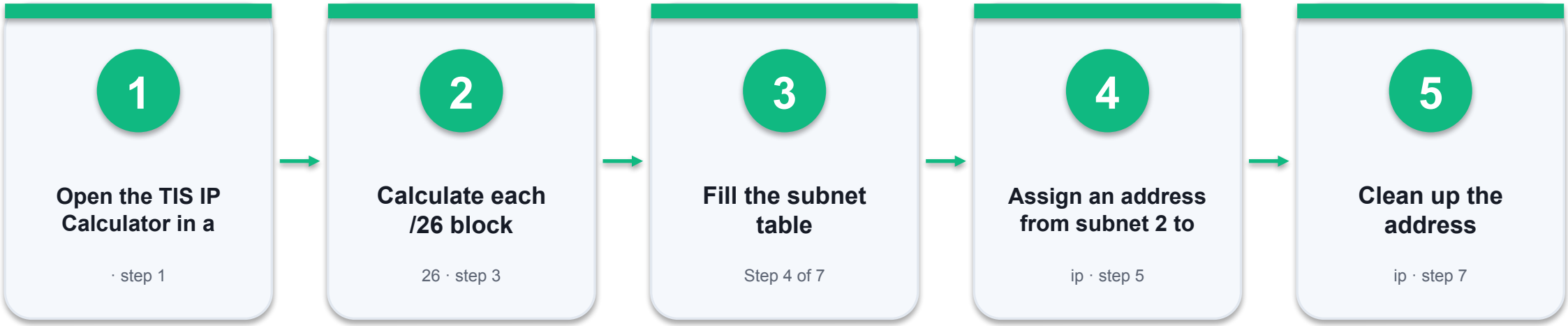
DONE WHEN

The calculator shows 62 usable hosts and unique network/broadcast per /26, and ip -4 addr show eth0 lists inet 192.168.10.65/26 on label eth0:1.

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LAB 2 · PROCESS MAP

How Lab 2 Runs



THE POINT

Split a /24 into four /26 subnets and verify the math with the TIS IP Calculator.

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LAB 2 · VERIFY

IPv4 Subnetting and CIDR

✅ Expected result

The calculator shows 62 usable hosts and unique network/broadcast per /26, and ip -4 addr show eth0 lists inet 192.168.10.65/26 on label eth0:1.

IF IT DOESN'T WORK

Calculator ranges look wrong

Confirm you entered prefix /26, not /24 — the prefix length drives every boundary.

RTNETLINK: File exists

The address is already assigned — remove it first with ip addr del 192.168.10.65/26 dev eth0.

Address missing after add

Run as root and check you used dev eth0 — Killercoda's primary NIC is eth0.

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Why IPv6

1

128-bit addresses

2^128 addresses end exhaustion — written as eight hex groups separated by colons.

2

Shortening rules

Drop leading zeros in each group; compress one run of zero groups with :: (once only).

3

No broadcast

IPv6 replaces broadcast with multicast (ff02::1 = all nodes) and anycast.

4

No NAT required

End-to-end addressing returns; firewalls, not address translation, enforce the boundary.

5

Efficient header

Fixed 40-byte header and no router fragmentation — faster forwarding.

6

2001:db8::/32

The documentation prefix — always safe for labs, examples and courseware.

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Shortening an IPv6 Address

Two rules: strip leading zeros per group, and compress one zero run with ::.

```
Full:
2001:0db8:0000:0000:0001:0000:0000:1234

Rule 1 - drop leading zeros:
2001:db8:0:0:1:0:0:1234

Rule 2 - compress ONE zero run:
2001:db8::1:0:0:1234

Wrong: 2001:db8::1::1234  (two ::)
```

Only one ::

Two compressions would be ambiguous — the parser could not count the missing groups.

Pick the longest run

When two zero runs tie, compress the leftmost — the convention resolvers use.

Exam trap

Expanding a shortened address back to eight groups is a favourite question.

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Key IPv6 Address Types

Type	Prefix	Role
Global unicast	2000::/3	Public, Internet-routable
Link-local	fe80::/10	Auto on every interface; one link only
Unique local	fc00::/7	Private — the RFC1918 analogue
Multicast	ff00::/8	ff02::1 all nodes, ff02::2 routers
Loopback	::1/128	Same role as 127.0.0.1

WHEN IT MATTERS Every IPv6 interface always holds a link-local address — even with no router or DHCP anywhere.

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SLAAC: An Address Without a Server



WHY IT MATTERS

A v6 host configures itself from router advertisements alone — DHCPv6 is optional, used only when you need central control or extra options like DNS.

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Interface IDs and Coexistence

1

EUI-64

Builds the 64-bit interface ID from the MAC: split it, insert FFFE, flip bit 7.

2

Privacy extensions

Random interface IDs defeat device tracking — the default on modern OSes.

3

Dual stack

Run IPv4 and IPv6 side by side — today's standard migration strategy.

4

Tunnels and NAT64

6to4/Teredo carry v6 inside v4; NAT64/DNS64 lets v6-only clients reach v4 servers.

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IPv6 Subnetting Conventions

1

Every LAN is a /64

SLAAC needs 64 bits of interface ID — never subnet a LAN smaller than /64.

2

A site gets a /48

That is 65,536 /64 subnets — plan generously, address conservation is over.

3

Nibble boundaries

Subnet in 4-bit steps so prefixes stay readable in hex.

4

Zone IDs

Link-local commands need %eth0 to choose the interface — exactly as in Lab 3.

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DOMAIN 1.0 · HANDS-ON

IPv6 Addressing and Link-Local

LAB 3

Plan an IPv6 /64, add a global address, and ping link-local using a zone ID.

Break down 2001:db8::/64 in the TIS IP Calculator's IPv6 mode, add a documentation-range global address to eth0, observe the auto-generated fe80:: link-local address, and ping across both.

YOU'LL BUILD

An interface holding both a link-local fe80:: address and a global 2001:db8::10/64, verified by ping.

TOOLCHAIN

TIS IP Calculator (alfredang.github.io/ipcalculator), iproute2, ping6, Killercoda

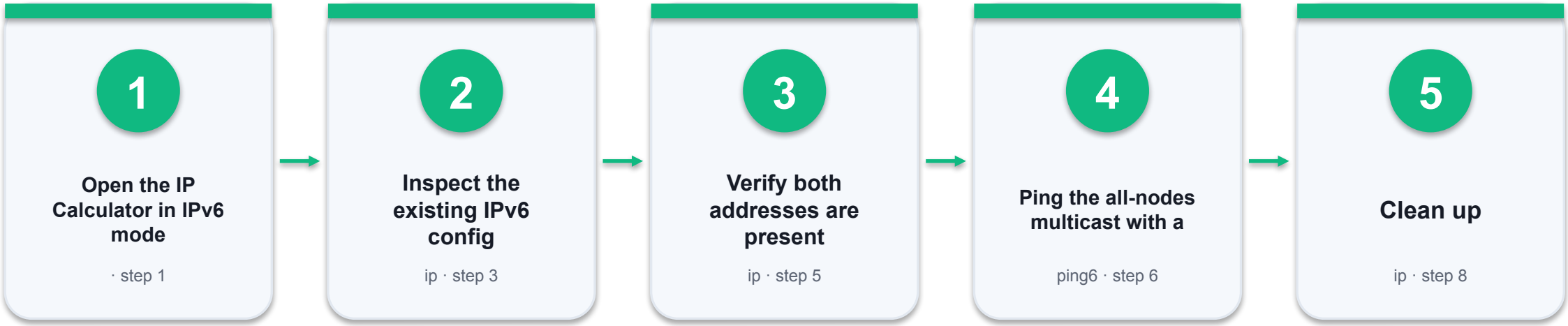
DONE WHEN

ip -6 addr show eth0 lists both fe80::... and 2001:db8::10/64, and both pings reply (the ff02::1 ping needs the %eth0 zone suffix).

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LAB 3 · PROCESS MAP

How Lab 3 Runs



THE POINT

Plan an IPv6 /64, add a global address, and ping link-local using a zone ID.

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LAB 3 · VERIFY

IPv6 Addressing and Link-Local

✅ Expected result

ip -6 addr show eth0 lists both fe80::... and 2001:db8::10/64, and both pings reply (the ff02::1 ping needs the %eth0 zone suffix).

IF IT DOESN'T WORK

ping6: Invalid argument

Link-local targets need a zone — use ff02::1%eth0 so the kernel knows the egress interface.

No fe80:: address on eth0

IPv6 is disabled — sysctl -w net.ipv6.conf.eth0.disable_ipv6=0, then re-check.

Global ping fails

Confirm the add worked: ip -6 addr show eth0 must list 2001:db8::10/64 before you ping.

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Port Numbers: The L4 Address

1

0-1023 well-known

Reserved for standard services — HTTP 80, SSH 22, DNS 53. Memorise these.

2

1024-49151 registered

Vendor-registered applications: RDP 3389, MySQL 3306, SIP 5060.

3

49152-65535 ephemeral

Random client-side source ports, recycled per connection.

4

The socket pair

Source IP:port plus destination IP:port uniquely identifies every conversation.

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TCP vs UDP

TCP — connection-oriented

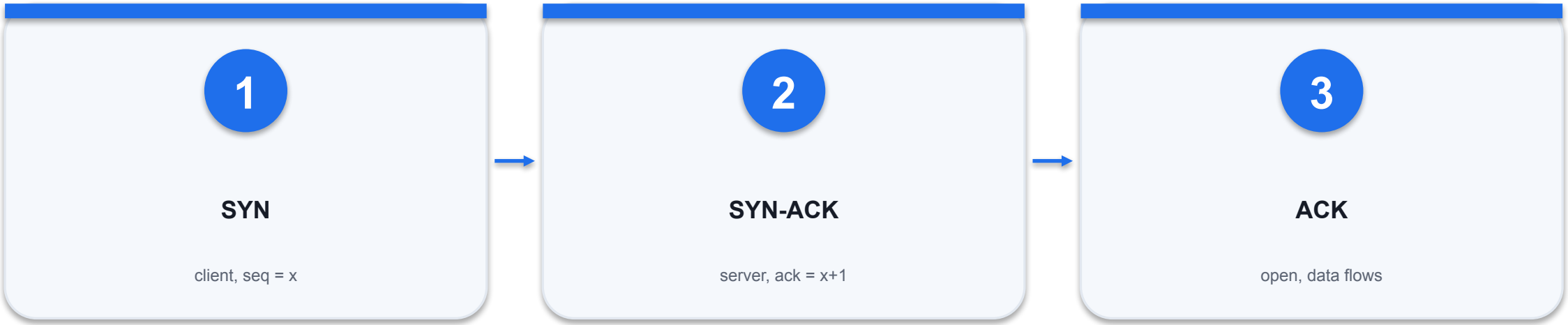
- Three-way handshake before any data
- Sequencing, ACKs and retransmission
- Flow and congestion control
- HTTP(S), SSH, SMTP, FTP, RDP

UDP — connectionless

- No handshake — just send
- No delivery guarantee, low overhead
- Ideal for real-time and tiny queries
- DNS, DHCP, TFTP, NTP, SNMP, RTP

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The TCP Three-Way Handshake



WHY IT MATTERS

The [S] flag you captured in Lab 1 is step one of this exchange — and half-open handshakes are how SYN floods and stealth scans work.

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TCP or UDP for Your Application?



YES — use TCP

Pay the handshake and header overhead to gain sequencing, acknowledgements and retransmission — right for web, mail, file transfer.

NO — use UDP

Skip the ceremony for speed: lost packets are ignored or handled by the app — right for voice, video, DNS lookups and telemetry.

QUIC (HTTP/3) builds TCP-like reliability on top of UDP — know that it exists.

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ICMP, IGMP, GRE and IPsec

1

ICMP

Layer 3 control messages — echo (ping), destination unreachable, TTL exceeded. No ports.

2

IGMP

Hosts join and leave IPv4 multicast groups in 224.0.0.0/4.

3

GRE

A generic tunnel that wraps any protocol inside IP — no encryption of its own.

4

IPsec AH

Authentication Header: integrity and origin proof, but no confidentiality.

5

IPsec ESP

Encapsulating Security Payload: encryption — the workhorse of site-to-site VPNs.

6

IKE

Negotiates IPsec keys and security associations over UDP 500/4500.

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Ports: File Transfer and Remote Access

Port	Protocol	Transport	Security note
20/21	FTP data / control	TCP	Cleartext — avoid
22	SSH and SFTP	TCP	Encrypted channel
23	Telnet	TCP	Cleartext — use SSH
69	TFTP	UDP	LAN boot/config only
445	SMB	TCP	Windows file sharing
3389	RDP	TCP	TLS-protected desktop

WHEN IT MATTERS Pairs to contrast on the exam: 21 vs 22 (SFTP), 23 vs 22 (SSH), 69 vs 21.

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Ports: Web, Mail and Name Services

Port	Protocol	Transport	Notes
80 / 443	HTTP / HTTPS	TCP	443 adds TLS
25 / 587	SMTP / submission	TCP	587 = STARTTLS sending
110 / 995	POP3 / POP3S	TCP	Download-and-delete mail
143 / 993	IMAP / IMAPS	TCP	Server-side mailboxes
53	DNS	UDP + TCP	TCP for zone transfers
67 / 68	DHCP	UDP	Server 67, client 68

WHEN IT MATTERS Mail flows: SMTP sends between servers, POP3/IMAP retrieve to clients — the S-suffixed ports add TLS.

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Securing E-mail: The Port Story

1

Legacy trio

SMTP 25, POP3 110, IMAP 143 — all cleartext, all still on the exam.

2

Implicit TLS

Wrapped-in-TLS variants: SMTPS 465, POP3S 995, IMAPS 993.

3

STARTTLS on 587

The modern client-submission port: starts plain, upgrades to TLS in-band.

4

Anti-spoofing

SPF, DKIM and DMARC live in DNS TXT records — covered again in the DNS lab.

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Ports: Infrastructure Services

Port	Protocol	Transport	Notes
123	NTP	UDP	Clock sync, stratum tree
161 / 162	SNMP / traps	UDP	Monitoring; v3 adds crypto
389 / 636	LDAP / LDAPS	TCP	Directory queries
514	Syslog	UDP	Central log collection
5060 / 5061	SIP / SIP-TLS	UDP/TCP	VoIP call signalling
500 / 4500	IKE / NAT-T	UDP	IPsec key negotiation

WHEN IT MATTERS These six rows plus the previous two tables cover every port listed in objective 1.4.

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Reading ss and nmap Output

Two commands cover both directions: what you expose, and what a target exposes.

```
# ss -tulnp          (your own listeners)
tcp  LISTEN  0.0.0.0:22    sshd
udp  UNCONN  127.0.0.53:53 systemd-resolve

# nmap -sT -p 1-1024 scanme.nmap.org
22/tcp  open      ssh
80/tcp  open      http
25/tcp  filtered  smtp
```

LISTEN vs open

ss shows sockets bound locally; nmap shows what a remote host answers on.

filtered

A firewall silently dropped the probe — no RST ever came back.

Stay numeric

The -n flags keep raw port numbers visible so you learn numbers, not names.

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DOMAIN 1.0 · HANDS-ON

Common Ports and Protocols

LAB 4

Connect protocol names to well-known port numbers using ss and nmap.

Enumerate the listening TCP/UDP sockets on your own host with ss, scan the well-known port range of scanme.nmap.org with nmap, and map every open port you find to the N10-009 protocol list.

YOU'LL BUILD

A completed port-to-protocol map covering your own listeners and the remote scan results.

TOOLCHAIN

nmap, ss, net-tools (netstat), getent, Killercoda Ubuntu

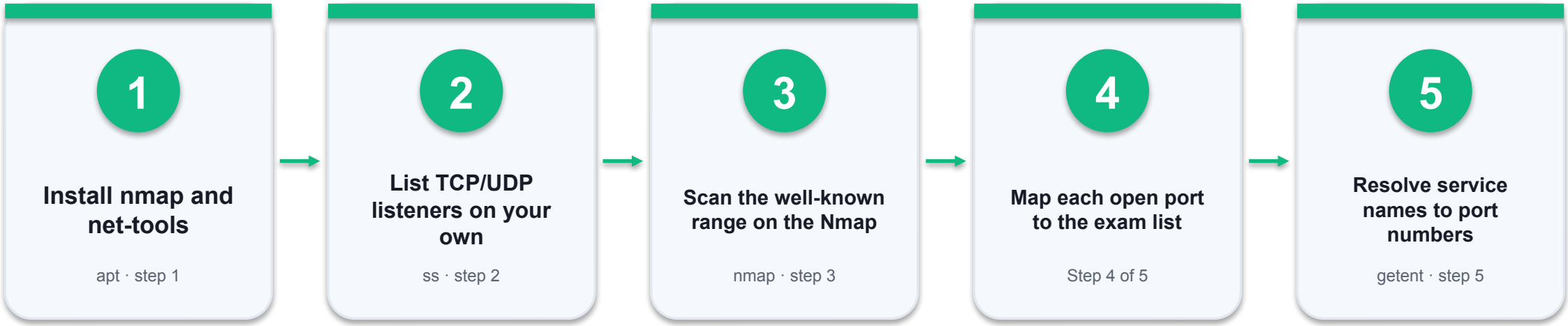
DONE WHEN

ss shows local listeners with numeric ports; nmap reports open ports on scanme.nmap.org (expect 22 and 80); getent services https returns 443/tcp.

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LAB 4 · PROCESS MAP

How Lab 4 Runs



THE POINT

Connect protocol names to well-known port numbers using ss and nmap.

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LAB 4 · VERIFY

Common Ports and Protocols

✅ Expected result

ss shows local listeners with numeric ports; nmap reports open ports on scanme.nmap.org (expect 22 and 80); getent services https returns 443/tcp.

IF IT DOESN'T WORK

nmap scan returns nothing

Check egress with ping scanme.nmap.org — playground networks can be slow; re-run the scan.

ss: command not found

Use netstat -tulnp from net-tools instead — both list the same listening sockets.

Ports show names, not numbers

Keep the -n flag (ss -tulnp) so output stays numeric and you learn the numbers.

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DNS: The Internet's Directory

1

FQDN

host.domain.tld — www.example.com is read right to left, ending at the root.

2

Hierarchy

Root servers -> TLD servers (.com, .sg) -> authoritative servers for each zone.

3

Recursive resolver

Your configured server (e.g. 8.8.8.8) chases referrals and caches the answers.

4

UDP/TCP 53

Queries ride UDP 53; zone transfers and oversized answers switch to TCP 53.

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A DNS Lookup, Step by Step

1. Client asks its recursive resolver
2. Cache hit? Answer returns instantly
3. Miss: resolver asks root, then .com TLD
4. TLD refers to the authoritative server
5. A record returned and cached per TTL

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DNS Server Roles and Variants

1

Authoritative

Holds the zone file and gives the definitive answer for its domains.

2

Recursive / caching

Resolves on behalf of clients and remembers answers for the TTL.

3

Internal DNS

Private zones for corp resources — split horizon keeps them invisible outside.

4

Dynamic DNS

Clients or DHCP update records automatically as addresses change.

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DNS Record Types on the Exam

Record	Maps	Example / note
A / AAAA	Name -> IPv4 / IPv6	The bread-and-butter records
CNAME	Name -> another name	An alias — never points to an IP
MX	Domain -> mail host	Priority number: lower wins
NS / SOA	Zone -> servers / authority	SOA holds serial and timers
PTR	IP -> name	Reverse zones; mail server checks
TXT / SRV	Text / service + port	SPF and DKIM live in TXT

WHEN IT MATTERS Lab 5 queries every one of these with dig against live domains.

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dig in Action

dig +short strips the noise and prints only the answer — perfect for learning records.

```
$ dig +short A google.com
142.250.4.100

$ dig +short MX google.com
10 smtp.google.com.

$ dig +short -x 8.8.8.8
dns.google.

$ dig +short CNAME www.github.com
github.com.
```

MX priority

The leading 10 is the preference — lower numbers are tried first.

-x reverses

Turns 8.8.8.8 into its PTR name from the in-addr.arpa reverse zone.

CNAME chains

www.github.com is an alias for github.com — the resolver follows the chain.

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DOMAIN 1.0 · HANDS-ON

DNS Record Types

LAB 5

Query every N10-009 DNS record type with dig and interpret the answers.

Use dig to query A, AAAA, MX, TXT, NS, CNAME, PTR, SOA and SRV records against live domains, reading addresses, mail priorities, aliases and zone-authority data from the answers.

YOU'LL BUILD

A worked set of dig queries — one per record type on the exam — with each answer interpreted.

TOOLCHAIN

dig, nslookup, host (dnsutils), Killercoda Ubuntu

DONE WHEN

Every dig query returns an answer: A gives IPv4, AAAA gives IPv6, MX shows priorities, -x 8.8.8.8 returns dns.google, and the SOA lists the primary NS and serial.

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LAB 5 · PROCESS MAP

How Lab 5 Runs



THE POINT

Query every N10-009 DNS record type with dig and interpret the answers.

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LAB 5 · VERIFY

DNS Record Types

✅ Expected result

Every dig query returns an answer: A gives IPv4, AAAA gives IPv6, MX shows priorities, -x 8.8.8.8 returns dns.google, and the SOA lists the primary NS and serial.

IF IT DOESN'T WORK

dig: command not found

Install the tools: apt update && apt install -y dnsutils.

Empty answer for CNAME

Not every name has one — query an alias such as www.github.com, not the bare domain.

SERVFAIL or timeout

Query an explicit resolver: dig @8.8.8.8 +short A google.com to bypass the local stub.

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DORA: How a Lease Happens



WHY IT MATTERS

Discover and Request are broadcasts on UDP 67/68 — which is why every broadcast domain needs its own DHCP server or a relay pointing to one.

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DHCP Server Concepts

1

Scope / pool

The range of addresses the server may hand out on a subnet.

2

Lease time

Addresses are borrowed — clients try to renew at 50% of the lease.

3

Reservation

A fixed IP tied to a MAC address — printers and servers stay stable.

4

Exclusion range

Addresses inside the scope the server must never assign.

5

Options

The lease also carries mask, gateway, DNS servers, NTP and more.

6

Relay / IP helper

Forwards client broadcasts to a central DHCP server across subnets.

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Static vs Dynamic vs APIPA

Static assignment

- Manually configured per host
- For routers, servers, printers
- No dependency on DHCP
- Risk: typos, duplicates, drift

Dynamic (DHCP)

- Automatic from a scope
- Self-documenting via leases
- Scales to thousands of hosts
- One server per broadcast domain

APIPA fallback

- 169.254.0.0/16 self-assigned
- Means DHCP was unreachable
- Link-local talk only, no gateway
- A diagnostic clue, not a fix

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Watching DORA in tcpdump

Filter on 'port 67 or port 68' and renew a lease — all four messages appear.

```
# tcpdump -i eth0 -nn -v 'port 67 or port 68'
0.0.0.0.68 > 255.255.255.255.67:
   DHCP-Message: Discover
10.244.0.1.67 > 255.255.255.255.68:
   DHCP-Message: Offer  (your-ip 10.244.0.5)
0.0.0.0.68 > 255.255.255.255.67:
   DHCP-Message: Request
10.244.0.1.67 > 255.255.255.255.68:
   DHCP-Message: ACK  (lease-time 3600)
```

Broadcasts

The client has no address yet, so Discover and Request come from 0.0.0.0.

The Offer

Carries the proposed IP, mask, gateway, DNS and lease time in DHCP options.

ACK seals it

Only after the ACK may the client actually use the address.

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When DHCP Goes Wrong

1

Exhausted scope

More devices than addresses — shrink lease times or widen the scope.

2

Rogue DHCP server

An unauthorised server hands out bad leases — find it, then enable DHCP snooping.

3

Missing relay

Clients across a router never reach the server — configure an IP helper address.

4

IPAM

IP address management tools track scopes, leases and reservations across sites.

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DOMAIN 1.0 · HANDS-ON

DHCP Lease Capture (DORA)

LAB 6

Capture all four DHCP DORA messages during a live lease renewal.

Run tcpdump against UDP ports 67/68 while releasing and re-requesting your DHCP lease with dhclient, then identify the Discover, Offer, Request and Acknowledge messages in the capture.

YOU'LL BUILD

A capture showing the full DORA exchange and a freshly renewed lease on eth0.

TOOLCHAIN

isc-dhcp-client (dhclient), tcpdump, Killercoda Ubuntu

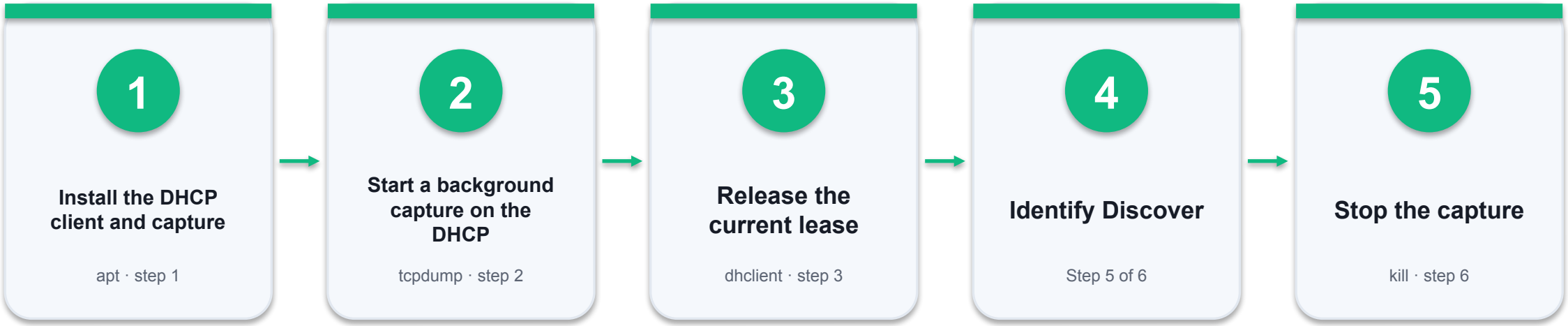
DONE WHEN

The capture shows DHCPDISCOVER broadcast from 0.0.0.0 to 255.255.255.255, followed by Offer, Request and Ack; dhclient reports the address it bound.

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LAB 6 · PROCESS MAP

How Lab 6 Runs



THE POINT

Capture all four DHCP DORA messages during a live lease renewal.

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LAB 6 · VERIFY

DHCP Lease Capture (DORA)

✅ Expected result

The capture shows DHCPDISCOVER broadcast from 0.0.0.0 to 255.255.255.255, followed by Offer, Request and Ack; dhclient reports the address it bound.

IF IT DOESN'T WORK

Connectivity lost after release

Expected — dhclient -v eth0 re-acquires a lease; if the session breaks, reload the playground.

No DHCP packets in capture

Check the filter is 'port 67 or port 68' and start the capture before releasing.

dhclient: command not found

Install it: apt update && apt install -y isc-dhcp-client.

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OBJ 1.3 · CLOUD

The network is now software.

Hypervisors, NFV and cloud VPCs rebuild every concept in this domain — switches, routers, firewalls, subnets — as code you can deploy in seconds.

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Virtualization Foundations

1

Type 1 hypervisor

Bare-metal: ESXi, Hyper-V, KVM — the production data center standard.

2

Type 2 hypervisor

Hosted on a desktop OS: VirtualBox, VMware Workstation — labs and testing.

3

Why virtualize

Consolidation, power savings, snapshots and fast recovery of whole systems.

4

vSwitch

A software switch connecting VM NICs — configure it like a physical switch.

5

Virtual disks

VHD/VHDX/VMDK files hold entire machines — portable and snapshottable.

6

Not emulation

Virtualization shares real CPU hardware; emulation imitates different hardware.

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NFV and SDN

1

NFV

Network functions — router, firewall, load balancer — shipped as software appliances.

2

SDN

Splits the control plane from the data plane; the network is programmed centrally.

3

The planes

Data forwards packets, control decides routes, management configures it all.

4

Distributed switching

One logical switch spans every host — configure once, applied everywhere.

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Cloud Service Models

Model	Provider manages	You manage	Example
IaaS	Hardware, hypervisor	OS, apps, data	AWS EC2
PaaS	Up to the runtime	Code and data	Heroku, App Service
SaaS	The whole stack	Usage and config	Microsoft 365
DaaS	Virtual desktops	Profiles, policy	Windows 365

WHEN IT MATTERS The more the provider manages, the less you control — pick per workload, not per company.

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OBJ 1.3 · CLOUD

Cloud Deployment Models

Model	Who uses it	Trade-off
Public	Anyone, shared infrastructure	Lowest cost, least control
Private	One organisation	Full control, full cost
Hybrid	A mix of both	Flexibility + complexity
Community	A shared-interest group	Split compliance costs

WHEN IT MATTERS Hybrid is the enterprise norm: sensitive systems stay private, elastic workloads burst to public.

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Cloud Characteristics

1

Scalability

Grow capacity on demand without buying or racking hardware.

2

Elasticity

Grow and shrink automatically with load — pay only for what runs.

3

Multitenancy

Many customers share the provider's infrastructure, isolated logically.

4

Least privilege

Cloud IAM grants every identity only the access it needs — nothing more.

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Inside a VPC

1. **VPC**
   Your private software-defined network in a cloud region, with a CIDR block you choose.

2. **Subnets**
   Carve the VPC CIDR per availability zone; public subnets route to an internet gateway.

3. **Route tables**
   Decide where each subnet's traffic goes — the VPC's built-in routers.

4. **Security groups**
   Stateful per-instance firewalls — allow rules only, replies pass automatically.

5. **Network ACLs**
   Stateless subnet-level filters supporting both allow and deny rules.

6. **Gateways**
   Internet, NAT and VPN gateways connect the VPC to everything outside it.

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Security Groups vs Network ACLs

Security group (instance level)

- Stateful — return traffic auto-allowed
- Allow rules only
- Attached to instances / NICs
- First line for app-tier isolation

Network ACL (subnet level)

- Stateless — reply needs its own rule
- Allow and deny rules, numbered
- Applies to everything in the subnet
- Broad guardrails and IP blacklists

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A Packet Leaves Your VPC



WHY IT MATTERS

Cloud troubleshooting is this pipeline in order: instance firewall, subnet ACL, route, gateway — the same layered thinking as the OSI model.

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Connecting to the Cloud

1

VPN over Internet

IPsec tunnels to a cloud VPN gateway — quick, cheap, encrypted.

2

Direct Connect

Private circuits (ExpressRoute, Direct Connect) — predictable latency and bandwidth.

3

Transit gateways

Hub-and-spoke interconnect for many VPCs and on-prem sites.

4

NAT gateway

Lets private subnets reach out to the Internet without being reachable inbound.

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Infrastructure as Code

1

IaC

Networks described in files (Terraform, CloudFormation) — versioned like source code.

2

Automation

Provisioning without manual steps: consistent, repeatable, reviewable.

3

Orchestration

Chains automated tasks into whole-environment deployments.

4

Why it wins

A VPC built from code can be rebuilt identically in minutes — Lab 7 does it by hand once.

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Public, Private or Hybrid?



YES — private or hybrid

Keep regulated data on private infrastructure and burst stateless workloads to public cloud through a hybrid design.

NO — public first

Start public for elasticity and low capital cost; revisit only if scale, latency or new rules demand it.

Cost surprises come from egress traffic — model data flows before you commit.

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DOMAIN 1.0 · HANDS-ON

Simulating a Cloud VPC with Namespaces

LAB 7

Build a VPC-style virtual network from namespaces, a bridge and veth pairs.

Recreate cloud VPC building blocks on one Linux host: two network namespaces act as VMs, a Linux bridge acts as the virtual switch, and veth pairs cable them together on subnet 10.0.1.0/24.

YOU'LL BUILD

Two isolated 'VMs' (web and db) pinging each other through a Linux bridge — a working VPC subnet.

TOOLCHAIN

iproute2 (ip netns, ip link, veth, bridge), Killercoda Ubuntu

DONE WHEN

ip netns exec web ping -c2 10.0.1.20 returns two replies — two isolated hosts sharing one subnet through a bridge, with no router involved.

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LAB 7 · PROCESS MAP

How Lab 7 Runs



THE POINT

Build a VPC-style virtual network from namespaces, a bridge and veth pairs.

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LAB 7 · VERIFY

Simulating a Cloud VPC with Namespaces

✅ Expected result

ip netns exec web ping -c2 10.0.1.20 returns two replies — two isolated hosts sharing one subnet through a bridge, with no router involved.

IF IT DOESN'T WORK

Network is unreachable

Bring interfaces up in both namespaces, including lo, and re-check the 10.0.1.x addresses.

Destination Host Unreachable

Verify both bridge-side ends are enslaved and up: ip link set br-web master br0 (and br-db).

Cannot find device veth-web

That end now lives inside the namespace — manage it via ip netns exec web ip link ...

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Domain 1 Exam-Day Anchors

1

OSI in order

Please Do Not Throw Sausage Pizza Away — and PDUs: bits, frame, packet, segment, data.

2

Port table cold

FTP 21, SSH 22, DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, NTP 123, RDP 3389.

3

RFC1918 ranges

10/8, 172.16/12, 192.168/16 — private, NATed, never routed publicly.

4

Subnet math

Block size = 256 minus the mask's last octet; hosts = 2^host-bits minus 2.

5

DORA + SLAAC

DHCP is Discover-Offer-Request-Ack; IPv6 can self-configure from an RA.

6

Cloud layers

SG then NACL then route then gateway — the packet pipeline of every VPC.

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NEXT · DOMAIN 2

Concepts are the map. Everything else is the road.

Domain 2 puts these concepts to work: routing, switching, VLANs and wireless — built on the addressing and models you now own.

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Recap — Networking Concepts

- You can now: Capture live traffic and map each protocol header to its OSI layer.
- You can now: Split a /24 into four /26 subnets and verify the math with the TIS IP Calculator.
- You can now: Plan an IPv6 /64, add a global address, and ping link-local using a zone ID.
- You can now: Connect protocol names to well-known port numbers using ss and nmap.
- You can now: Query every N10-009 DNS record type with dig and interpret the answers.
- You can now: Capture all four DHCP DORA messages during a live lease renewal.
- You can now: Build a VPC-style virtual network from namespaces, a bridge and veth pairs.

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2.0

DOMAIN 2.0 · 20% OF THE EXAM

Network Implementation

Routing technologies · switching technologies · wireless devices · physical installations

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WHAT YOU'LL DO

Hands-On Labs — Network Implementation

Labs 8–10

- Static Routing Between Two Subnets
- 802.1Q VLAN Tagging
- Link Aggregation (LACP Bond)

Labs 11–13

- Wireless Concepts and Channel Survey
- NAT and PAT with iptables
- DHCP Server with isc-dhcp-server

Lab 14

- Authoritative DNS with BIND9

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20% OF THE EXAM

Domain 2 — Network Implementation.

Routing, switching, wireless and physical installations — 20% of the N10-009 exam and the heart of the build-it job role.

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Routers Connect Network IDs

1

Routers care about destinations

A router reads the destination network ID, looks it up in its routing table and forwards out the matching interface.

2

Any medium

Routing is Layer 3 — a router joins Ethernet, fiber and wireless segments because IP is independent of the link below.

3

Every router has a routing table

The table maps destination networks to a next hop or exit interface; no match and no default route means the packet is dropped.

4

Default route

0.0.0.0/0 is the gateway of last resort — everything without a more specific match goes there, usually upstream.

5

Gateway router

The router end hosts point at (their default gateway); it moves traffic between the LAN and everything beyond.

6

Upstream router

The next router towards the ISP and Internet core — SOHO gateways simply default-route everything upstream.

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Reading a Routing Table

1

Destination + prefix

The network this row matches, e.g. 10.2.2.0/24 — the prefix length says how many leading bits must match.

2

Next hop (via)

The neighbouring router's IP the packet is handed to; it must sit on a directly connected subnet.

3

Interface (dev)

Which NIC the frame leaves on; connected routes appear automatically when an IP is configured.

4

Metric

Tie-breaker between equal routes from the same protocol — lower wins.

5

See it yourself

Linux: ip route. Windows: route print or netstat -r. Cisco: show ip route.

6

Loopback & multicast

127.0.0.0/8 stays on-box; 224.0.0.0/4 (class D) is multicast — both appear as special rows.

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Metrics and Convergence — Key Terms

1

Hop count

Routers crossed to reach the destination — RIP's only measure, blind to link speed.

2

Bandwidth / cost

OSPF derives cost from interface bandwidth, so a 3-hop 10G path beats a 1-hop 100M link.

3

Latency / delay

EIGRP factors delay into its composite metric; satellite links get penalised.

4

MTU

Maximum transmission unit per link — mismatches cause fragmentation or silent drops (Lab 33).

5

Convergence

The state where every router's table reflects the current topology; faster is better.

6

Convergence window

Between failure and re-convergence, packets can loop or blackhole — protocols race to shrink it.

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Static vs Dynamic Routing

Static routing

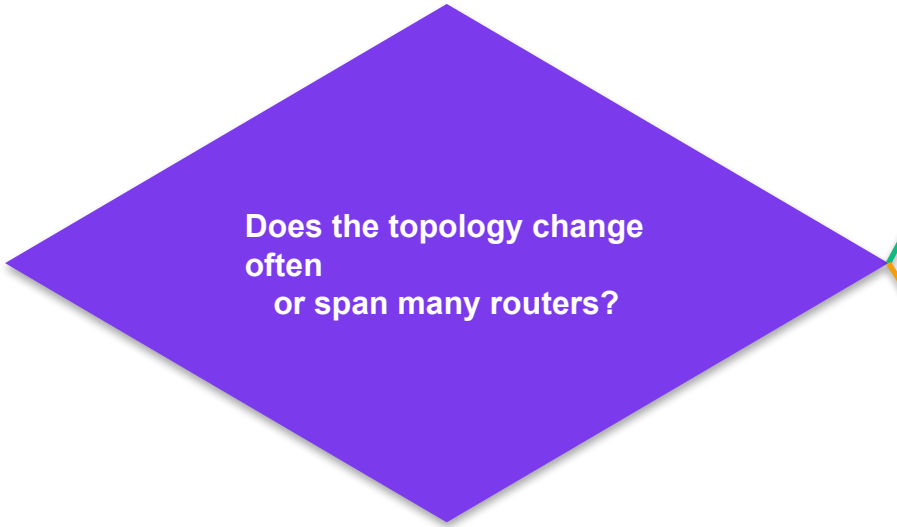
- Admin types each route by hand — persistent until removed
- Zero protocol overhead: no hellos, no updates, no CPU
- Predictable and secure — nothing advertises a bad route
- Does not adapt: a failed link means manual repair
- Best for stubs, labs and default routes

Dynamic routing

- Routers exchange reachability via a routing protocol
- Learns new subnets and routes around failures itself
- Costs CPU, bandwidth and a convergence window
- Needs securing (authentication) against rogue peers
- Best for anything beyond a handful of routers

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Choosing a Routing Approach



YES — DYNAMIC ROUTING

Run OSPF (open) or EIGRP (Cisco); routers learn and re-converge around failures automatically. Use BGP between autonomous systems.

NO — STATIC ROUTES

A handful of fixed routes is simpler, uses zero CPU or bandwidth and cannot be poisoned by a rogue peer — ideal for stub networks.

Most real networks mix both: dynamic in the core, static at stubs and for the default route.

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IGP vs EGP — Where Protocols Live

1

Autonomous System (AS)

One organisation's network under a single routing policy, identified by an AS number — your ISP is one, Google is another.

2

IGP — interior

Runs INSIDE one AS: OSPF, EIGRP, RIP, IS-IS. Optimises for fast paths and quick convergence.

3

EGP — exterior

Runs BETWEEN ASes. BGP is the only EGP in use — it optimises for policy and reachability, not speed.

4

Distance vector

Routers tell neighbours their whole table periodically ('routing by rumour') — RIP.

5

Link state

Routers flood link advertisements so everyone builds the same map, then run Dijkstra — OSPF, IS-IS.

6

Path vector

BGP advertises the full AS path per prefix; a router seeing its own AS number detects the loop.

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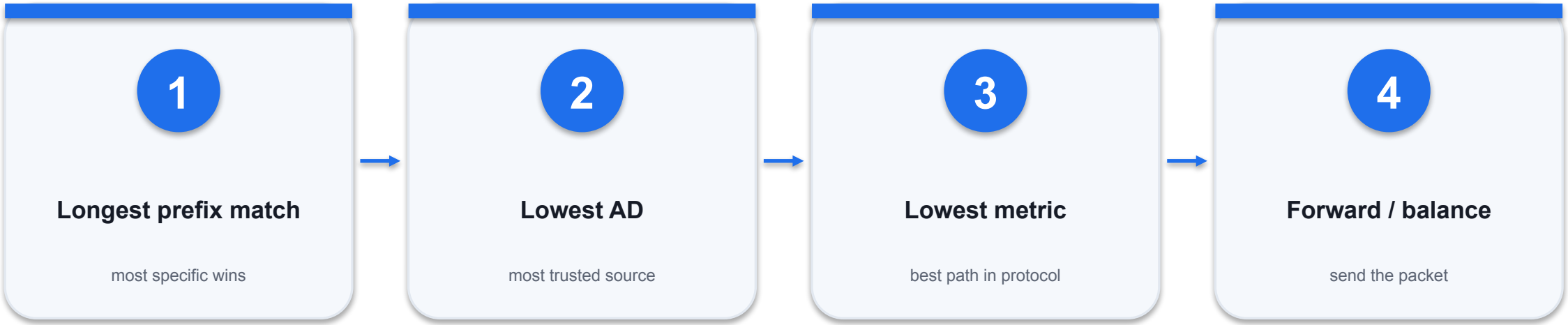
Administrative Distance — Who Do You Trust?

Route source	AD	Trust level
Connected	0	Directly attached — always wins
Static	1	Manually configured
eBGP	20	External BGP peers
EIGRP	90	Cisco IGP
OSPF	110	Standards-based IGP
RIP	120	Legacy — least trusted

WHEN IT MATTERS When two protocols offer the SAME prefix, the lower AD wins before any metric is compared.

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Route Selection — The Order That Matters



WHY IT MATTERS

A /32 static route beats an OSPF /16 every time — prefix length is compared before AD or metric, a classic exam trap.

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Routing Protocols Compared

Protocol	Type	Metric	AD	Role
RIP	Distance vector	Hop count (max 15)	120	Legacy small LANs
OSPF	Link state	Cost (bandwidth)	110	Enterprise IGP
EIGRP	Adv. distance vector	Bandwidth + delay	90	Cisco networks
BGP	Path vector	AS-path + attributes	20 eBGP	The Internet's EGP

WHEN IT MATTERS Know the AD column cold — it is the most-tested fact on this slide.

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RIP — Routing Information Protocol

1

Distance vector

Advertises its full table to neighbours every 30 seconds; routes by hop count alone.

2

Max 15 hops

Hop 16 means unreachable — a hard diameter limit that rules RIP out of large networks.

3

RIPv1 vs RIPv2

v1 was classful (no masks in updates); v2 added masks, multicast updates and authentication.

4

Slow convergence

Minutes to stabilise after a failure, with loop-prevention hacks (split horizon, poison reverse).

5

Where you'll see it

Legacy gear and tiny stub networks; on the exam it is the 'AD 120, hop-count, max 15' answer.

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OSPF — Open Shortest Path First

1

Link state

Floods LSAs so every router shares an identical topology map, then computes shortest paths (SPF/Dijkstra).

2

Cost metric

Cost = reference bandwidth / link bandwidth — the protocol prefers fast links, not few hops.

3

Areas

Area 0 is the backbone; other areas attach to it, shrinking each router's database and SPF workload.

4

Neighbors & adjacency

Routers hello on 224.0.0.5, become neighbours, then synchronise databases to reach FULL.

5

Fast convergence

Seconds after a change — triggered updates, no periodic full-table floods.

6

Open standard

RFC-standard, multi-vendor, AD 110 — the default enterprise IGP choice.

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EIGRP — Enhanced Interior Gateway

1

Advanced distance vector

Cisco's hybrid: neighbour-based updates like distance vector, but fast and loop-free via DUAL.

2

Composite metric

Bandwidth + delay by default (load and reliability optional) — far smarter than hop count.

3

Feasible successor

A pre-computed backup route lets EIGRP fail over almost instantly without recomputing.

4

Partial updates

Sends only changes, only to affected neighbours — very low steady-state overhead.

5

AD 90

More trusted than OSPF (110) on Cisco boxes; now an open RFC but still mostly Cisco shops.

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BGP — The Internet's Routing Protocol

1

Path vector EGP

The routing protocol of the Internet — every ISP exchanges reachability with BGP.

2

Autonomous systems

Prefixes are advertised with the full AS path; policy decides which path wins, not speed.

3

eBGP vs iBGP

eBGP peers between different ASes (AD 20); iBGP carries those routes inside your AS (AD 200).

4

Attributes over metrics

Selection walks attributes in order: weight, local preference, AS-path length, MED...

5

Slow and steady

Over a million Internet routes; convergence takes time — stability and policy beat speed here.

6

TCP 179

BGP peers over a TCP session — a filter or reachability problem shows up as a dropped peering.

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Worked: Checking an OSPF Neighbor

Two commands tell you whether OSPF is healthy and what it learned.

```
R1# show ip ospf neighbor

Neighbor ID   Pri  State     Address     Interface
2.2.2.2       1    FULL/DR   10.0.12.2   Gi0/0
3.3.3.3       1    FULL/BDR  10.0.13.2   Gi0/1

R1# show ip route ospf
O  10.2.2.0/24 [110/2] via 10.0.12.2, Gi0/0
```

FULL state

Adjacency complete — databases synchronised; anything stuck in EXSTART or 2-WAY needs checking

DR / BDR

On multi-access LANs routers elect a Designated Router to cut LSA flooding

[110/2]

AD 110 (OSPF) / metric 2 — cost summed from interface bandwidths along the path

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First Hop Redundancy — VRRP & HSRP

1

The problem

Hosts have ONE default gateway; if that router dies the whole subnet is offline while a second router sits idle.

2

FHRP

First Hop Redundancy Protocols let two or more routers share one virtual IP that hosts use as their gateway.

3

VRRP

Open standard: one master answers the VIP, backups take over on failure — election by priority.

4

HSRP

Cisco's equivalent: active and standby routers exchange hellos; preempt lets the preferred router reclaim.

5

Virtual IP + MAC

The VIP and a virtual MAC float to whichever router is active — hosts never notice the failover.

6

GLBP

Cisco's load-balancing FHRP — multiple gateways forward simultaneously.

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SOHO vs Enterprise Routers

SOHO router

- Router + switch + WAP + firewall in one box
- Web GUI; default IP, username, password — change them
- NAT and a DHCP server enabled by default
- WAN port is usually a DHCP client
- Sized for a handful of devices

Enterprise router

- Dedicated routing hardware, modular interfaces
- CLI-driven OS (e.g. Cisco IOS); config as files
- Runs OSPF/BGP, FHRP and QoS at line rate
- 100x the throughput and connection capacity
- Managed via SSH/console with AAA

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Worked: Static Routes on Linux

The exact commands Lab 8 uses to route between 10.1.1.0/24 and 10.2.2.0/24.

```
# on the router r1
sysctl -w net.ipv4.ip_forward=1

# on host A (10.1.1.2)
ip route add 10.2.2.0/24 via 10.1.1.1

# on host B (10.2.2.2)
ip route add 10.1.1.0/24 via 10.2.2.1

ip route        # verify the table
```

ip_forward

A Linux box only forwards packets between interfaces when this kernel switch is on

via 10.1.1.1

The next hop — must be on a directly connected subnet the host can already reach

Symmetry

Both sides need a route; a one-way route gives replies no path back

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DOMAIN 2.0 · HANDS-ON

Static Routing Between Two Subnets

LAB 8

Build a two-subnet topology and route between them with static routes.

Create three network namespaces (host A, router R1, host B), wire them with veth pairs, enable IP forwarding on R1 and add the static routes that let 10.1.1.0/24 reach 10.2.2.0/24.

YOU'LL BUILD

A working routed path A -> R1 -> B with hand-written static routes on both end hosts.

TOOLCHAIN

iproute2, ip netns, sysctl

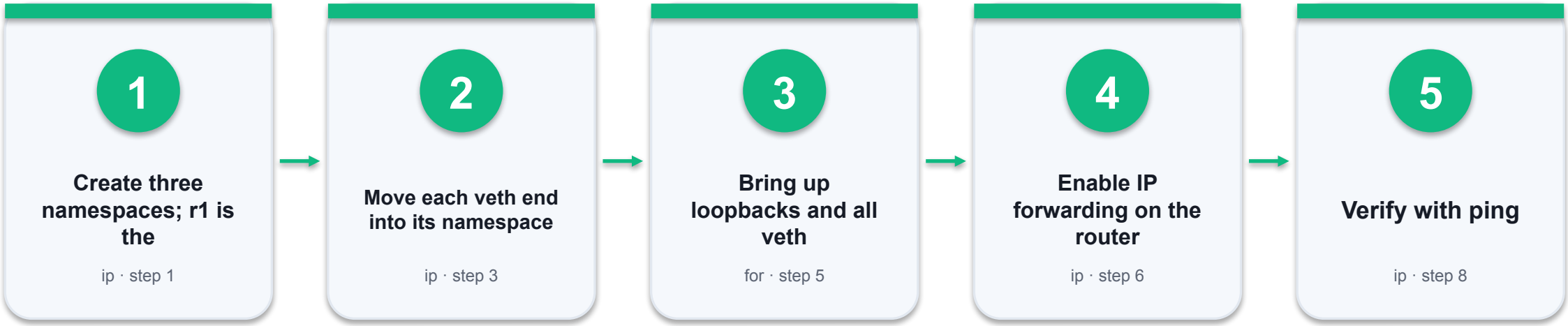
DONE WHEN

ip netns exec a ping -c2 10.2.2.2 succeeds, and ip route inside namespace a shows 10.2.2.0/24 via 10.1.1.1.

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LAB 8 · PROCESS MAP

How Lab 8 Runs



THE POINT

Build a two-subnet topology and route between them with static routes.

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LAB 8 · VERIFY

Static Routing Between Two Subnets

✅ Expected result

ip netns exec a ping -c2 10.2.2.2 succeeds, and ip route inside namespace a shows 10.2.2.0/24 via 10.1.1.1.

IF IT DOESN'T WORK

Ping A to B fails

Check forwarding: ip netns exec r1 sysctl net.ipv4.ip_forward must return 1

Network is unreachable

A static route is missing — re-add: ip route add 10.2.2.0/24 via 10.1.1.1

Interface shows DOWN

Bring up BOTH ends of each veth pair with ip link set <if> up

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Managed Switches and Switch Management

1

Unmanaged switch

Plug and play, zero configuration, no VLANs — fine for a desk, invisible to the network team.

2

Managed switch

Configurable: VLANs, trunks, STP, LACP, mirroring, SNMP. Everything in this domain assumes one.

3

Management IP / SVI

Give a management VLAN interface an address; your PC must be able to reach that subnet to manage it.

4

Change the defaults

Default admin passwords on switches are public knowledge — change them before the switch carries traffic.

5

Console access

A console/rollover cable gives CLI access with no network at all — your lifeline when config breaks.

6

Out-of-band mgmt

A separate management network keeps control access alive when the production LAN is down.

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VLAN Fundamentals

1

What a VLAN is

One physical switch split into multiple logical switches — each VLAN is its own broadcast domain.

2

Why segment

Smaller broadcast domains, failure containment, and policy boundaries between departments and device types.

3

VLAN ID

12-bit field: 1-4094 usable. VLAN 1 is the default everywhere — move production traffic off it.

4

Data VLAN

Ordinary user traffic — PCs, laptops and servers grouped by function or department.

5

Voice VLAN

IP phones get their own tagged VLAN for QoS and clean troubleshooting; the PC daisy-chains through the phone.

6

Crossing VLANs

Traffic between VLANs must be ROUTED — a Layer 3 hop, covered two slides on.

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Inside the 802.1Q Tag

1

4-byte tag

802.1Q inserts 4 bytes into the Ethernet header after the source MAC — Lab 9 captures one live.

2

TPID 0x8100

The first 2 bytes announce 'this frame is tagged' so receivers parse the rest of the tag.

3

VLAN ID (12 bits)

Which VLAN the frame belongs to — switches use it to keep traffic separated on trunks.

4

PCP (3 bits)

Priority Code Point — the Class of Service marking QoS uses to prioritise voice and video.

5

Tag lifecycle

Added when a frame enters a trunk, stripped when it exits an access port — hosts normally never see tags.

6

Linux view

A sub-interface eth0.10 tags everything VID 10 on the way out — exactly how Lab 9 models a trunk.

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Access Ports vs Trunk Ports

Access port

- Carries a single VLAN, untagged
- Connects end devices: PCs, printers, cameras
- switchport mode access + access vlan 10
- Voice VLAN rides alongside, tagged with CoS
- Protect with port security and BPDU guard

Trunk port

- Carries many VLANs between switches/routers
- Every frame tagged 802.1Q except the native VLAN
- switchport mode trunk + allowed vlan list
- Native VLAN must match both ends (not VLAN 1)
- Linux models it as eth0.<id> sub-interfaces

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Native VLAN, Voice VLAN and VTP

1

Native VLAN

The one VLAN a trunk carries UNTAGGED. Frames arriving untagged on the trunk land here.

2

Mismatch danger

If the two ends disagree on the native VLAN, frames silently leak between VLANs — CDP/LLDP will warn.

3

Best practice

Set the native VLAN to an unused ID (not 1) with no user traffic — blunts VLAN-hopping attacks.

4

Voice VLAN config

switchport voice vlan 20: phone traffic tagged with CoS; the PC behind the phone stays untagged.

5

VTP

Cisco's VLAN Trunking Protocol syncs the VLAN database between switches — convenient, and dangerous if a stale switch joins.

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InterVLAN Routing — Three Designs

1

Why needed

VLANs are separate broadcast domains — without a router, SALES can never reach the printer in OPS.

2

Router-on-a-stick

One router interface split into 802.1Q subinterfaces (g0/0.10, g0/0.20) — one per VLAN, each that VLAN's gateway.

3

Subinterface config

encapsulation dot1q 10 plus an IP in VLAN 10's subnet — logical interfaces on one trunk cable.

4

SVI

On a Layer 3 switch, 'interface vlan 10' is a virtual routed interface — no external router needed.

5

Layer 3 switch

Routes between SVIs in hardware at wire speed — the standard enterprise design.

6

Which to pick

Router-on-a-stick for labs and small sites (the one cable is the bottleneck); SVIs everywhere else.

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Worked: VLAN + Trunk on Cisco IOS

Reading the config that puts a PC in VLAN 10 and uplinks over a trunk.

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config)# interface gi0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface gi0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk native vlan 99
```

access vlan 10

End-device ports carry one untagged VLAN; the PC never sees a tag

mode trunk

The uplink carries every VLAN, each frame tagged 802.1Q with its VLAN ID

native vlan 99

Untagged trunk frames fall into the native VLAN — move it off VLAN 1 for security

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Interface Configuration: Speed & Duplex

1

Speed & duplex

Modern ports autonegotiate 10/100/1000/10G full duplex; hard-coding only one side breaks it.

2

Duplex mismatch

One side full, one half: late collisions, CRC errors, terrible throughput — a classic exam symptom.

3

Autonegotiation rule

Both sides auto, or both sides forced identically — never mixed.

4

Runts & giants

Frames under 64 bytes (runts) or oversize (giants) usually signal duplex, cable or MTU faults.

5

show interface

Errors, drops, and the speed/duplex actually negotiated — first stop for a slow-port complaint.

6

Err-disabled

Violations (port security, BPDU guard) shut the port to err-disabled; it stays down until recovered.

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MTU and Jumbo Frames

1

MTU 1500

The Ethernet default payload size — every hop must handle a packet's size or fragment it.

2

Jumbo frames

MTU up to ~9000 bytes: fewer frames per gigabyte and less CPU — used on storage and backup networks.

3

End-to-end rule

EVERY device in the path (switches and both hosts) must agree on jumbo MTU or traffic mysteriously fails.

4

Fragmentation

Routers fragment oversized IPv4 packets unless DF is set; IPv6 never fragments in transit.

5

Black holes

If DF is set and ICMP 'frag needed' is filtered, big packets vanish silently — Lab 33 reproduces this.

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DOMAIN 2.0 · HANDS-ON

802.1Q VLAN Tagging

LAB 9

Create 802.1Q VLAN sub-interfaces and capture a tagged frame.

Load the 8021q kernel module, create eth0.10 and eth0.20 sub-interfaces each with its own subnet, verify the VLAN IDs with ip -d link, then capture a live tagged frame with tcpdump -e.

YOU'LL BUILD

Two VLAN sub-interfaces on one NIC — a Linux model of a trunk port — plus a captured 802.1Q tag.

TOOLCHAIN

iproute2, 8021q module, tcpdump

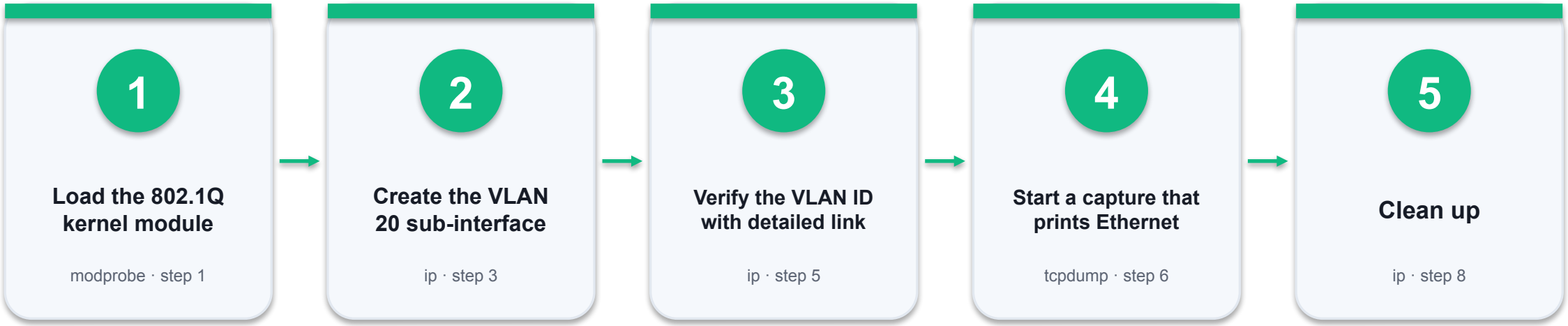
DONE WHEN

ip -d link show eth0.10 reports 'vlan id 10', and the tcpdump output shows the 802.1Q tag (vlan 10) in the Ethernet header.

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LAB 9 · PROCESS MAP

How Lab 9 Runs



THE POINT

Create 802.1Q VLAN sub-interfaces and capture a tagged frame.

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LAB 9 · VERIFY

802.1Q VLAN Tagging

✅ Expected result

ip -d link show eth0.10 reports 'vlan id 10', and the tcpdump output shows the 802.1Q tag (vlan 10) in the Ethernet header.

IF IT DOESN'T WORK

modprobe: module not found

apt update first; the 8021q module ships with standard Ubuntu kernels

No vlan tag in tcpdump output

Capture on the parent eth0 with -e; tags are already stripped on the sub-interface

RTNETLINK: File exists

The sub-interface already exists — ip link del eth0.10, then recreate it

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Loops, Storms and Spanning Tree

1

The loop problem

Ethernet frames have no TTL — two links between switches and one broadcast circulates forever.

2

Broadcast storm

The loop amplifies every broadcast; CPU and links saturate and the LAN 'melts' until a cable is pulled.

3

MAC flapping

The same source MAC arrives on two ports alternately; the MAC table thrashes and delivery breaks.

4

STP (802.1D)

Switches elect a root bridge and block redundant ports, pruning the mesh into a loop-free logical tree.

5

BPDU

Bridge Protocol Data Units — hello frames exchanged every 2 s to build and maintain the tree.

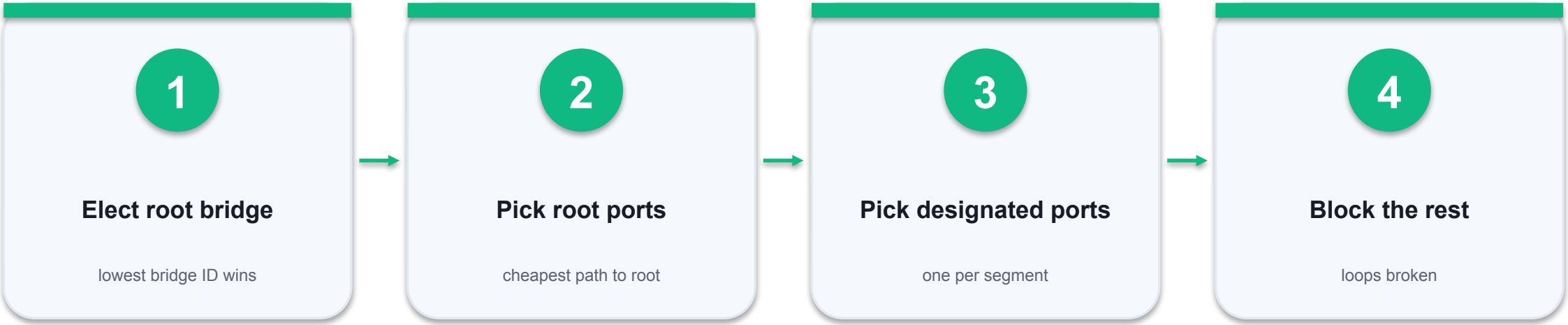
6

Bridge ID

Priority + MAC; lowest wins root. Set priority so YOUR core is root, not a random access switch.

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How Spanning Tree Converges



WHY IT MATTERS

One logical tree, no loops: a cabling mistake no longer melts the LAN in a broadcast storm — and RSTP re-converges in about a second.

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STP Port States — 802.1D vs RSTP

802.1D state	Learns MACs	Forwards	RSTP (802.1w)
Blocking	No	No	Discarding (alt/backup)
Listening	No	No	Discarding
Learning	Yes	No	Learning
Forwarding	Yes	Yes	Forwarding

WHEN IT MATTERS Legacy STP takes 30-50 s to converge; RSTP cuts it to about a second — always prefer RSTP/MSTP.

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Protecting Switch Ports

1

Port security

Limit which and how many MACs may use an access port; violations err-disable it.

2

BPDU guard

Access ports should never receive BPDUs — if one arrives (rogue switch!), kill the port.

3

Root guard

Stops a downstream switch from hijacking the root role — protects your tree design.

4

Loop guard

Blocks a port that stops hearing BPDUs instead of trusting it — belt and braces against loops.

5

DHCP snooping

The switch tracks legit DHCP servers per port; rogue OFFERs from untrusted ports are dropped.

6

Dynamic ARP inspection

Uses the snooping table to drop spoofed ARP replies — kills on-LAN man-in-the-middle.

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Link Aggregation and LACP

1

Port bonding / aggregation

Bundle 2-8 physical links into one logical port-channel — more bandwidth AND redundancy.

2

LACP (802.3ad / 802.1AX)

The standard negotiation protocol: both ends agree the bundle — mismatches stay down safely.

3

Modes

active/active and active/passive form a channel; passive/passive never negotiates — exam favourite.

4

Hashing, not magic

Each flow hashes onto ONE member link — a single big transfer never exceeds one link's speed.

5

NIC teaming

The server-side twin: bond NICs for redundancy (active-backup) or throughput (802.3ad to the switch).

6

Linux bonding

modprobe bonding mode=802.3ad — /proc/net/bonding/bond0 shows partner and slave state (Lab 10).

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Port Mirroring (SPAN)

1

SPAN / port mirror

The switch copies frames from source ports or a VLAN to a destination port for analysis.

2

Why

Switches deliver frames only to their destination — without a mirror your sniffer sees almost nothing.

3

Setup

Choose the source (port or VLAN; rx, tx or both) and the destination port where Wireshark or an IDS listens.

4

Remote SPAN

RSPAN carries the mirror across a dedicated VLAN to a sniffer on another switch.

5

TAP alternative

A hardware test access point copies traffic passively — no switch CPU load, no drops under load.

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DOMAIN 2.0 · HANDS-ON

Link Aggregation (LACP Bond)

LAB 10

Bundle interfaces into an LACP (802.3ad) bond and inspect its state.

Install ifenslave, load the bonding driver in 802.3ad mode, create bond0, enslave a NIC and read the LACP aggregator, partner MAC and per-slave link state from /proc/net/bonding/bond0.

YOU'LL BUILD

A bond0 interface running LACP with aggregator and slave state visible in /proc/net/bonding/bond0.

TOOLCHAIN

ifenslave, bonding module, iproute2

DONE WHEN

cat /proc/net/bonding/bond0 shows 'Bonding Mode: IEEE 802.3ad Dynamic link aggregation' plus the slave's MII status and aggregator ID.

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LAB 10 · PROCESS MAP

How Lab 10 Runs



THE POINT

Bundle interfaces into an LACP (802.3ad) bond and inspect its state.

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LAB 10 · VERIFY

Link Aggregation (LACP Bond)

✅ Expected result

cat /proc/net/bonding/bond0 shows 'Bonding Mode: IEEE 802.3ad Dynamic link aggregation' plus the slave's MII status and aggregator ID.

IF IT DOESN'T WORK

Playground disconnects

Enslaving the only NIC drops Killercoda — rebuild the playground or use a multi-NIC VM

bond0 shows no partner MAC

The peer must speak LACP too; without an 802.3ad partner the aggregator stays down

Cannot enslave eth0

Set the interface down first (ip link set eth0 down), then set master bond0

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Quality of Service and Traffic Shaping

1

Why QoS

When a link congests something must be dropped — QoS decides what suffers so voice and video do not.

2

Classify & mark

Traffic is identified and marked (802.1p CoS at L2, DSCP at L3) so every hop honours the same policy.

3

Queuing

Priority queues drain first — a VoIP packet jumps the line ahead of a backup transfer.

4

Traffic shaping

Buffers and smooths traffic to a target rate; policing simply drops the excess.

5

SOHO QoS

Home routers offer simple per-protocol or per-device priorities — the same idea, one checkbox.

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IDS vs IPS

IDS — detect and alert

- Out-of-band: watches a COPY (port mirror or TAP)
- Detects and alerts; humans respond
- Cannot block, so it cannot break traffic
- False positives cost analyst time only
- Example: Suricata in IDS mode (Lab 26)

IPS — detect and act

- In-band (inline): real traffic flows through it
- Drops or resets malicious flows automatically
- A false positive can block good traffic
- Needs throughput matching the link
- A firewall filters, an IDS notifies, an IPS acts

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Proxy Servers — Three Flavours

Forward proxy

- Sits in front of clients
- Caching and content filtering
- Hides clients from servers
- Common in schools and offices

Transparent proxy

- No client configuration
- Must sit inline in the path
- Intercepts traffic silently
- Often at the network edge

Reverse proxy

- Sits in front of servers
- Load balancing plus caching
- TLS offload / crypto acceleration
- Absorbs DoS attacks
- Hides the server fleet

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Load Balancing

1

What it does

Spreads client requests across a pool of servers — capacity AND availability behind one VIP.

2

Round robin

Each request goes to the next server in the list; DNS round robin is the zero-hardware version.

3

Least connections

Smarter schedulers send work to the server that is actually least busy right now.

4

Health checks

Dead servers are pulled from rotation automatically — users never see the failure.

5

Persistence

Sticky sessions pin a client to one backend when the application keeps state locally.

6

Where it lives

Dedicated appliances, reverse proxies (nginx, HAProxy) or cloud load-balancer services.

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Device Placement — Edge to Core

1

Screened subnet (DMZ)

Public-facing servers live in an isolated zone between two firewall layers — a breach there stays there.

2

Edge firewall

Faces the Internet: allows unsolicited traffic only to published screened-subnet services.

3

Interior firewall

Second layer between the DMZ and LAN — nothing from the DMZ initiates into the private network.

4

Inline IPS

Sits in-band just inside the edge so it can act on attacks before they spread.

5

Internal segmentation

Extra firewalls and ACLs around HR, finance and OT — blast-radius control inside the domain.

6

SOHO 'DMZ host'

A home router's DMZ setting forwards ALL ports to one host — outside protection, not a real DMZ.

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OBJ 2.3 · WIRELESS DEVICES

No cables. Same rules.

Wireless swaps copper for radio — but VLANs, DHCP, security and troubleshooting all still apply on the far side of the access point.

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802.11 Building Blocks

1

Access point (WAP)

A Layer 2 bridge between 802.11 radio and 802.3 Ethernet — wireless clients join the wired LAN through it.

2

SSID

The network name a human selects — a word or phrase up to 32 bytes.

3

BSSID

The MAC address of one AP's radio — the identifier your NIC actually associates with.

4

ESSID

Several APs sharing one SSID form an Extended Service Set — clients roam between BSSIDs seamlessly.

5

CSMA/CA

Radios cannot detect collisions, so they AVOID them: listen first, back off randomly, then transmit.

6

Half duplex on air

Only one station talks per channel at a time — airtime is the shared resource to protect.

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Wireless Network Types

1

Infrastructure

Clients talk through an AP wired into the LAN — the normal enterprise mode.

2

Ad hoc (IBSS)

Client-to-client with no AP — a quick file share between two laptops, nothing more.

3

Mesh

APs relay traffic wirelessly to each other — coverage without pulling cable to every node.

4

Point-to-point

Two directional antennas bridge buildings — a wireless leased line.

5

Wireless bridge

Joins two wired segments over radio — often the role at each end of a point-to-point link.

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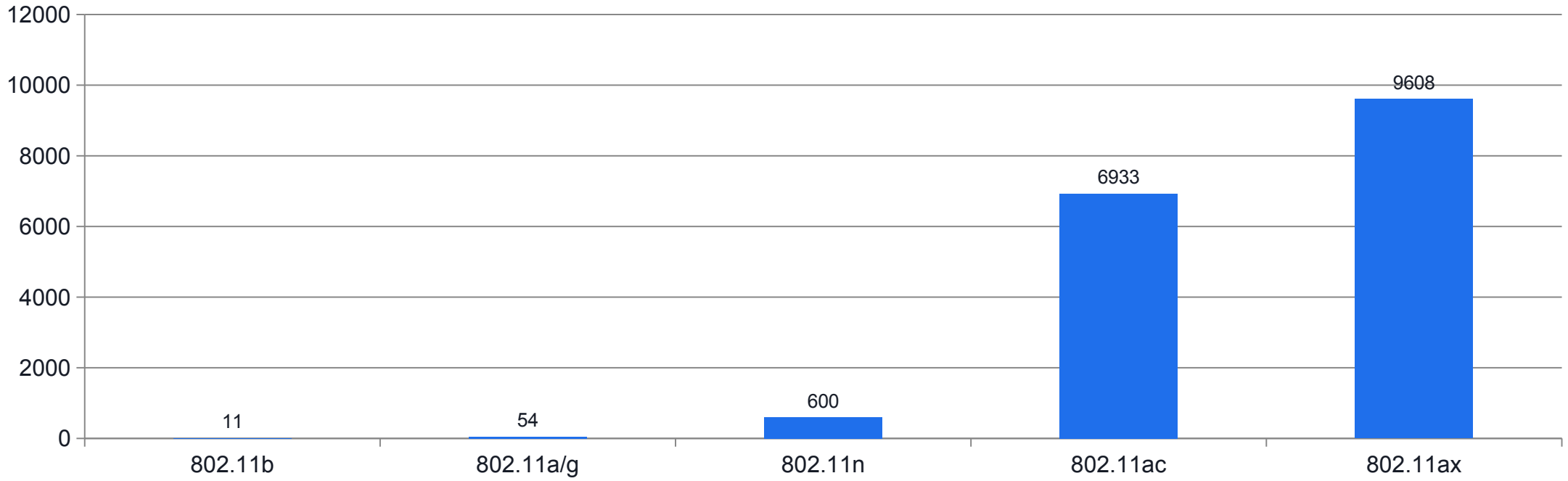
802.11 Standards at a Glance

Standard	Wi-Fi name	Band	Max PHY rate
802.11b	-	2.4 GHz	11 Mbps
802.11a	-	5 GHz	54 Mbps
802.11g	-	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps (MIMO)
802.11ac	Wi-Fi 5	5 GHz	6.9 Gbps (MU-MIMO)
802.11ax	Wi-Fi 6 / 6E	2.4 / 5 / 6 GHz	9.6 Gbps (OFDMA)

WHEN IT MATTERS n introduced MIMO; ac added MU-MIMO; ax adds OFDMA and, as 6E, the 6 GHz band.

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Wireless Ceilings by Generation



WHAT THE DATA SHOWS

Each generation multiplies the ceiling — nearly 900x from b to ax. Real throughput is roughly half the PHY rate; airtime and interference decide the rest.

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Frequency Bands: 2.4 vs 5 vs 6 GHz

Band	Clean channels	Widths	Notes
2.4 GHz	1, 6, 11 only	20 MHz	Long range; congested
5 GHz	36-165 (DFS mid)	20/40/80/160	More channels; radar DFS
6 GHz (6E)	PSC channels	20-160+	WPA3 mandatory; short range

WHEN IT MATTERS Lower frequency travels further and penetrates better; higher frequency carries more data over less distance.

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Channels and Channel Width

1

Channel

The slice of the band an AP transmits on — neighbouring APs must use different, non-overlapping channels.

2

2.4 GHz reality

Channels overlap heavily; only 1, 6 and 11 are clean of each other — the classic honeycomb plan.

3

Channel width

20 MHz standard; 40/80/160 MHz bond channels for speed but eat spectrum and invite interference.

4

DFS

Dynamic Frequency Selection: mid-band 5 GHz shares with radar — the AP must vacate on detection.

5

6 GHz PSC

Preferred Scanning Channels help clients discover 6E networks in the huge new band.

6

Wider is not better

In dense offices many 20/40 MHz cells beat a few 160 MHz cells — airtime reuse wins.

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Antenna Types and Patterns

Antenna	Pattern	Gain	Typical use
Omni / dipole	360-degree doughnut	Low	Ceiling APs, open offices
Patch	Half-sphere from a wall	Medium	Exterior walls, corridors
Yagi	Narrow directional beam	High	Long halls, building links
Parabolic	Very narrow beam	Very high	Point-to-point bridges

WHEN IT MATTERS Gain (dBi) trades coverage width for reach — match the radiation pattern to the room, not the other way round.

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Power over Ethernet Standards

Standard	Name	Power (PSE)	Typically powers
802.3af	PoE	15.4 W	VoIP phones, basic WAPs
802.3at	PoE+	30 W	Wi-Fi 6 WAPs, PTZ cameras
802.3bt	PoE++	60-90 W	Displays, laptops, kiosks

WHEN IT MATTERS Power comes from a PoE switch or a midspan injector — no mains socket needed at the AP's ceiling mount.

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Wireless Security: WEP to WPA3

Mode	Encryption	Key exchange	Notes
WEP	RC4, static keys	None	Broken — never use
WPA	TKIP (RC4)	PSK 4-way	Stopgap; deprecated
WPA2-PSK	AES-CCMP	4-way handshake	PMKID/offline attacks
WPA2-Enterprise	AES-CCMP	802.1X / RADIUS	Per-user credentials
WPA3-SAE	AES-CCMP/GCMP	SAE (Dragonfly)	Forward secrecy
WPA3-Ent 192-bit	GCMP-256	802.1X	Government grade

WHEN IT MATTERS Deploy WPA3 where clients allow; WPA2/WPA3 transition mode bridges the gap. Wi-Fi 6E mandates WPA3.

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PSK or Enterprise Authentication?



YES — WPA3-ENTERPRISE

802.1X with RADIUS (EAP-TLS or PEAP): per-user authentication, per-session keys, revoke one account without touching the rest.

NO — WPA3-SAE (PSK)

One shared passphrase, zero infrastructure. SAE resists offline guessing — but anyone with the phrase gets in; rotate on staff exit.

Guest traffic belongs on its own SSID with client isolation and a captive portal.

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802.1X — Who Does What

1

Supplicant

The client software asking to join — built into every modern OS.

2

Authenticator

The AP (or switch) that relays credentials; it never sees the password itself.

3

RADIUS server

Checks credentials against AD/LDAP and returns accept or reject plus keying material (UDP 1812).

4

EAP methods

EAP-TLS (certificates, strongest) and PEAP/MS-CHAPv2 (username + password) are the common choices.

5

Per-session keys

Every user gets unique encryption keys — one compromised laptop never exposes others' traffic.

6

Same on wires

802.1X also gates Ethernet ports — network access control for the whole access layer.

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Guest Networks and Captive Portals

1

Separate SSID

Guests get their own SSID mapped to its own VLAN — never the corporate network.

2

Captive portal

A web page intercepts the first request: accept terms, enter a voucher or log in before Internet access.

3

Client isolation

Guests cannot see each other's devices — blocks lateral snooping inside the hotspot.

4

Internet-only ACL

The guest VLAN routes to the Internet and NOTHING internal — enforced at the firewall.

5

Bandwidth limits

Rate-limit the guest SSID so visitors cannot starve production airtime.

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Hardening the WLAN

1

Change defaults

Default admin credentials and IPs are in every manual online — the first thing an attacker tries.

2

Disable WPS

The 8-digit PIN brute-forces in hours; turn WPS off everywhere.

3

Management access

Disable remote/WAN management; administer APs from a wired admin VLAN over HTTPS or SSH.

4

SSID hiding & MAC filters

Both are trivially sniffed — they stop only casual users; treat as tidiness, not security.

5

Firmware updates

AP vulnerabilities (KRACK and friends) are fixed in firmware — patch APs like servers.

6

Strong passphrase

For PSK networks a long random passphrase resists offline cracking; rotate on staff exit.

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Threats to Your Wireless Network

1

Rogue AP

An unauthorised AP on your LAN — often an employee's convenience — punches a hole in the perimeter.

2

Evil twin

An attacker clones your SSID with a stronger signal; victims associate and hand over their traffic.

3

Deauth attack

Spoofed deauthentication frames kick clients off — DoS, or bait for the evil twin. 802.11w (PMF) blocks it.

4

Jamming

Illegal RF noise floods the band — nothing associates until the source is found with a spectrum analyser.

5

Wardriving

Attackers map weak networks in bulk — WPA3 and monitoring make you a poor target.

6

WIPS

Wireless IPS sensors detect rogues, twins and deauth floods — and can contain them automatically.

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Enterprise Wireless Architecture

1

Autonomous AP

Each AP configured on its own — fine for a handful, unmanageable at fifty.

2

Lightweight AP

A thin AP pulls config and firmware from a controller and tunnels traffic to it (CAPWAP).

3

WLC

The Wireless LAN Controller pushes SSIDs, channels, power and security to every AP centrally.

4

Cloud-managed

Controller-in-the-cloud (Meraki-style) — the same model with a dashboard instead of an on-prem box.

5

Roaming

Controllers coordinate fast handoff (802.11r) so a VoIP call survives a walk between APs.

6

RF management

The controller auto-tunes channels and power across APs to minimise co-channel interference.

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Site Surveys and AP Placement

1

Site survey

Walk the space with a Wi-Fi analyser: map coverage, noise and neighbours BEFORE and after installing.

2

Interference

Microwaves, wireless peripherals, neighbouring APs — 2.4 GHz suffers most; prefer 5 or 6 GHz.

3

Absorption & reflection

Concrete, metal and water absorb or bounce signal; glass reflects — placement is physics.

4

SNR

Signal-to-noise ratio matters more than raw signal: aim for 25 dB or better, more for voice.

5

AP placement

Central, ceiling-mounted, away from metal; match the antenna pattern to the room's shape.

6

Test and iterate

Antenna placement is part art — measure, adjust, re-measure. Lab 11 is your first survey.

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Wireless Concepts and Channel Survey

LAB 11

Survey nearby wireless networks and map channels, bands and security modes.

Use iw on a wireless laptop (or WiFi Analyzer / Wireless Diagnostics) to scan nearby APs, recording each SSID's channel, signal and security mode, then map channels to bands and compare WEP through WPA3.

YOU'LL BUILD

A site-survey table of nearby APs: SSID, band, channel, signal (dBm) and security mode.

TOOLCHAIN

iw, wavemon, WiFi Analyzer (Windows), Wireless Diagnostics (macOS)

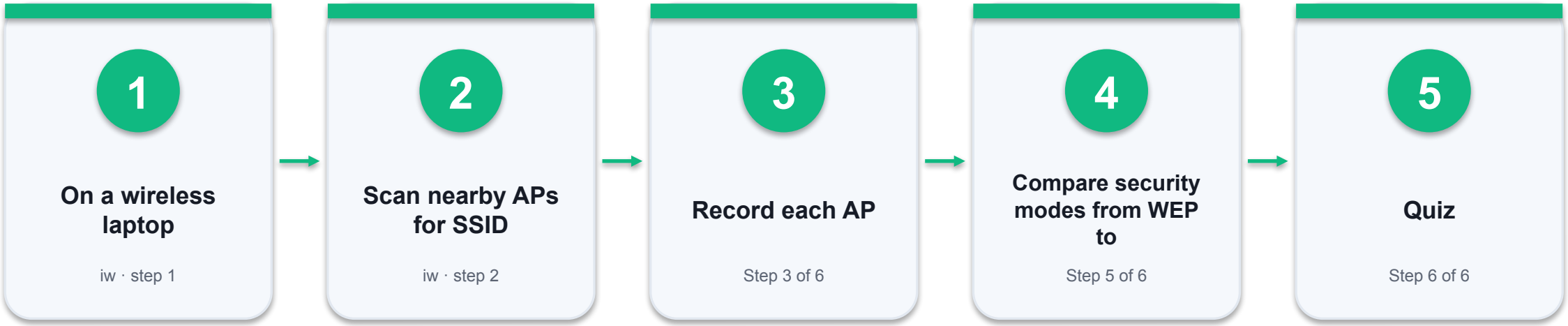
DONE WHEN

The scan output lists SSIDs with freq, signal and RSN/WPA details, and you can explain why 1, 6 and 11 are the only non-overlapping 2.4 GHz channels.

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LAB 11 · PROCESS MAP

How Lab 11 Runs



THE POINT

Survey nearby wireless networks and map channels, bands and security modes.

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LAB 11 · VERIFY

Wireless Concepts and Channel Survey

✅ Expected result

The scan output lists SSIDs with freq, signal and RSN/WPA details, and you can explain why 1, 6 and 11 are the only non-overlapping 2.4 GHz channels.

IF IT DOESN'T WORK

iw: command not found

apt install iw wavemon on Linux; on Windows/macOS use the free analyzer apps instead

Scan returns nothing

Run as root, check the interface name with iw dev, and rfkill unblock wifi

No 6 GHz networks visible

The adapter must support Wi-Fi 6E; older NICs only report 2.4/5 GHz BSSs

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NAT Comes in Three Flavours

Static NAT (SNAT)

- One private IP to one public IP
- The mapping never changes
- Publishes inside servers
- Wastes public addresses

Dynamic NAT

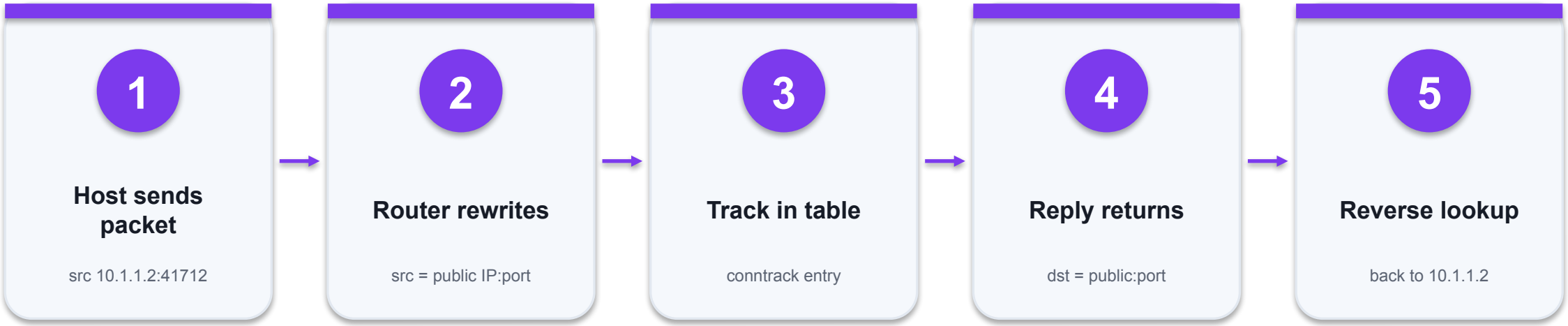
- Pool of public addresses
- First come, first mapped
- Pool exhausts under load
- Still 1-to-1 while held

PAT / NAT overload

- Many private to ONE public IP
- Tracks flows by source port
- What every SOHO router does
- Linux: iptables MASQUERADE

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How PAT Translates a Flow



WHY IT MATTERS

Thousands of private hosts share one public IPv4 address — PAT is why IPv4 survives, and why unsolicited inbound needs port forwarding.

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Port Forwarding, Triggering and DMZ

1

Port forwarding

The router maps inbound public:port to a chosen inside host:port — how a NATed server goes public.

2

Port range forwarding

Forward a whole span of ports at once for applications that need many.

3

Port triggering

An outbound connection on the trigger port opens a temporary inbound port — classic for active FTP.

4

SOHO DMZ host

Forwards EVERYTHING to one host — that machine now lives outside the router's protection. Avoid.

5

NAT and servers

Without a forward rule PAT has no mapping for unsolicited inbound — connections die at the router.

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Worked: Reading a MASQUERADE Rule

The one iptables line behind Lab 12, and the conntrack evidence it leaves.

```
iptables -t nat -A POSTROUTING \
  -s 10.1.1.0/24 -o eth0 -j MASQUERADE

# live translation table
conntrack -L | head -2
tcp ... src=10.1.1.2 dst=8.8.8.8 sport=41712
        src=8.8.8.8 dst=<public-ip> dport=41712
```

POSTROUTING

Source NAT happens after the routing decision, just before the packet leaves eth0

MASQUERADE

Dynamic SNAT that always uses the outgoing interface's current address — ideal for DHCP WANs

sport tracking

PAT maps each inside flow to a unique public port so replies find their way back

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DOMAIN 2.0 · HANDS-ON

NAT and PAT with iptables

LAB 12

Configure PAT with iptables MASQUERADE and verify live translations in conntrack.

Rebuild a namespace client behind the host, enable forwarding, add a MASQUERADE rule on eth0, prove the source address is rewritten with curl ifconfig.me, and inspect the live translation table with conntrack -L.

YOU'LL BUILD

A namespace reaching the Internet through the host's IP — many-to-one PAT exactly as a SOHO router does it.

TOOLCHAIN

iptables, conntrack, iproute2, curl

DONE WHEN

curl -s ifconfig.me from inside the namespace prints the host's public IP, not 10.1.1.2, and conntrack -L shows the translated flow with its ports.

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LAB 12 · PROCESS MAP

How Lab 12 Runs



THE POINT

Configure PAT with iptables MASQUERADE and verify live translations in conntrack.

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LAB 12 · VERIFY

NAT and PAT with iptables

✅ Expected result

curl -s ifconfig.me from inside the namespace prints the host's public IP, not 10.1.1.2, and conntrack -L shows the translated flow with its ports.

IF IT DOESN'T WORK

Ping 8.8.8.8 fails from namespace

Check ip_forward=1 and that the default route inside a points at 10.1.1.1

curl still shows 10.1.1.2

MASQUERADE rule missing or wrong -o interface; list with iptables -t nat -L -v

conntrack: command not found

apt install -y conntrack, then rerun conntrack -L

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IP Tunneling and VPNs

1

Tunneling

Encapsulate one protocol inside another — usually to carry unencrypted traffic through an encrypted channel.

2

VPN

A tunnel that makes a remote machine or site behave as part of the local network.

3

Client-to-site

A laptop VPNs into HQ — remote-access VPN terminated on a concentrator or headend at the edge.

4

Site-to-site

Two offices' routers keep a permanent tunnel — one logical network across the Internet.

5

Protocols

IPsec/IKEv2 and WireGuard dominate today; OpenVPN where TLS-friendliness matters; GRE tunnels without encryption.

6

Split vs full tunnel

Full sends ALL traffic via HQ; split sends only corporate subnets — bandwidth vs inspection trade-off.

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WAN Technologies Compared

Technology	Medium	Speed class	Notes
DSL / VDSL	Phone line (RJ-11)	To ~100 Mbps	Asymmetric; needs filters
Cable (DOCSIS)	Coax, F-type	To ~1 Gbps	Shared neighbourhood node
Fiber / Metro	Single-mode fiber	1-100 Gbps	DWDM multiplies capacity
Satellite	RG-6 to dish	25-220 Mbps	High latency (GEO)
Cellular 5G	Licensed radio	0.1-10 Gbps	Low/mid/high bands

WHEN IT MATTERS Match the technology to latency and symmetry needs, not just headline speed — satellite latency kills VoIP.

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MPLS, SD-WAN and Metro Ethernet

1

MPLS

Provider routers switch on short labels instead of IP lookups — predictable performance, traffic engineering.

2

Label edge

The provider edge pushes and pops labels; your traffic rides a private label-switched path, not the raw Internet.

3

SD-WAN

Software steers many transports (MPLS + broadband + LTE) as one WAN — app-aware paths, central policy.

4

Why SD-WAN won

Commodity Internet links at a fraction of MPLS cost, with encryption and failover built in.

5

Metro Ethernet

Carrier Ethernet across a city — a simple Layer 2 handoff at each site.

6

mGRE / DMVPN

Dynamic multipoint tunnels build a site mesh from one hub configuration.

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Remote Access Tools

1

RDP

Microsoft Remote Desktop Protocol, TCP 3389 — a full graphical session; never expose it raw to the Internet.

2

VNC

Cross-platform desktop sharing on TCP 5900 — weakly protected alone, so tunnel it.

3

SSH

Secure Shell, TCP 22 — encrypted CLI, file transfer (SFTP) and port tunnels; keys beat passwords (Lab 23).

4

Jump box / bastion

One hardened, audited entry host placed in front of all management interfaces.

5

In-band vs out-of-band

Manage over the production network, or over a dedicated path or console that survives outages.

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DORA — How a Lease Happens



WHY IT MATTERS

The first messages are broadcast because the client has no address yet — which is why remote subnets need a DHCP relay (IP helper).

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Designing a DHCP Scope

1

Scope / pool

The subnet plus the range of leasable addresses — keep infrastructure IPs outside the range.

2

Options

Alongside the lease the server hands out router (gateway), DNS servers and domain — one wrong option breaks every client.

3

Lease times

Short leases (minutes) suit guest churn; long leases (days) suit stable LANs — Lab 13 uses 600/7200 s.

4

Reservations

Pin a MAC to a fixed IP centrally — printers get stable addresses without static config.

5

Relay / IP helper

DHCP broadcasts do not cross routers; a relay on each subnet unicasts to the central server.

6

Rogue DHCP

A second server handing out bad gateways breaks a subnet — DHCP snooping is the defence.

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Worked: Reading dhcpd.conf

The exact scope Lab 13 serves on dum0 — every line has a job.

```
default-lease-time 600;
max-lease-time 7200;
authoritative;

subnet 10.50.0.0 netmask 255.255.255.0 {
  range 10.50.0.100 10.50.0.150;
  option routers 10.50.0.1;
  option domain-name-servers 1.1.1.1, 8.8.8.8;
  option broadcast-address 10.50.0.255;
}
```

range

The pool the server may lease from — keep static servers outside it

option routers

Becomes each client's default gateway; a wrong value strands every client

authoritative

The server NAKs bogus lease requests instead of staying silent — one authority per subnet

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DHCP Server with isc-dhcp-server

LAB 13

Configure isc-dhcp-server to lease addresses on a private subnet.

Install isc-dhcp-server, create a dummy interface at 10.50.0.1/24, write a scope with range, router, DNS and lease-time options, bind the daemon to dum0 only, start it and inspect the leases file.

YOU'LL BUILD

A running authoritative DHCP server scoped to 10.50.0.100-150 on dum0, leaving eth0 untouched.

TOOLCHAIN

isc-dhcp-server, iproute2, systemd

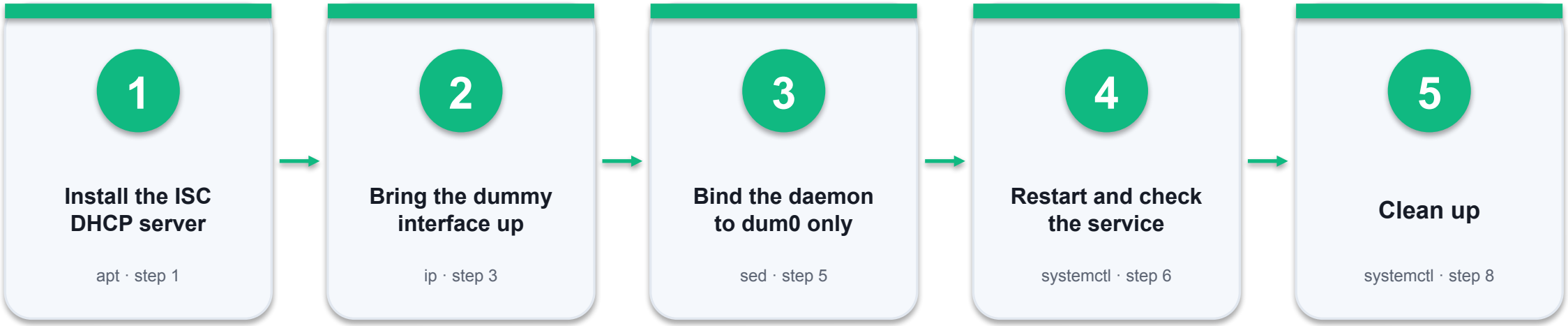
DONE WHEN

systemctl status isc-dhcp-server shows active (running) bound to dum0, and /var/lib/dhcp/dhcpd.leases exists, ready to record client leases.

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LAB 13 · PROCESS MAP

How Lab 13 Runs



THE POINT

Configure isc-dhcp-server to lease addresses on a private subnet.

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LAB 13 · VERIFY

DHCP Server with isc-dhcp-server

✅ Expected result

systemctl status isc-dhcp-server shows active (running) bound to dum0, and /var/lib/dhcp/dhcpd.leases exists, ready to record client leases.

IF IT DOESN'T WORK

Service fails to start

Syntax-check with dhcpd -t; every option line in dhcpd.conf must end with a semicolon

Daemon not listening on dum0

Confirm INTERFACESv4="dum0" in /etc/default/isc-dhcp-server, then restart

Leases file is empty

Normal until a client DISCOVERs on 10.50.0.0/24; leases are appended as granted

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Running Authoritative DNS

1

Authoritative server

Holds the zone's actual records and answers with authority — what Lab 14 builds with BIND9.

2

Recursive resolver

Chases referrals from root to TLD to authoritative on behalf of clients — the other half of DNS.

3

Zone

A delegated slice of the namespace (lab.local) stored as resource records in a zone file.

4

Primary / secondary

The primary holds the writable zone; secondaries pull copies (AXFR) whenever the serial rises.

5

named.conf

BIND's configuration: each zone block names the zone, its type (master/slave) and its file.

6

named-checkzone

Validate syntax BEFORE reloading — a broken zone otherwise takes your whole domain offline.

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Anatomy of the SOA Record

SOA field	Lab 14 value	Meaning
Serial	2026051001	Zone version — bump on every edit
Refresh	3600	Secondary re-check interval (s)
Retry	1800	Re-poll after a failed refresh
Expire	604800	Secondary gives up after this
Minimum	86400	Negative-caching TTL

WHEN IT MATTERS Forget to bump the serial and your secondaries keep serving the OLD zone — a classic silent DNS failure.

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Worked: Reading a BIND Zone File

The db.lab.local file from Lab 14, line by line.

```
$TTL 86400
@   IN  SOA ns.lab.local. admin.lab.local. (
        2026051001 ; serial
        3600 1800 604800 86400 )
@       IN  NS    ns.lab.local.
ns      IN  A     127.0.0.1
www     IN  A     10.0.0.10
@       IN  MX 10 mail.lab.local.
```

Trailing dot

ns.lab.local. is absolute; omit the dot and BIND appends the zone name again

Serial

Secondaries only transfer the zone when the serial increases — bump it on every edit

MX 10

The preference number — lower wins when several mail exchangers exist

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DOMAIN 2.0 · HANDS-ON

Authoritative DNS with BIND9

LAB 14

Host an authoritative DNS zone in BIND9 and query it with dig.

Install bind9, write a zone file for lab.local with SOA, NS, A and MX records, register the zone in named.conf.local, validate the syntax with named-checkzone and query your own server with dig @127.0.0.1.

YOU'LL BUILD

A local BIND9 server answering authoritatively for lab.local, with validated zone syntax.

TOOLCHAIN

bind9, dnsutils (dig), named-checkzone

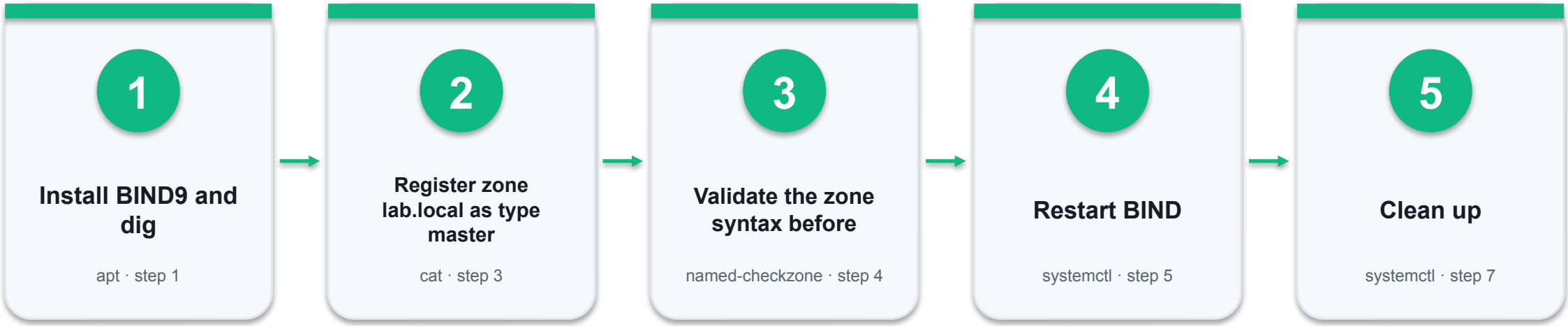
DONE WHEN

dig @127.0.0.1 www.lab.local returns A 10.0.0.10 with the aa (authoritative answer) flag, and dig @127.0.0.1 MX lab.local returns 'MX 10 mail.lab.local.'

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LAB 14 · PROCESS MAP

How Lab 14 Runs



THE POINT

Host an authoritative DNS zone in BIND9 and query it with dig.

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LAB 14 · VERIFY

Authoritative DNS with BIND9

✅ Expected result

dig @127.0.0.1 www.lab.local returns A 10.0.0.10 with the aa (authoritative answer) flag, and dig @127.0.0.1 MX lab.local returns 'MX 10 mail.lab.local.'

IF IT DOESN'T WORK

named-checkzone reports errors

Check the SOA parentheses and the trailing dots on FQDNs like ns.lab.local.

dig times out

BIND is not running — systemctl status bind9; verify config with named-checkconf

SERVFAIL for lab.local

The zone failed to load; journalctl -u named shows the parse error — fix, restart

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MDF, IDF and the Rack

1

MDF

Main Distribution Frame — the building's core room: WAN handoff, core switches, routers, servers.

2

IDF

Intermediate Distribution Frames per floor or area — access switches within 90 m of their users.

3

Demarc

Where the carrier's responsibility ends and yours begins — the demarcation point, usually in the MDF.

4

Rack units

Equipment is measured in U (1U = 1.75 in); plan space, airflow and growth before install day.

5

Patch panels

Horizontal runs terminate on panels; short patch leads jump to switch ports — never re-terminate walls.

6

Locked & dedicated

Distribution rooms are locked and badge-controlled, free of storage and water pipes — security at layer 0.

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Cable Management Done Right

1

Structured pathways

Ladder racks, trays and vertical managers — no cable stress, no spaghetti, airflow preserved.

2

Label everything

Both ends of every cable, every panel port, every drop — future-you troubleshoots at 2 a.m.

3

Bend radius & slack

Respect the minimum bend radius (especially fiber); service loops allow moves without re-pulls.

4

Separation

Keep UTP away from fluorescent lights, motors and power runs — EMI is a Layer 1 gremlin.

5

Velcro, not zip ties

Over-tightened ties crush pairs and change cable geometry — velcro wraps are the standard.

6

Documentation

Port maps and as-built diagrams live with the rack — updated with every change.

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Power: UPS, PDU, Load and Voltage

1

UPS

The battery bridge over outages and brownouts — line-interactive, or online double-conversion for clean power.

2

Runtime vs capacity

Size the UPS in VA/watts for the load, then check minutes of runtime at that load — both matter.

3

PDU

Rack power distribution — metered and switched PDUs report draw and can remote-cycle an outlet.

4

Dual feeds

Dual power supplies on separate circuits and PDUs (A + B feeds) survive a tripped breaker.

5

Voltage & circuits

Match equipment to the circuit (230 V in Singapore); dedicated circuits per rack, no daisy-chained strips.

6

Generator

For real continuity a generator backs the UPS: battery rides the seconds, diesel rides the hours.

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Environment: HVAC, Humidity, Fire

1

HVAC

Racks emit constant heat; dedicated cooling with hot/cold-aisle layouts keeps intake air cold.

2

Temperature

Target roughly 18-27 C at the equipment intake; monitor per rack, not per room.

3

Humidity

40-60% relative humidity — too dry invites static discharge, too damp invites condensation and corrosion.

4

Fire suppression

Clean-agent gas (FM-200, Novec) or pre-action sprinklers — water and live electronics do not mix.

5

Sensors

Temperature, humidity, leak and door sensors feed monitoring — alerts before hardware cooks.

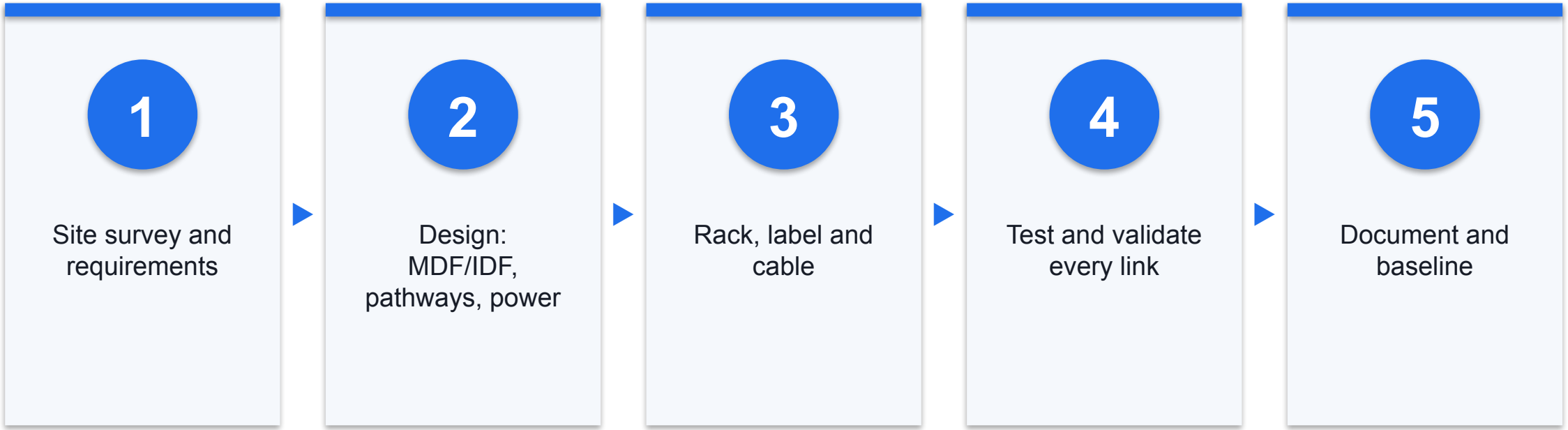
6

Grounding

Bond racks to earth ground and use ESD straps on boards — invisible damage kills gear later.

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A Physical Installation, Start to Finish



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NEXT: NETWORK OPERATIONS

You built the network. Now run it.

Domain 3 turns to operations: monitoring, documentation and the disaster-recovery numbers that keep what you just built alive.

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Recap — Network Implementation

- You can now: Build a two-subnet topology and route between them with static routes.
- You can now: Create 802.1Q VLAN sub-interfaces and capture a tagged frame.
- You can now: Bundle interfaces into an LACP (802.3ad) bond and inspect its state.
- You can now: Survey nearby wireless networks and map channels, bands and security modes.
- You can now: Configure PAT with iptables MASQUERADE and verify live translations in conntrack.
- You can now: Configure isc-dhcp-server to lease addresses on a private subnet.
- You can now: Host an authoritative DNS zone in BIND9 and query it with dig.

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3.0

DOMAIN 3.0 · 19% OF THE EXAM

Network Operations

Documentation · life-cycle · change & configuration management · monitoring · DR · network services · access & management

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WHAT YOU'LL DO

Hands-On Labs — Network Operations

Labs 15–16

- SNMP Monitoring
- NetFlow Analysis with nfdump

Labs 17–18

- Centralised Logging with Rsyslog
- Time Synchronisation with Chrony

Labs 19–20

- Configuration Backup with Git
- Performance Baseline with iperf3

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19% OF THE EXAM

Domain 3 — Network Operations.

19% of the N10-009 exam: documentation, monitoring, disaster recovery, the network services everyone depends on, and secure access to your own infrastructure.

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Domain 3 Roadmap

1

OBJ 3.1 — Documentation & processes

Physical and logical diagrams, rack diagrams, IPAM, asset inventory, life-cycle, change and configuration management.

2

OBJ 3.2 — Network monitoring

SNMP, flow data, packet capture, baselines, log aggregation and SIEM — knowing what the network is doing.

3

OBJ 3.3 — Disaster recovery

RPO/RTO, MTTR/MTBF, cold/warm/hot sites, active-active vs active-passive, and testing the plan.

4

OBJ 3.4 — Network services

DHCP, SLAAC, DNS and time services (NTP, PTP, NTS) — the quiet plumbing every network leans on.

5

OBJ 3.5 — Access & management

Site-to-site and client VPNs, SSH, APIs, console, out-of-band and jump boxes.

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OBJ 3.1 · DOCUMENTATION

Documentation is the network's memory.

The network that exists is the one in production; the network you can repair at 3 a.m. is the one in the documentation. Keeping them identical is an operations discipline.

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Documenting the Network

1

Physical diagram

Where things ARE — rooms, racks, cable runs, patch panels. Follows the actual hardware and wiring.

2

Logical diagram

How data FLOWS — subnets, VLANs, routing paths, firewall zones. Independent of physical location.

3

Rack diagram

Elevation view of one rack: every rack unit numbered, each device drawn at its mounted position.

4

Cable map

Maps each patch-panel port to its wall jack and switch port so a tech can trace any link in seconds.

5

Asset inventory

Hardware, software, licensing and warranty per device — what you own, what it runs, when support ends.

6

IPAM

IP address management — tracks every subnet, assignment and DHCP scope so no address is allocated twice.

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Physical vs Logical Diagrams

Physical (Layer 1)

- Rooms, racks, cable runs, patch panels
- MDF and IDF locations and wiring
- Follows the copper and the fibre
- Used for moves, adds, changes, repairs

Logical (Layer 2/3)

- Subnets, VLANs and IP addressing
- Routing paths and firewall zones
- Follows the traffic flow
- Used for design and troubleshooting

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Network Diagrams by Layer

Layer 1 — physical

- Cables, ports and patch panels
- Rack elevations and RU positions
- Media types and cable labels
- TAP and mirror points marked

Layer 2 — switching

- Switches, VLANs and trunks
- STP topology and port channels
- MAC-level adjacencies
- Link speeds and duplex

Layer 3 — routing

- Subnets and IP addressing
- Routers, SVIs and firewalls
- Routing protocols and next hops
- NAT boundaries and VPN tunnels

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MDF, IDF and the Rack Diagram

1

MDF

Main distribution frame — the primary wiring hub where WAN circuits land and core equipment lives.

2

IDF

Intermediate distribution frame — the per-floor closet linking local drops back to the MDF.

3

Rack diagram

One column per rack, numbered in rack units (1U = 1.75 in); shows exactly what is mounted where.

4

Port maps

Patch panel and switch port labelling ties every user drop to its IDF termination — no more toning cables.

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The Documentation Set

Document	Shows	Used when
Floor plan	Rooms, racks, aisles, outlets	Planning space and power
Rack diagram	Devices by rack unit	Installing or replacing gear
Wiring diagram	Terminations, runs, standards	Tracing a faulty link
Logical diagram	Subnets, VLANs, routing	Design and troubleshooting
Heat map	Wi-Fi coverage by colour	Fixing dead zones

WHEN IT MATTERS Exam questions give a task — pick the document that answers it: physical for 'where', logical for 'how it flows'.

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Asset Inventory

1

Hardware

Make, model, serial, location and owner for every device — the base record for audits and refresh planning.

2

Software

Installed OS and application versions per host — the record patch management works from.

3

Licensing

Which licences you hold, what they cover, when they renew — non-compliance is a legal and audit risk.

4

Warranty & support

Contract and warranty end dates drive refresh budgets and flag devices approaching end of support.

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IPAM — IP Address Management

1

Subnet tracking

Every prefix with its VLAN, gateway and purpose in one authoritative database — not a spreadsheet.

2

Assignment records

Static assignments and DHCP scopes recorded together so no address is ever allocated twice.

3

Discovery & reconciliation

IPAM tools scan the network and reconcile what is actually on the wire against the plan.

4

DDI integration

DNS + DHCP + IPAM suites keep records, scopes and leases in sync automatically.

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Wireless Surveys & Heat Maps

1

Predictive survey

Software models coverage from floor plans and wall materials before a single AP is mounted.

2

On-site survey

Walk the site measuring real signal, interference and attenuation with the actual AP placement.

3

Heat map

Colour-coded floor plan of signal strength — dead zones and channel overlap visible at a glance.

4

Validation survey

Repeat after deployment to prove the design targets (RSSI, SNR, throughput) are actually met.

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OBJ 3.1 · DOCUMENTATION

Business Documents — SLA, MOU, SOW

Document	Parties	What it defines
SLA	Provider + customer	Service scope, uptime targets, penalties
MOU	Two partners	Agreed duties where a contract is premature
SOW	Vendor + customer	Deliverables, timeline, milestones
NDA	Org + staff / vendor	Keeps sensitive information confidential
AUP	Org + users	What users may do on org systems

WHEN IT MATTERS The SLA's uptime figure drives your redundancy design — 99.99% cannot be met with single points of failure.

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Life-cycle Milestones

Milestone	Vendor stops	Your action
End of sale	Selling the product	Stop new purchases
End of life (EOL)	Feature development	Plan and budget replacement
End of support (EOS)	Patches, fixes, RMA	Decommission by this date

WHEN IT MATTERS EOS is the hard deadline: gear past EOS gets no security patches, so every CVE against it stays open forever.

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EOL vs EOS — Why the Difference Matters

End of Life (EOL)

- Vendor stops selling / developing
- Existing units keep receiving fixes
- Trigger to plan the replacement
- Often years before EOS

End of Support (EOS)

- No more patches — even for CVEs
- Failing units cannot be RMA'd
- Unpatched gear = standing audit finding
- Hard deadline to decommission

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Software Management

1

OS updates

The most frequent patch type — schedule through change management, never ad hoc on network gear.

2

Firmware

Low-level device code; fixes hardware bugs and vulnerabilities — verify the image checksum before flashing.

3

Patch discipline

Research the release notes, test in a lab, back up the config, then deploy in a maintenance window.

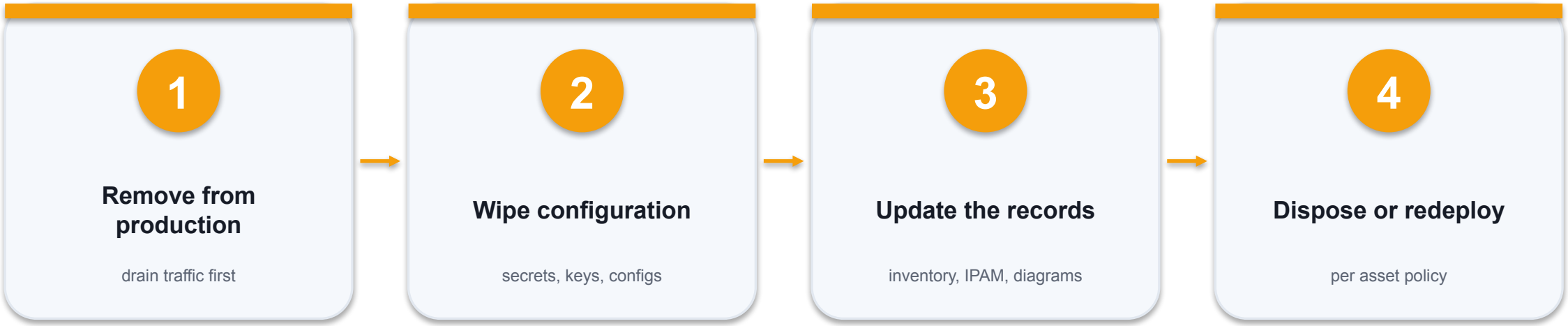
4

Rollback plan

Every update ships with a tested path back — keep the previous image and config within reach.

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Decommissioning a Device



WHY IT MATTERS

A decommissioned switch with its config intact is a free map of your network — and its credentials — for whoever buys it second-hand.

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Change Management

1. **Service request**
   Every change starts as a tracked ticket — who wants what, why, and by when.

2. **Impact assessment**
   What breaks if this goes wrong? Which users, which services, which SLAs are exposed?

3. **Approval**
   A change advisory board weighs risk against benefit before any window is booked.

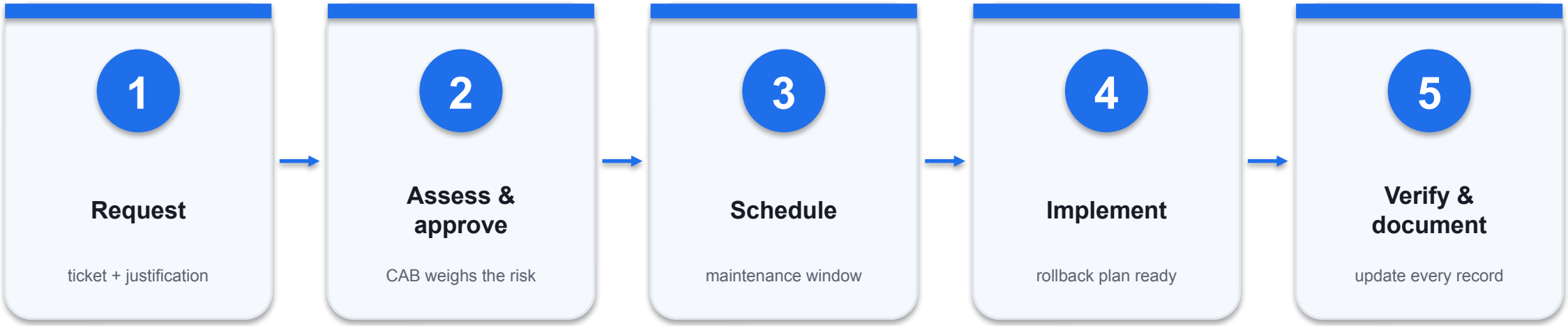
4. **Maintenance window**
   Changes land in agreed low-impact hours, announced to stakeholders in advance.

5. **Rollback plan**
   Written before the change: the exact steps that restore the last known-good state.

6. **Documentation**
   The final step — diagrams, configs and the ticket updated to match the new reality.

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The Change Workflow



WHY IT MATTERS

Most outages are self-inflicted — change management turns risky edits into reviewed, reversible, documented events.

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Standard vs Emergency Changes

Standard change

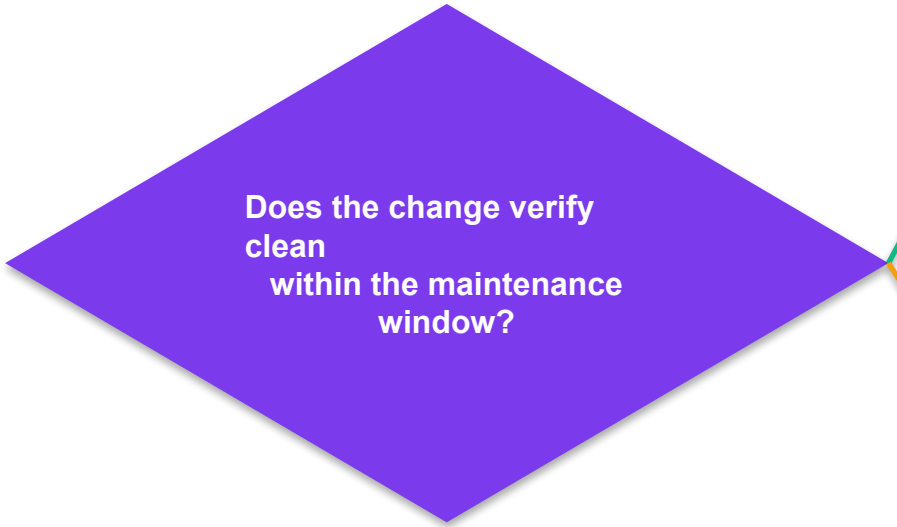
- Pre-approved, low-risk, repeatable
- Follows the full request workflow
- Lands in a planned window
- Example: adding a VLAN to a trunk

Emergency change

- Fixes an active outage or exploit
- Approval fast-tracked, never skipped
- Implemented immediately
- Documented retrospectively — always

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Post-Change Verification



YES — close the ticket

Record the result, update diagrams, configs and baselines, and communicate completion to stakeholders.

NO — roll back

Execute the pre-written rollback to the last known-good configuration; never troubleshoot past the window's end.

The rollback trigger is agreed before the change starts — not negotiated at 2 a.m.

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OBJ 3.2 · NETWORK MONITORING

If you cannot see it, you cannot manage it.

Network monitoring layers device polling (SNMP), traffic summaries (flow data), full packet capture and centralised logs into one operational picture.

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Network Monitoring Methods

1

SNMP polling

The NMS asks each device for counters on a schedule — CPU, memory, interface octets, errors.

2

SNMP traps

The device pushes an alert the moment something happens — no waiting for the next poll.

3

Flow data

Routers export per-conversation summaries (NetFlow/IPFIX) — who talked to whom, and how much.

4

Packet capture

The full contents of every frame — the deepest view, used surgically via port mirroring.

5

Log aggregation

Syslog from every device lands on one collector for searching and correlation.

6

API integration

Modern gear exposes REST APIs — monitoring pulls structured JSON instead of screen-scraping CLIs.

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SNMP Building Blocks

1

Agent

Runs on the managed device, listens on UDP/161 and answers reads from the manager.

2

NMS / manager

The monitoring station — Zabbix, LibreNMS, PRTG — that polls agents and graphs the results.

3

MIB

Management Information Base — the tree-structured schema of everything a device can report.

4

OID

Object identifier — a dotted path into the MIB; 1.3.6.1.2.1.1.3.0 is sysUpTime.

5

GET vs WALK

snmpget reads one OID; snmpwalk traverses an entire subtree of related objects.

6

Trap (UDP/162)

Unsolicited notification from agent to manager when a threshold or event fires.

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SNMP Versions

Version	Security	Verdict
v1	Community string, cleartext	Legacy — do not use
v2c	Community string, cleartext	Adds bulk gets; still unauthenticated
v3	User auth + encryption	The only choice for production

WHEN IT MATTERS Exam favourite: a community string is a password sent in cleartext — v3 (authPriv) adds real authentication and privacy.

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SNMPv3 Security Levels

Level	Authentication	Encryption
noAuthNoPriv	Username only	None
authNoPriv	SHA/MD5 digest	None
authPriv	SHA/MD5 digest	AES/DES — full privacy

WHEN IT MATTERS authPriv is the deployment target: the manager is authenticated and the polled data is encrypted in transit.

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Traps vs Polls

Polling (GET, UDP/161)

- Manager pulls on a fixed schedule
- Regular series — ideal for graphs
- Misses events between polls
- Predictable load on devices

Traps (UDP/162)

- Device pushes the moment events fire
- Instant notification of failures
- Fire-and-forget — can be lost
- Informs add acknowledgement

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Reading snmpwalk Output

A v2c walk of the system subtree against the lab agent.

```
$ snmpwalk -v2c -c public 127.0.0.1 system
SNMPv2-MIB::sysDescr.0 = STRING: Linux ubuntu
    5.15.0-91-generic x86_64
SNMPv2-MIB::sysObjectID.0 = OID:
    NET-SNMP-MIB::netSnmpAgentOIDs.10
DISMAN-EVENT-MIB::sysUpTimeInstance =
    Timeticks: (8642100) 1 day, 0:00:21.00
SNMPv2-MIB::sysContact.0 = STRING: root@lab
SNMPv2-MIB::sysName.0 = STRING: ubuntu
```

-v2c -c public

Version 2c with community 'public' — the credential, and it crosses the wire in cleartext.

MIB::name.0

With MIBs installed, numeric OIDs render as names — sysDescr.0 instead of 1.3.6.1.2.1.1.1.0.

Timeticks

sysUpTime counts hundredths of a second — a sudden reset here means the device rebooted.

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SNMP Monitoring

LAB 15

Configure an SNMP agent and poll MIB objects with snmpwalk and snmpget (OBJ 3.2).

Install snmpd on Ubuntu, configure a read-only v2c community string, then walk the system and IF-MIB subtrees to read uptime, interface names and live traffic counters — the raw data every NMS graphs.

YOU'LL BUILD

A polling-ready SNMP agent on UDP/161 answering v2c queries, plus walked system and interface OIDs.

TOOLCHAIN

snmpd, snmpwalk, snmpget, snmp-mibs-downloader

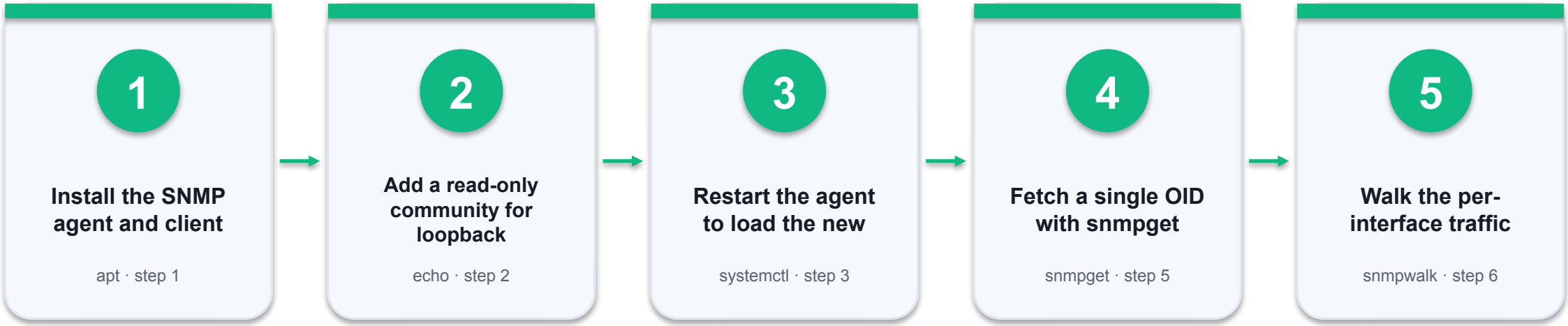
DONE WHEN

snmpwalk returns sysDescr/sysUpTime values and IF-MIB::ifInOctets shows a rising octet counter per interface — rerun it and watch the numbers grow.

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LAB 15 · PROCESS MAP

How Lab 15 Runs



THE POINT

Configure an SNMP agent and poll MIB objects with snmpwalk and snmpget (OBJ 3.2).

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LAB 15 · VERIFY

SNMP Monitoring

✅ Expected result

snmpwalk returns sysDescr/sysUpTime values and IF-MIB::ifInOctets shows a rising octet counter per interface — rerun it and watch the numbers grow.

IF IT DOESN'T WORK

Timeout: No Response from 127.0.0.1
Restart snmpd and confirm agentaddress in snmpd.conf includes udp:127.0.0.1:161.

Output shows numeric OIDs only

Install snmp-mibs-downloader and comment out the 'mibs :' line in /etc/snmp/snmp.conf.

Authentication/community failure

Community mismatch — use -c public and confirm the rocommunity line was appended to snmpd.conf.

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Flow Data — NetFlow & IPFIX

1

What a flow is

One record per conversation: the 5-tuple plus byte, packet and duration counters — metadata, no payload.

2

Exporter

The router, switch or probe (fprobe in the lab) that watches traffic and emits flow records.

3

Collector

Receives and stores the records — nfcapd listening on UDP/9995 in the lab.

4

Analyser

nfdump, ntopng or a SIEM turns stored flows into top-talker and capacity reports.

5

Versions

NetFlow v5 (fixed fields), v9 (templates), IPFIX (the IETF standard), sFlow (packet sampling).

6

vs packet capture

Flows are tiny to store and keep weeks of history; capture keeps payloads but only for minutes.

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Reading an nfdump Report

Top source IPs summarised from the lab collector's stored flows.

```
$ nfdump -R /tmp/flows -s srcip -n 3
Top 3 Src IP Addr ordered by flows:
Date first seen    Src IP Addr  Flows  Bytes
09:12:04 2026      10.0.0.15      142  8.1 M
09:12:31 2026      10.0.0.22       57  1.2 M
09:13:02 2026      8.8.8.8         12   4560
Summary: total flows: 211, bytes: 9.4 M
```

-s srcip

Aggregate by source IP — an instant top-talkers list for capacity or incident work.

Flows vs Bytes

Many flows with few bytes suggests scanning; few flows with huge bytes means bulk transfer.

-R /tmp/flows

Reads every rolled capture file in the directory — flow history is just files on disk.

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Packet Capture & Port Mirroring

1

Port mirroring (SPAN)

The switch copies frames from chosen ports or VLANs to an analyser port — capture without touching the path.

2

Network TAP

Inline hardware splits the signal to a monitor port — sees every bit, including errors SPAN may drop.

3

Full capture

tcpdump and Wireshark record whole packets, payloads included — storage-hungry, deepest evidence.

4

Choosing the tool

Flows for trends and history, capture for deep dives — mirror only what you must, only as long as you must.

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DOMAIN 3.0 · HANDS-ON

NetFlow Analysis with nfdump

LAB 16

Export, collect and analyse flow data to find top talkers and busiest ports (OBJ 3.2).

Run a flow exporter (fprobe) and collector (nfcapd) on one VM, generate web and ICMP traffic, then use nfdump to build top-talker and destination-port reports from the captured flow records.

YOU'LL BUILD

A working exporter-to-collector NetFlow v5 pipeline with nfdump top-N reports of your own traffic.

TOOLCHAIN

fprobe, nfcapd, nfdump, curl, ping

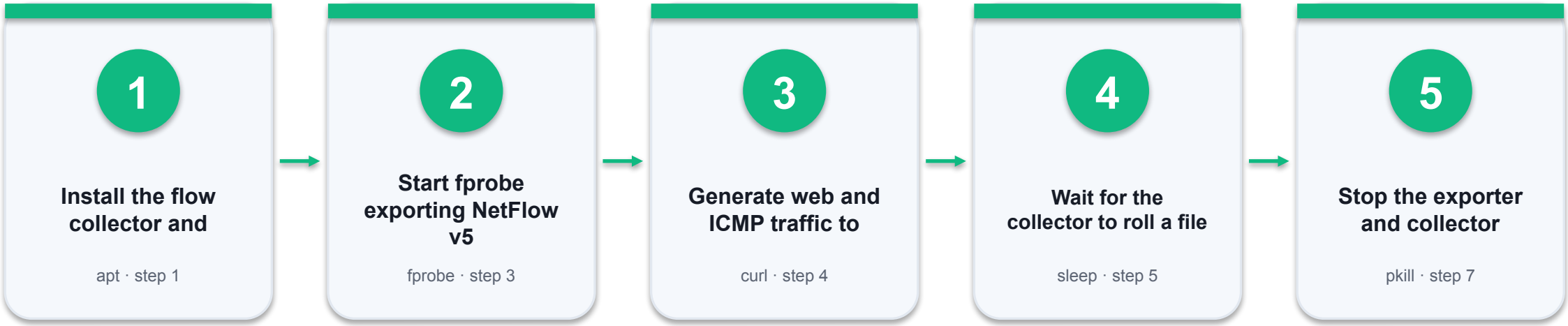
DONE WHEN

nfdump -R /tmp/flows -s srcip lists your VM as a top talker and -s dstport shows ports 443 and ICMP traffic from the curl and ping you generated.

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LAB 16 · PROCESS MAP

How Lab 16 Runs



THE POINT

Export, collect and analyse flow data to find top talkers and busiest ports (OBJ 3.2).

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LAB 16 · VERIFY

NetFlow Analysis with nfdump

✅ Expected result

nfdump -R /tmp/flows -s srcip lists your VM as a top talker and -s dstport shows ports 443 and ICMP traffic from the curl and ping you generated.

IF IT DOESN'T WORK

nfdump: no matched flows

Wait the full 60 s — nfcapd only rolls a readable file every rotation interval.

Collector directory is empty

Confirm fprobe points at 127.0.0.1:9995 and nfcapd is running with -p 9995 (check with ps).

fprobe: no such device eth0

List interfaces with ip link and start fprobe against the VM's actual NIC name.

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Log Aggregation with Syslog

1

Why centralise

Logs on a failed or compromised device are gone; on a collector they survive and can be correlated.

2

Syslog transport

UDP/514 classic, TCP/514 for reliable delivery, TLS on TCP/6514 for confidentiality.

3

Facility.severity

Every message carries both tags — auth.warning, local0.info — and selectors route on them.

4

Collector

rsyslog or syslog-ng receives, filters and stores; the SIEM ingests from there.

5

Retention & WORM

Keep logs long enough for compliance and incident timelines; write-once storage resists tampering.

6

Test the pipeline

logger injects an event end-to-end — prove forwarding works before an incident depends on it.

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Syslog Severity Levels 0-7

Level	Keyword	Meaning
0	emergency	System unusable — all hands
1 / 2	alert / critical	Act immediately / critical fault
3 / 4	error / warning	Failures and conditions to review
5 / 6	notice / info	Significant-but-normal / routine events
7	debug	Verbose troubleshooting output

WHEN IT MATTERS Lower number = more severe. Logging 'at level 4' collects 0-4; debug is switched on only while you troubleshoot.

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SIEM — Aggregate, Correlate, Alert

1

Aggregation

One searchable store for logs, flows and alerts from every source in the estate.

2

Correlation

Links related events across systems — one failed login is noise; 500 across 20 hosts is an attack.

3

Alerting

Configurable triggers and thresholds notify the on-call the moment a pattern fires.

4

WORM storage

Write once, read many — evidence-grade log storage that cannot be quietly edited.

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Working the Aggregated Log

1

grep / grep -E

First-line analysis — filter the collector's log by host, tag, severity or pattern.

2

Regular expressions

Match classes of events, not single strings — build and test them in the TIS Regex Generator, then run with grep -E.

3

logger

Injects labelled test events (-p local0.info) so you can verify routing rules safely.

4

Structured logging

JSON output and APIs feed parsers directly, replacing fragile text scraping.

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DOMAIN 3.0 · HANDS-ON

Centralised Logging with Rsyslog

LAB 17

Build a syslog forwarding pipeline and filter aggregated events by severity and pattern (OBJ 3.2).

Configure rsyslog to receive on TCP/514 and forward to itself over loopback, inject test events with logger, then verify and filter the aggregated log — including a regex built in the TIS Regex Generator.

YOU'LL BUILD

A send-and-receive rsyslog collector on TCP/514 with verified test events and regex-filtered output.

TOOLCHAIN

rsyslog, logger, ss, grep -E, TIS Regex Generator

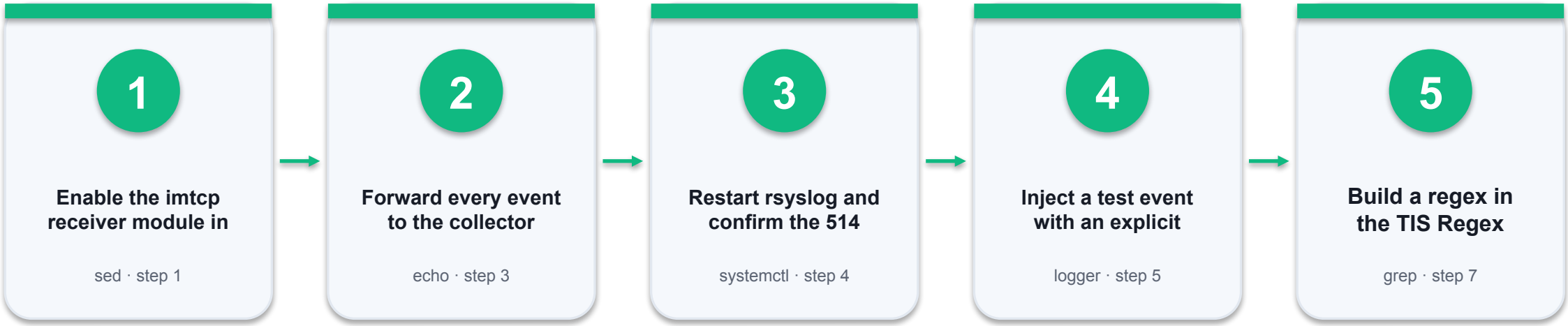
DONE WHEN

ss shows rsyslog listening on :514, grep finds the injected 'Network+ Lab 17 test event' in /var/log/syslog, and your grep -E regex from the TIS Regex Generator matches it too.

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LAB 17 · PROCESS MAP

How Lab 17 Runs



THE POINT

Build a syslog forwarding pipeline and filter aggregated events by severity and pattern (OBJ 3.2).

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LAB 17 · VERIFY

Centralised Logging with Rsyslog

✅ Expected result

ss shows rsyslog listening on :514, grep finds the injected 'Network+ Lab 17 test event' in /var/log/syslog, and your grep -E regex from the TIS Regex Generator matches it too.

IF IT DOESN'T WORK

Nothing listening on port 514

Both sed edits must apply — recheck rsyslog.conf uncommented imtcp lines, then restart rsyslog.

Test event missing from syslog

Run logger without -p first; then check /etc/rsyslog.d/50-forward.conf uses @@127.0.0.1:514.

grep -E returns no matches

Test the regex in the TIS Regex Generator against a pasted log line before using it with grep.

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OBJ 3.4 · NETWORK SERVICES

The quiet services every network leans on.

DHCP hands out addresses, DNS resolves names, NTP keeps every clock honest — and when any of them wobbles, everything else looks broken.

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DHCP Essentials

1

Scope

The pool of addresses the server may lease on one subnet, minus any exclusions.

2

Lease

An address is borrowed, not owned — the client renews at 50% of the lease time.

3

Reservation

A fixed IP bound to a MAC address — printers and servers get the same address every time.

4

Options

Extras pushed with the lease: router (3), DNS (6), domain name (15), TFTP for phones (66).

5

Relay / IP helper

A router forwards the local DHCP broadcast as unicast to a central server on another subnet.

6

Exclusions

Addresses inside the range the server must never hand out — statics live here.

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DORA — the DHCP Handshake



WHY IT MATTERS

DORA runs as broadcast — without a relay agent (IP helper) it can never cross a router to reach a centralised DHCP server.

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Reading dhcpd.conf

A minimal ISC DHCP scope with one reservation.

```
subnet 192.168.10.0 netmask 255.255.255.0 {
  range 192.168.10.100 192.168.10.199;
  option routers 192.168.10.1;
  option domain-name-servers 9.9.9.9;
  default-lease-time 86400;
}
host printer1 {
  hardware ethernet 00:11:22:33:44:55;
  fixed-address 192.168.10.50;
}
```

range

The scope: only .100-.199 are leased; .2-.99 stay free for statics and reservations.

option lines

DHCP options 3 and 6 — the default gateway and DNS server pushed with every lease.

host block

A reservation: the client with this MAC always receives 192.168.10.50.

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Reservation vs Static Assignment

DHCP reservation

- Configured centrally on the server
- Device still runs plain DHCP
- Options update from one place
- Survives device re-imaging

Static configuration

- Typed into the device itself
- Works with no DHCP server at all
- Every change is a device visit
- Right for routers, core switches

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SLAAC — Addressing Without a Server

1

Router Advertisement

IPv6 routers advertise the /64 prefix; hosts solicit and listen for RAs at startup.

2

Self-generated address

The host builds its own address from the prefix plus EUI-64 or a random identifier.

3

DAD

Duplicate Address Detection confirms no other node already holds the address.

4

DHCPv6 still exists

Stateful DHCPv6 or the RA 'other config' flag supplies DNS and options when SLAAC alone is not enough.

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DNS Architecture

1

Authoritative server

Holds the zone data and answers for it definitively — the source of truth for a domain.

2

Recursive resolver

Chases the answer through root, TLD and authoritative servers on the client's behalf, then caches it.

3

Primary zone

The writable copy of the zone, edited on the primary server.

4

Secondary zone

Read-only replicas kept in sync by zone transfer (AXFR/IXFR) for redundancy.

5

Zone file & SOA

The records for one domain; the SOA serial number versions every change.

6

TTL

How long resolvers may cache an answer — long TTLs cut load but slow down migrations.

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Authoritative vs Recursive

Authoritative

- Owns the zone's records
- Answers only for its own domains
- Run where your zones live
- Compromise = forged records

Recursive

- Owns no records — resolves and caches
- Answers anything for its clients
- Run near users (ISP, 1.1.1.1, 8.8.8.8)
- Compromise = poisoned cache

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DNS Record Types — Recap

Record	Maps	Note
A / AAAA	Name to IPv4 / IPv6	The workhorse records
CNAME	Alias to canonical name	Never at the zone apex
MX	Domain to mail server	Priority number picks order
PTR	IP back to name	Reverse DNS; mail filters check it
TXT	Name to free text	SPF, DKIM, site verification
NS / SOA	Zone to its servers	Delegation and zone authority

WHEN IT MATTERS Domain 1 covered the records; Domain 3 asks who answers — authoritative vs recursive, primary vs secondary.

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Time Services — NTP, PTP, NTS

1

Why time matters

Logs, certificates, Kerberos and SIEM correlation all break when clocks drift apart.

2

NTP (UDP/123)

Polls reference servers and gradually disciplines the local clock toward true time.

3

Stratum hierarchy

Distance from the reference clock — each hop from stratum 0 hardware adds one.

4

Multiple sources

Clients compare several upstream servers and vote out 'falsetickers' before trusting any.

5

PTP (IEEE 1588)

Precision Time Protocol — hardware timestamping for sub-microsecond accuracy in finance and telecom.

6

NTS

Network Time Security wraps NTP key exchange in TLS — stops attackers spoofing your time source.

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NTP Strata

Stratum	What it is	Example
0	The reference clock itself	GPS, caesium / atomic clock
1	Server wired to a stratum 0	Time appliance in the DC
2-15	Each network hop adds one	Enterprise and pool servers
16	Unsynchronised	Clock not valid — do not trust

WHEN IT MATTERS Lower stratum = closer to the reference. Ported from the legacy deck: NTP operates as a hierarchy of clock strata over UDP/123.

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Reading chronyc sources

Which server is my clock actually following?

```
$ chronyc sources -v
MS Name/IP address    Stratum Poll Reach Offset
^* time.cloudflare.com    3    6   377  +142us
^- ntp.ubuntu.com         2    6   377 -1250us
^? 10.0.0.99              0    6     0    +0ns
```

^* / ^- / ^?

* is the selected source, - a viable backup candidate, ? unreachable or untested.

Stratum

Cloudflare here is stratum 3 — three hops from a hardware reference clock.

Reach 377

Octal bitmask of the last eight polls — 377 means all eight answered.

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DOMAIN 3.0 · HANDS-ON

Time Synchronisation with Chrony

LAB 18

Configure NTP time sync with chrony and interpret stratum, offset and drift (OBJ 3.4).

Install chrony (the modern ntpd replacement), inspect the selected time sources and stratum hierarchy, add an extra NTP server with iburst, and force an immediate clock step.

YOU'LL BUILD

A VM synchronised over UDP/123 with multiple NTP sources and a readable chronyc tracking report.

TOOLCHAIN

chrony, chronyc

DONE WHEN

chronyc sources -v shows time.cloudflare.com in the list with a ^* or ^- marker, and chronyc tracking reports a stratum below 16 with a sub-second system time offset.

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LAB 18 · PROCESS MAP

How Lab 18 Runs



THE POINT

Configure NTP time sync with chrony and interpret stratum, offset and drift (OBJ 3.4).

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LAB 18 · VERIFY

Time Synchronisation with Chrony

✅ Expected result

chronyc sources -v shows time.cloudflare.com in the list with a ^* or ^- marker, and chronyc tracking reports a stratum below 16 with a sub-second system time offset.

IF IT DOESN'T WORK

All sources show ^? unreachable

UDP/123 egress may be blocked — wait for retries, or verify with chronyc activity and tracking.

New server never appears

The server line must be in /etc/chrony/chrony.conf and chrony restarted — check for typos.

Stratum shows 16

16 means unsynchronised — run chronyc -a makestep and give the sources a minute to be selected.

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Configuration Management

1

Production config

The running configuration on the live device — what the network actually does right now.

2

Backup config

A stored copy taken on schedule and before every change — your rollback path.

3

Baseline config

The approved golden template a device of that role should match — any drift is a defect.

4

Golden images

Known-good firmware and OS images kept with checksums for clean rebuilds.

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Config Backup Discipline

1. Back up before every change
2. Store versioned, off the device
3. Diff running vs baseline regularly
4. Alert on unauthorised drift
5. Test restores — a backup is a claim

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Version Control for Configurations

1

History

Every change recorded with author, timestamp and message — an audit trail for free.

2

Diff

See exactly what changed between any two versions — the fastest root-cause tool there is.

3

Blame

Line-by-line attribution answers 'who set this, and when?' in one command.

4

Revert

Undo one bad change without losing everything committed after it.

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DOMAIN 3.0 · HANDS-ON

Configuration Backup with Git

LAB 19

Put network configuration under version control for history, diff, blame and rollback (OBJ 3.1).

Initialise a git repository in /etc, commit a baseline of the critical network configs, make and commit a change, inspect the audit trail with log and show, then roll the change back with revert.

YOU'LL BUILD

A version-controlled /etc with a baseline commit, a tracked change, and a demonstrated one-command rollback.

TOOLCHAIN

git

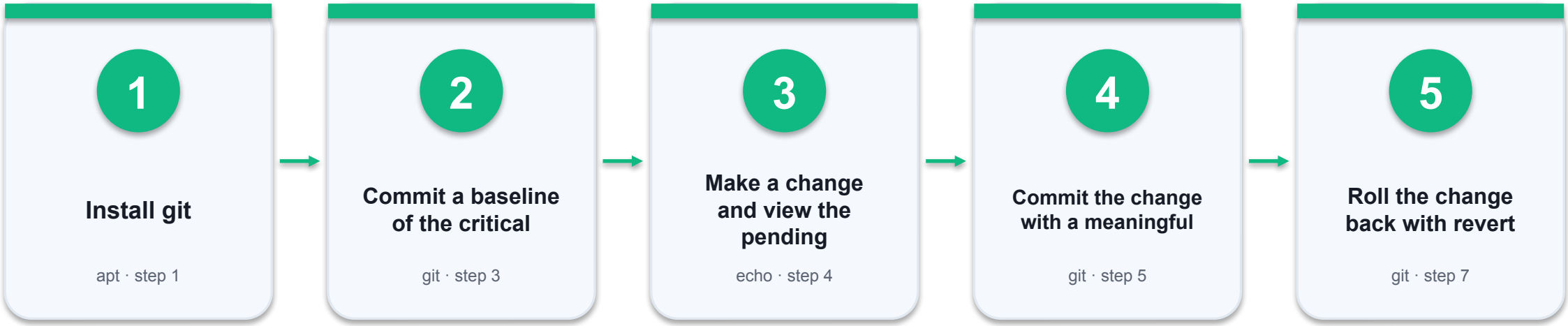
DONE WHEN

git log --oneline shows the baseline, the change and the revert as three commits, and cat /etc/hosts confirms the lab-vm line was removed by the rollback.

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LAB 19 · PROCESS MAP

How Lab 19 Runs



THE POINT

Put network configuration under version control for history, diff, blame and rollback (OBJ 3.1).

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LAB 19 · VERIFY

Configuration Backup with Git

✅ Expected result

git log --oneline shows the baseline, the change and the revert as three commits, and cat /etc/hosts confirms the lab-vm line was removed by the rollback.

IF IT DOESN'T WORK

Please tell me who you are

Set both identity values: git config user.email and git config user.name before committing.

Nothing added to commit

Run the git add with explicit filenames from inside /etc — pathspec errors mean a wrong cwd.

Revert did not change hosts

Revert undoes the LAST commit — check git log to confirm HEAD was the hosts change, not another edit.

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Baselines & Anomaly Alerting

1

Baseline

Measurements taken while the network is healthy — the operational definition of 'normal'.

2

Performance metrics

Bandwidth, throughput, latency, jitter and loss — trended per link and per service.

3

Availability monitoring

Up/down state and uptime percentage, tracked against the SLA target.

4

Anomaly alerting

Alert on deviation from the baseline, not just fixed thresholds — it catches the subtle failures.

5

Interface counters

Errors, discards, CRCs, giants and runts — rising counters flag layer 1-2 trouble.

6

Utilisation

Sustained load near capacity means congestion is coming — plan upgrades from the trend line.

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Interface Statistics to Watch

1. **Link state**
   Up/down and flapping history — a link that bounces hourly is a failing SFP or cable.

2. **Speed / duplex**
   A 100/half port on a gigabit switch is usually a failed autonegotiation, not a slow NIC.

3. **CRC errors**
   Frames failing the checksum — cabling, connectors or EMI corrupting bits in flight.

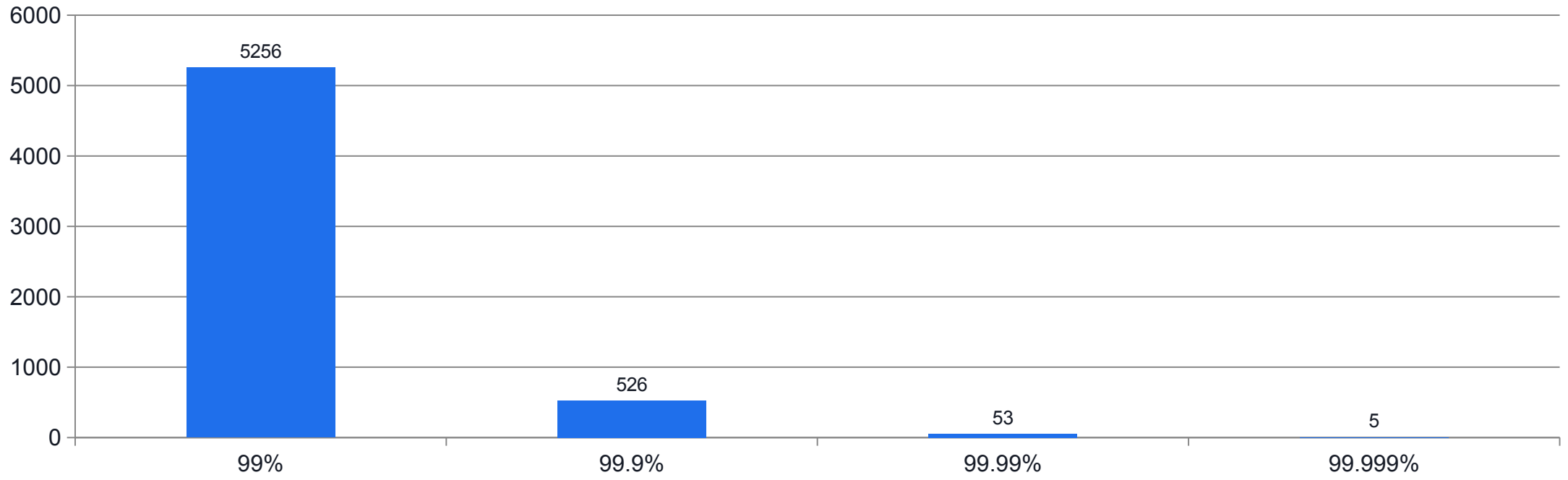
4. **Giants & runts**
   Frames beyond or below legal Ethernet sizes — MTU mismatches, bad NICs, collisions.

5. **Discards / drops**
   Good frames thrown away for lack of buffer — the signature of congestion.

6. **Byte counts**
   Send/receive octets feed the utilisation graphs every capacity decision rests on.

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The Cost of a Nine



WHAT THE DATA SHOWS

Each extra nine cuts permitted downtime tenfold — 99.9% allows almost nine hours a year, 99.99% under an hour. The SLA figure dictates the redundancy budget.

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Measuring Throughput with iperf3

1

Client / server model

One end listens (-s), the other transmits (-c) — measure the path between them, not a guess.

2

TCP vs UDP tests

TCP reports achievable goodput; UDP at a set rate (-b) exposes jitter and datagram loss.

3

Parallel streams

-P opens several flows at once — separates TCP window limits from true bandwidth limits.

4

Reverse mode

-R sends from server to client — the test for asymmetric paths like DOCSIS upstream.

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Performance Baseline with iperf3

LAB 20

Baseline throughput, jitter and loss with iperf3 TCP, UDP, parallel and reverse tests (OBJ 3.2).

Run an iperf3 server as a daemon and test against it: a 10-second TCP throughput test, four parallel streams, a 100 Mbps UDP test that reports jitter and loss, and a reverse test for asymmetric paths.

YOU'LL BUILD

A recorded performance baseline: TCP bitrate, parallel-stream totals, UDP jitter/loss and reverse-direction results.

TOOLCHAIN

iperf3

DONE WHEN

The TCP test reports a multi-Gbps loopback Bitrate, the UDP test shows jitter in ms with near-zero lost datagrams, and the -P 4 run sums four streams in the [SUM] row.

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LAB 20 · PROCESS MAP

How Lab 20 Runs



THE POINT

Baseline throughput, jitter and loss with iperf3 TCP, UDP, parallel and reverse tests (OBJ 3.2).

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LAB 20 · VERIFY

Performance Baseline with iperf3

✅ Expected result

The TCP test reports a multi-Gbps loopback Bitrate, the UDP test shows jitter in ms with near-zero lost datagrams, and the -P 4 run sums four streams in the [SUM] row.

IF IT DOESN'T WORK

Connection refused to 127.0.0.1

The server is not running — start it again with iperf3 -s -D and confirm with pgrep iperf3.

Server busy — another test running

Only one client at a time per server; wait for the previous run or pkill iperf3 and restart.

UDP shows heavy datagram loss

Lower the offered rate (-b 10M) — loss at a set bandwidth is the finding, not an error.

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OBJ 3.3 · DISASTER RECOVERY

Plan for the bad day before it arrives.

Disaster recovery quantifies what you can afford to lose (RPO), how long you can be down (RTO), and rehearses the switchover before it is ever real.

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DR Metrics — RPO, RTO, MTTR, MTBF

Metric	Question it answers	Example
RPO	How much data can we lose?	Backups every 4 h = RPO 4 h
RTO	How long until service is back?	Restore must finish in 2 h
MTTR	Average time to repair a failure	Ticket open to fixed: 45 min
MTBF	Average time between failures	PSU rated at 100,000 h

WHEN IT MATTERS RPO sizes your backup frequency; RTO sizes your recovery infrastructure. Both are business decisions before they are technical ones.

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Recovery Sites — Cold, Warm, Hot

Site	Readiness	Cost	Failover time
Cold	Space and power only — no gear	$	Days
Warm	Hardware in place, data loaded on demand	$$	Hours
Hot	Live replica with data in sync	$$$	Minutes
Cloud	Elastic DR-as-a-service, pay on use	$-$$	Design-dependent

WHEN IT MATTERS Pick the cheapest site that still meets the RTO — a hot site protecting a tolerant workload is money burned.

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Active-Active vs Active-Passive

Active-active

- All nodes serve traffic at once
- Full capacity used in normal running
- A failure just shrinks capacity
- Needs state sync between nodes

Active-passive

- Standby waits idle for failover
- Simpler state management
- Brief pause while standby takes over
- Hardware that mostly sits idle

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High Availability Building Blocks

1

Single point of failure

One component whose failure stops the whole workflow — HA design exists to remove these.

2

Redundancy

Duplicate power, links, supervisors and paths so one failure never becomes an outage.

3

Clustering

Multiple servers act as one logical service and survive the loss of members.

4

Load balancing

Spreads requests across the cluster and steers around failed members automatically.

5

Virtual IP / FHRP

VRRP or HSRP float one gateway address across routers — the default gateway survives a router death.

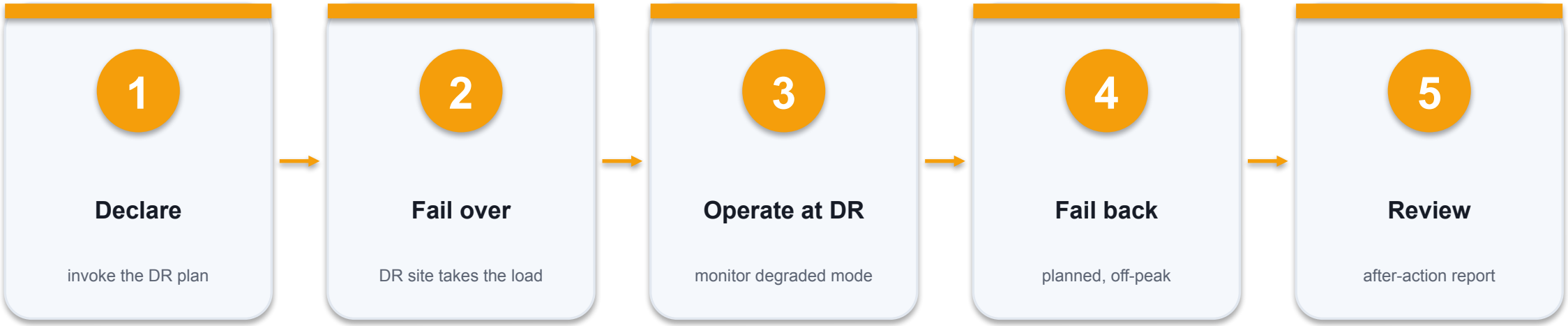
6

Critical assets & nodes

Know which systems keep production alive — they get the redundancy budget first.

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DR Failover — and Failback



WHY IT MATTERS

Failback is the forgotten half — returning to the primary site is itself a risky, planned change with its own window and rollback.

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Testing the DR Plan

1

Tabletop exercise

Walk the scenario around a table — cheap, and it finds the gaps in roles, contacts and assumptions.

2

Validation test

Actually restore the backups and fail services over — proves the runbooks, not the intentions.

3

Full failover drill

Run production from the DR site for a period — the only complete proof the plan works.

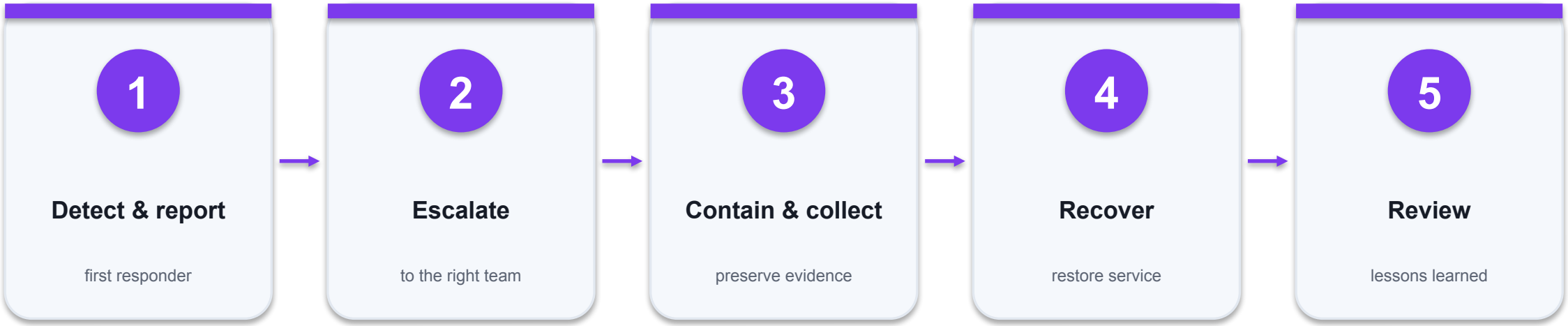
4

After-action report

Every exercise ends with documented findings and concrete plan updates.

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Incident Response



WHY IT MATTERS

Forensic discipline — secure the area, document the scene, keep chain of custody — is what keeps evidence admissible and the timeline reconstructable.

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Backups That Actually Restore

1

3-2-1 rule

Three copies, two media types, one offsite — survives device loss, site loss and ransomware.

2

Snapshots

Point-in-time images for fast local rollback — useful, but never a substitute for offsite copies.

3

Config + state data

Back up device configurations AND application data; the RPO applies to both.

4

Restore testing

An untested backup is a hope, not a plan — schedule restore drills like any other maintenance.

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VPN Types

Type	Connects	Typical use
Site-to-site	Two networks via gateways	Branch to HQ, always on
Client-to-site	One device to the network	Remote workers, on demand
Full tunnel	ALL client traffic via VPN	Maximum inspection and control
Split tunnel	Only corporate subnets via VPN	Saves bandwidth, less visibility

WHEN IT MATTERS Site-to-site tunnels are transparent to users; client-to-site needs software (or a clientless HTML5 portal) on each device.

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Split or Full Tunnel?



YES — full tunnel

Route everything through HQ: complete visibility and filtering, at the cost of bandwidth, latency and user experience.

NO — split tunnel

Send only corporate destinations through the VPN; internet traffic goes direct — faster, but invisible to your controls.

Document the choice either way — auditors ask, and users notice.

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Management Access Methods

1. **SSH (TCP/22)**
   Encrypted CLI — the default for device administration; prefer key-based auth over passwords.

2. **GUI**
   Web management over HTTPS — convenient, but restrict it to the management VLAN.

3. **API**
   REST and NETCONF programmatic control — the foundation of automation and infrastructure as code.

4. **Console**
   Direct serial access — works when the network is down; the recovery path of last resort.

5. **Out-of-band**
   A separate management network or LTE modem, reachable when production is not.

6. **Jump box**
   One hardened, logged host is the only door into the management network.

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In-band vs Out-of-band Management

In-band

- Management rides the production net
- Cheap — no extra infrastructure
- Lost exactly when you need it most
- Fine for day-to-day monitoring

Out-of-band

- Dedicated network, serial or LTE path
- Reachable during outages
- Extra cost, another network to secure
- Essential for core and remote sites

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The Jump Box Access Path

1. Admin authenticates with MFA
2. Lands on the hardened jump box
3. Session is logged and recorded
4. SSH to device on management VLAN
5. Change tracked against a ticket

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DOMAIN 3 COMPLETE

Operations is where designs meet reality.

Document it, monitor it, back it up, rehearse the disaster, and control every management door — that is Domain 3, and 19% of your exam.

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Recap — Network Operations

- You can now: Configure an SNMP agent and poll MIB objects with snmpwalk and snmpget (OBJ 3.2).
- You can now: Export, collect and analyse flow data to find top talkers and busiest ports (OBJ 3.2).
- You can now: Build a syslog forwarding pipeline and filter aggregated events by severity and pattern (OBJ 3.2).
- You can now: Configure NTP time sync with chrony and interpret stratum, offset and drift (OBJ 3.4).
- You can now: Put network configuration under version control for history, diff, blame and rollback (OBJ 3.1).
- You can now: Baseline throughput, jitter and loss with iperf3 TCP, UDP, parallel and reverse tests (OBJ 3.2).

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4.0

DOMAIN 4.0 · 14% OF THE EXAM

Network Security

Logical & physical security · deception · terminology · audits & compliance · segmentation · attacks · defence features

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WHAT YOU'LL DO

Hands-On Labs — Network Security

Labs 21–23

- Host Firewall with UFW
- TLS Certificate with OpenSSL and Nginx
- SSH Hardening with Key Authentication

Labs 24–26

- Site-to-Site VPN with WireGuard
- Port Scanning and Service Hardening
- Intrusion Detection with Suricata

Lab 27

- Building an ACL with iptables

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DOMAIN 4 · 14% OF THE EXAM

Domain 4 - Network Security.

14% of the exam: encryption, identity, attacks, hardening, segmentation and physical defence - how we keep the network confidential, intact and available.

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The CIA Triad

Confidentiality

- Only authorised eyes see the data
- Encryption in transit and at rest
- Access controls and least privilege

Integrity

- Data arrives exactly as it was sent
- Hashes detect tampering (SHA-256)
- Digital signatures prove the origin

Availability

- Systems stay reachable when needed
- Redundancy, backups, DDoS defence
- Balance security against usability

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Security Terminology

1

Threat

Any actor or event with the potential to harm the network - from a ransomware crew to a burst pipe above the server room.

2

Vulnerability

A weakness that could be abused - an unpatched service, a default password, an open port nobody remembers.

3

Exploit

The technique or code that weaponises a vulnerability to gain access or disrupt service.

4

Risk

Likelihood times impact of a threat exploiting a vulnerability - what budgets and priorities are set against.

5

CVE

Common Vulnerabilities and Exposures - the public catalogue of disclosed flaws, each with an ID like CVE-2021-44228 (Log4Shell).

6

Zero-day

A flaw attackers find before the vendor does - no patch exists yet, so layered defence is the only cover.

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From Threat to Impact



WHY IT MATTERS

Break any link and the attack fails - patching removes vulnerabilities, hardening blocks exploits, monitoring and response limit impact.

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Audits and Regulatory Compliance

1

Security audit

An independent check that controls exist and work - configs, logs, access reviews, penetration tests.

2

Data locality

Laws that dictate WHERE data may be stored and processed - pins your cloud regions and backup sites.

3

PCI DSS

Card-industry standard: anyone storing or transmitting card data must segment, encrypt and log that environment.

4

GDPR

EU personal-data regulation: privacy by design, breach notification within 72 hours, heavy fines for leaks.

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Compliance Regimes at a Glance

Regime	Scope	Network requirement
PCI DSS	Cardholder data environments	Segment the CDE, encrypt, log all access
GDPR	Personal data of EU residents	Protect by design; report breaches fast
Data locality	Where data may physically live	Keep regulated data in approved regions
PDPA (SG)	Singapore personal data	Reasonable security arrangements required

WHEN IT MATTERS Audits verify these on paper AND on the wire - segmentation and logs are the evidence.

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Defense in Depth

1

Perimeter

Edge firewalls, screened subnet, DDoS scrubbing - the first line, never the only line.

2

Network

Segmentation, VLANs, ACLs, IDS/IPS - contain whatever gets past the edge.

3

Host / endpoint

Host firewalls, EDR, patching, hardening - every machine defends itself.

4

Application

Input validation, secure APIs, WAF - the layer attackers actually talk to.

5

Data

Encryption at rest, backups, DLP - the final prize gets its own armour.

6

Separation of duties

No single person holds enough rights to abuse the system alone.

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Trusted vs Untrusted Zones

Trusted zone

- Internal LANs and data-centre subnets
- Users and devices are authenticated
- Servers holding business data
- Still segmented - trust is never absolute

Untrusted / screened

- The Internet - assume hostile
- Guest and BYOD networks
- Screened subnet (DMZ) for public services
- Traffic crossing zones must pass a firewall

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The Screened Subnet (DMZ)

1

Screened subnet

A buffer network between Internet and LAN hosting the public-facing servers - web, mail, DNS.

2

Bastion host

Any machine directly exposed to the Internet - hardened to the maximum, expected to be attacked.

3

Two-firewall design

Internet -> outer firewall -> DMZ -> inner firewall -> LAN; a DMZ breach still faces a wall.

4

Why it works

If a public server is compromised, the attacker is contained in the DMZ - not inside your LAN.

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Firewall Families

1

Network firewall

A dedicated device at zone boundaries filtering traffic for the whole network.

2

Host-based firewall

Software on each machine (ufw, Windows Defender FW) - the last line, per host.

3

Hardware vs software

Physical appliances at the edge; virtual/software firewalls in cloud and hosts.

4

UTM / NGFW

Unified threat management bundles firewall, IPS, AV and URL filtering in one box.

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Stateless, Stateful, Application-Aware

Type	Decides on	Strength	Limit
Stateless	Ports + IPs, per packet (ACL)	Fast, simple, cheap	No conversation context
Stateful	Tracks connection state	Return traffic auto-allowed	More CPU and memory
App-aware (NGFW)	Packet content via DPI	Sees apps inside port 443	Costly; TLS inspection

WHEN IT MATTERS Most modern firewalls combine all three - ACL speed plus stateful context plus DPI.

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OBJ 4.3 · THE GOLDEN RULE

Default deny. Allow by exception.

A firewall policy starts by blocking everything inbound, then permits only the specific ports, sources and services the business needs - the stance you build in Lab 21.

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DOMAIN 4.0 · HANDS-ON

Host Firewall with UFW

LAB 21

Configure a default-deny host firewall with UFW: global allows, source-restricted rules, rule management.

Set a deny-by-default inbound policy with UFW, permit SSH globally and MySQL only from 10.0.0.0/8, then list, delete and disable rules to manage the firewall lifecycle.

YOU'LL BUILD

A host firewall that denies all inbound traffic except SSH and source-restricted MySQL.

TOOLCHAIN

ufw, iptables

DONE WHEN

ufw status verbose shows Default: deny (incoming), allow (outgoing) plus your 22/tcp and 3306 rules; iptables -L shows the same rules generated underneath.

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LAB 21 · PROCESS MAP

How Lab 21 Runs



THE POINT

Configure a default-deny host firewall with UFW: global allows, source-restricted rules, rule management.

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LAB 21 · VERIFY

Host Firewall with UFW

✅ Expected result

ufw status verbose shows Default: deny (incoming), allow (outgoing) plus your 22/tcp and 3306 rules; iptables -L shows the same rules generated underneath.

IF IT DOESN'T WORK

Locked out of SSH

Always allow 22/tcp before ufw enable; from the console run ufw disable, re-add the rule, re-enable

Rule never matches

Check order with ufw status numbered - more specific source rules must sit above broad ones

ufw: command not found

Install it first: apt update && apt install -y ufw

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OBJ 4.1 · LOGICAL SECURITY

Making TCP/IP secure.

TCP/IP was designed for openness, not secrecy - confidentiality, integrity and availability are bolted on with encryption, hashes and authentication at every layer.

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Encryption: In Transit vs At Rest

Data in transit

- Moving across the network
- TLS 1.3 for web and APIs
- IPSec or WireGuard for site links
- SSH for management sessions

Data at rest

- Stored on disk or in the cloud
- Full-disk encryption (LUKS, BitLocker)
- Database and object-store encryption
- Key management decides real safety

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Symmetric Encryption

1

Cleartext

Any unencrypted data - readable by whoever captures the packet.

2

Algorithm + key

The algorithm (AES) is public and well studied; all the secrecy lives in the key.

3

Ciphertext

The scrambled output - useless to an eavesdropper without the key.

4

Same key both ways

Symmetric = one shared key encrypts AND decrypts - fast, but that key must travel safely.

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Symmetric vs Asymmetric Encryption

	Symmetric	Asymmetric
Keys	One shared secret key	Public + private key pair
Speed	Fast - bulk data encryption	Slow - small payloads only
Examples	AES-256, ChaCha20	RSA-2048, ECC / Curve25519
Key exchange	Hard - secret must travel	Easy - publish the public key
Used for	Session data after handshake	Handshake, signatures, certs

WHEN IT MATTERS TLS uses both: asymmetric crypto agrees a session key, symmetric crypto carries the data.

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Cryptographic Hashes

1

Fixed-size fingerprint

Any input, one fixed-length output - SHA-256 always yields 256 bits.

2

Integrity, not secrecy

A hash proves data is unchanged; it does not hide the data.

3

Avalanche effect

Change one bit of input and the entire hash changes - tampering is obvious.

4

MD5 / SHA-1 retired

Collisions have been engineered - use SHA-256 or SHA-3 today.

5

Password storage

Servers store salted hashes, never the password itself.

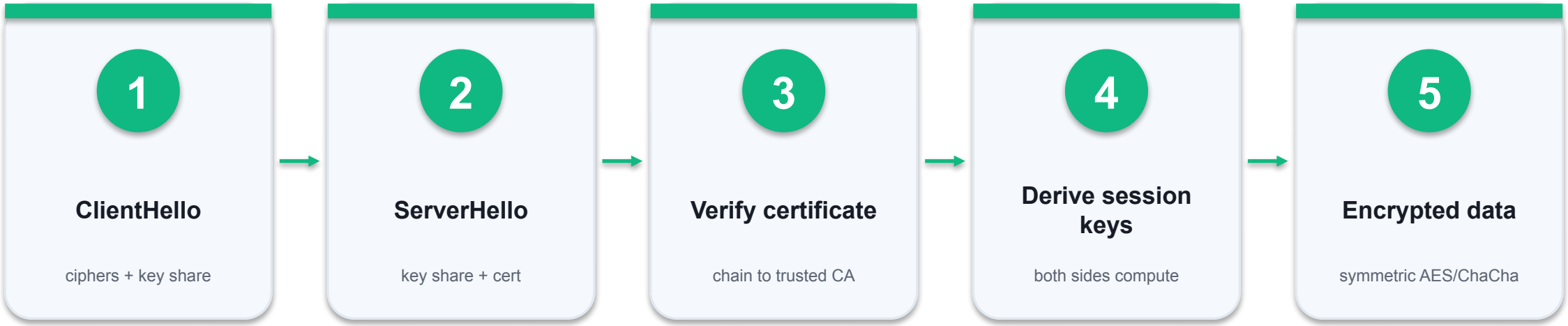
6

HMAC

A keyed hash proving both integrity and authenticity of a message.

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The TLS 1.3 Handshake



WHY IT MATTERS

TLS 1.3 completes in one round-trip: asymmetric crypto authenticates the server and agrees keys, then fast symmetric crypto protects the whole session.

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Certificates and PKI

1

X.509 certificate

Binds a public key to an identity (CN/SAN), signed by an issuer with validity dates.

2

Certificate authority

A trusted third party that signs certificates after verifying the requester's identity.

3

Chain of trust

Leaf cert -> intermediate CA -> root CA pre-installed in every OS and browser.

4

Self-signed

Signed by its own key - fine for labs and internal tools, but every browser warns.

5

CSR

Certificate signing request - your public key plus identity, sent to the CA for signing.

6

Let's Encrypt / ACME

Free automated CA - the ACME protocol issues and renews certs every 90 days.

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Reading a Certificate with openssl x509

The certificate you generate in Lab 22, decoded field by field.

```
$ openssl x509 -in lab.crt -noout -text
Certificate:
  Data:
    Serial Number: 5d:9c:41:...:2e
    Issuer: CN = lab.local
    Validity
      Not Before: Aug 19 02:00:00 2026 GMT
      Not After : Aug 19 02:00:00 2027 GMT
    Subject: CN = lab.local
    Subject Public Key Info:
      Public-Key: (2048 bit)
```

Issuer = Subject

Both read CN=lab.local - the certificate signed itself, so no browser will trust it.

Validity window

Outside Not Before / Not After the cert throws expired or not-yet-valid errors.

Public key

The RSA-2048 key clients use in the TLS handshake to agree session secrets.

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Certificate Error Scenarios

1

Self-signed warning

No CA in the chain - the browser shows ERR_CERT_AUTHORITY_INVALID.

2

Expired certificate

Past its Not After date - renew with the issuer; automate renewal with ACME.

3

Name mismatch

The CN/SAN does not match the URL's hostname - common after renames and moves.

4

Revoked certificate

The CA withdrew it - clients check live via OCSP or download CRL lists.

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DOMAIN 4.0 · HANDS-ON

TLS Certificate with OpenSSL and Nginx

LAB 22

Generate a self-signed X.509 certificate, serve HTTPS from nginx and inspect the TLS handshake.

Create an RSA-2048 self-signed certificate with openssl req -x509, configure nginx for HTTPS on 443, then verify the handshake and certificate fields from the client with curl, openssl x509 and s_client.

YOU'LL BUILD

An HTTPS server on 443 presenting your own certificate, plus a client-side view of the TLS 1.3 handshake.

TOOLCHAIN

openssl, nginx, curl

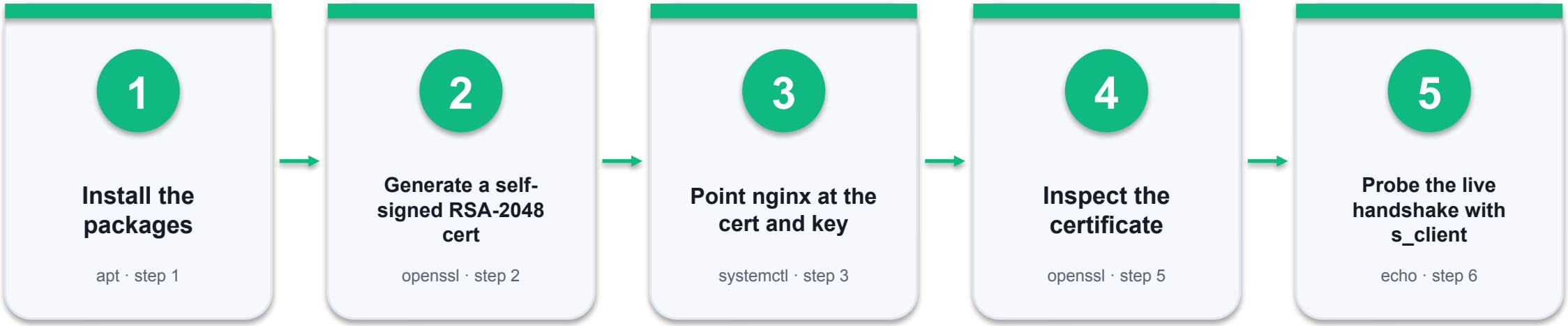
DONE WHEN

curl -kv https://localhost prints the TLS version, cipher suite and the CN=lab.local certificate; openssl x509 -text shows identical Issuer and Subject - the mark of a self-signed cert.

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LAB 22 · PROCESS MAP

How Lab 22 Runs



THE POINT

Generate a self-signed X.509 certificate, serve HTTPS from nginx and inspect the TLS handshake.

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LAB 22 · VERIFY

TLS Certificate with OpenSSL and Nginx

✅ Expected result

curl -kv https://localhost prints the TLS version, cipher suite and the CN=lab.local certificate; openssl x509 -text shows identical Issuer and Subject - the mark of a self-signed cert.

IF IT DOESN'T WORK

curl: SSL certificate problem

Expected for self-signed - pass -k to curl, or install a CA-signed certificate instead

nginx fails to restart

Run nginx -t to locate the config error; verify the ssl_certificate and key paths exist

Nothing listening on 443

Confirm listen 443 ssl in the server block, then check with ss -tlnp | grep 443

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OBJ 4.1 · IAM

Who are you, and what may you do?

Identity and access management is security's control plane: identification, authentication, authorization and accounting govern every login, port and API call.

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Identification to Accounting (IAM / AAA)

1. Identification - claim an identity (username)
2. Authentication - prove it (password + MFA)
3. Authorization - what you may do (RBAC)
4. Accounting - log what you did (audit trail)

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Multifactor Authentication (MFA)

1

Something you know

Password, PIN, passphrase - the weakest factor on its own.

2

Something you have

Phone TOTP app, smart card, hardware token (YubiKey).

3

Something you are

Biometrics - fingerprint, face, iris.

4

Somewhere you are

Location attribute - geofencing blocks logins from outside approved areas.

5

Something you do

Behavioural attributes - typing rhythm, signature dynamics.

6

MFA = 2+ DIFFERENT factors

Two passwords is not MFA; a password plus a TOTP code is.

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Time-Based and Location-Based Authentication

1

Time-based OTP (TOTP)

A shared secret plus the current time yields a 6-digit code valid for 30 seconds.

2

Geofencing

Access permitted only from defined geographic areas or IP ranges.

3

Push / passkeys

FIDO2 passkeys bind the login to the genuine site - phishing-resistant MFA.

4

Least privilege

Grant the minimum rights a role needs, and remove them when the role ends.

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Access Control Models

1

Role-based (RBAC)

Rights attach to roles/groups; users inherit by membership - the enterprise default.

2

Mandatory (MAC)

Labels and clearances fixed by policy; users cannot re-share - military and government.

3

Discretionary (DAC)

The resource owner decides who gets access - flexible, but permissions sprawl.

4

Least privilege lens

Whatever the model, grant minimum rights and review them regularly.

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Single Sign-On and Federation

1

Single sign-on (SSO)

Authenticate once, access many apps - one strong login to protect with MFA.

2

SAML

XML federation: an identity provider (IdP) asserts who you are to service providers.

3

LDAP / LDAPS

Directory lookups on TCP 389; LDAPS wraps them in TLS on 636 - how apps query AD.

4

Kerberos

Ticket-based SSO in Windows domains - time-stamped tickets issued by the KDC.

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RADIUS vs TACACS+

	RADIUS	TACACS+
Transport	UDP 1812/1813 (or 1645/1646)	TCP 49
Encryption	Password field only	Entire payload
AAA split	AuthN + AuthZ combined	All three separated
Typical use	Wi-Fi, VPN, 802.1X access	Admin access to network gear
Origin	Open standard	Cisco (now published)

WHEN IT MATTERS Exam favourite: RADIUS = network access over UDP; TACACS+ = device administration over TCP 49.

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EAP - the Authentication Framework

1

EAP

A framework, not one protocol - it carries many authentication methods inside 802.1X.

2

EAP-TLS

Certificates on both server and client - strongest, passwordless.

3

PEAP

The server's cert builds a TLS tunnel, credentials go inside - the campus Wi-Fi standard.

4

EAP-PSK / EAP-MD5

Legacy lightweight methods - avoid wherever certificates are possible.

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Device Hardening

1

Change default passwords

Factory admin credentials sit in every attacker's wordlist - change them on day one.

2

Disable unused services

Every listening daemon is attack surface - if nobody needs it, stop and disable it.

3

Disable unused ports

Shut idle switch ports; an unused wall jack must not be a live network drop.

4

Patch firmware

Known CVEs in network gear are scanned for within hours of disclosure.

5

Key management

Rotate SSH and API keys; revoke a leaver's keys the day they leave.

6

Secure the management plane

SSH not Telnet, HTTPS not HTTP, on a dedicated management VLAN.

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A Hardened sshd_config

The Lab 23 end-state: keys only, limited users, tested before reload.

```
# /etc/ssh/sshd_config - hardened
Port 2222
PermitRootLogin prohibit-password
PasswordAuthentication no
PubkeyAuthentication yes
MaxAuthTries 3
AllowUsers admin

# validate BEFORE reloading:
$ sshd -t && systemctl reload ssh
```

Keys only

PasswordAuthentication no removes brute force as an attack class - keys cannot be guessed.

Blast-radius limits

AllowUsers and MaxAuthTries shrink who may try to log in, and how often.

sshd -t

Test the config before reload - a syntax error must never lock you out of a remote box.

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DOMAIN 4.0 · HANDS-ON

SSH Hardening with Key Authentication

LAB 23

Harden SSH: Ed25519 key authentication, password login disabled, config validated before reload.

Generate an Ed25519 key pair, install the public key in authorized_keys, prove key login works, then disable PasswordAuthentication in sshd_config, validate with sshd -t and confirm password login is refused.

YOU'LL BUILD

An SSH daemon that accepts only public-key authentication, with root password login prohibited.

TOOLCHAIN

openssh-server, openssh-client, ssh-keygen

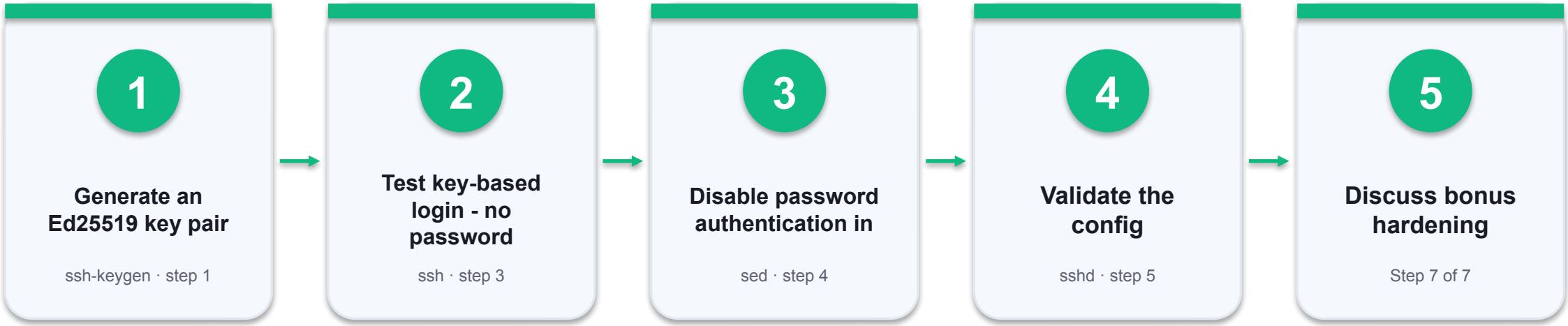
DONE WHEN

Key login returns root with no password prompt; forcing password authentication returns Permission denied (publickey).

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LAB 23 · PROCESS MAP

How Lab 23 Runs



THE POINT

Harden SSH: Ed25519 key authentication, password login disabled, config validated before reload.

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LAB 23 · VERIFY

SSH Hardening with Key Authentication

✅ Expected result

Key login returns root with no password prompt; forcing password authentication returns Permission denied (publickey).

IF IT DOESN'T WORK

Permission denied with the key

Fix permissions: chmod 700 ~/.ssh and chmod 600 ~/.ssh/authorized_keys - sshd refuses lax modes

sshd -t reports errors

Fix the reported line in /etc/ssh/sshd_config before reloading - a bad config can lock you out

Password login still works

Check override files in /etc/ssh/sshd_config.d/ for PasswordAuthentication yes, then reload ssh

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VPN - Extending the Trusted Zone

1

Site-to-site VPN

Routers or firewalls tunnel whole networks together across the Internet.

2

Client-to-site

Remote access - a laptop joins the corporate network from anywhere.

3

IPSec

AH/ESP with IKE key exchange - the classic standard: rich, interoperable, complex.

4

WireGuard

Modern and tiny: Curve25519 key pairs, no PKI needed - what you build in Lab 24.

5

Split vs full tunnel

Route everything through the VPN, or only the corporate subnets.

6

Why it matters

A VPN extends the trusted zone across an untrusted network - encryption in transit.

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VPN Technologies Compared

	IPSec	TLS VPN	WireGuard
Layer	Network (L3)	Session (TLS)	Network (L3)
Keys	IKE + certs / PSK	X.509 certificates	Static 32-byte key pairs
Codebase	Large, complex	Large (OpenVPN)	Tiny (~4k lines)
Speed	Good (HW offload)	Moderate	Excellent
Best for	Standards, site-to-site	Remote access via 443	Modern deployments

WHEN IT MATTERS WireGuard's allowed-ips doubles as a routing rule AND an inbound ACL - you will see this in Lab 24.

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DOMAIN 4.0 · HANDS-ON

Site-to-Site VPN with WireGuard

LAB 24

Build a site-to-site WireGuard tunnel between two network namespaces and verify the handshake.

Create two namespaces as VPN endpoints, generate Curve25519 key pairs, configure wg0 interfaces with peer public keys and allowed-ips, bring the tunnel up, ping across it and inspect handshake state with wg show.

YOU'LL BUILD

An encrypted point-to-point tunnel (10.99.0.1 <-> 10.99.0.2) between two namespaces acting as two sites.

TOOLCHAIN

wireguard, wireguard-tools, iproute2

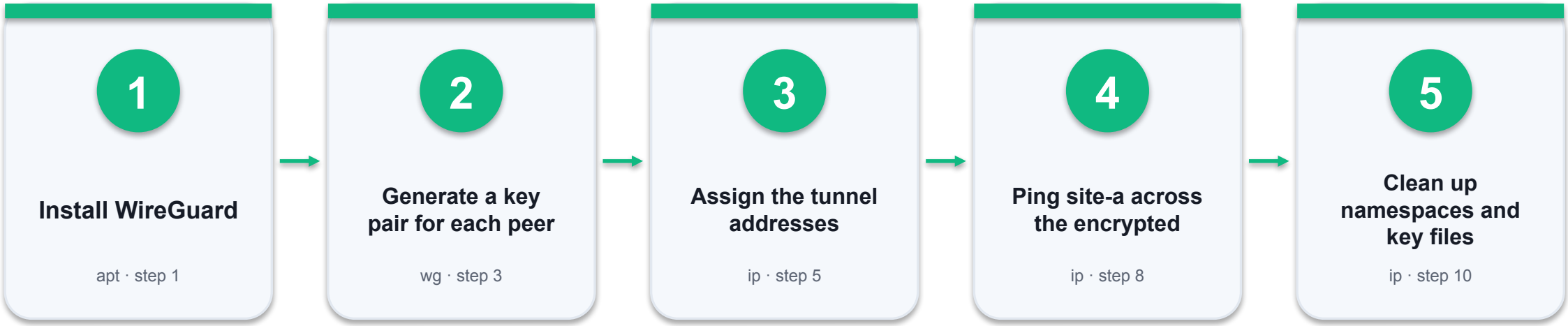
DONE WHEN

ping -c2 10.99.0.1 from site-b succeeds through the tunnel; wg show reports the peer public key, a recent handshake time and rising transfer counters.

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LAB 24 · PROCESS MAP

How Lab 24 Runs



THE POINT

Build a site-to-site WireGuard tunnel between two network namespaces and verify the handshake.

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LAB 24 · VERIFY

Site-to-Site VPN with WireGuard

✅ Expected result

ping -c2 10.99.0.1 from site-b succeeds through the tunnel; wg show reports the peer public key, a recent handshake time and rising transfer counters.

IF IT DOESN'T WORK

Ping fails across the tunnel

Check wg show for a handshake; verify each peer entry uses the other side's PUBLIC key

wg show shows no handshake

Confirm site-b's endpoint 127.0.0.1:51820 matches site-a's listen-port, and allowed-ips cover the peer /32

wg0 already exists

Stale run - delete with ip netns del site-a && ip netns del site-b and recreate

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OBJ 4.3 · ATTACK SURFACE

Every open port is a doorway.

Attack-surface reduction is the cheapest security win: enumerate what is listening, ask why it is there, and disable what nobody needs - before an attacker does the asking.

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Malware Taxonomy

1

Virus / worm

A virus infects files; a worm self-propagates across the network unaided.

2

Trojan / RAT

Hides inside legitimate-looking software; a RAT hands the attacker remote control.

3

Ransomware

Encrypts your data and demands payment - today's top business threat.

4

Spyware / keylogger

Silently collects keystrokes, credentials and screen contents.

5

Rootkit

Buries itself under the OS to hide processes and survive detection.

6

Logic bomb

Dormant code that triggers on a date or event - often an insider plant.

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Social Engineering

1

Phishing

Fraudulent mail and sites harvest credentials; spear phishing targets a person, whaling an exec.

2

Dumpster diving

Discarded printouts leak configs and passwords - shred everything.

3

Shoulder surfing

Watching screens and keypads - privacy filters and awareness defeat it.

4

Tailgating

Following a badge-holder through a secure door - vestibules stop it.

5

Impersonation

The fake IT-support call that just needs your password to help you.

6

Defence = education

Humans are part of the perimeter: train, phish-test, make reporting easy.

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Password Attacks

1

Brute force

Try every combination - length beats complexity; each extra character multiplies the work.

2

Dictionary

Try likely words and leaked passwords first - most humans are predictable.

3

Credential stuffing

Replay leaked username/password pairs on other sites - MFA breaks it.

4

Defences

Long passphrases, lockout policies, salted hashes, MFA everywhere.

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Vulnerability Management Loop

1. Scan the estate for exposed services
2. Prioritise by CVSS score and exploitability
3. Patch on a defined cadence with change control
4. Re-scan to verify the exposure is closed

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DOMAIN 4.0 · HANDS-ON

Port Scanning and Service Hardening

LAB 25

Enumerate exposed services with nmap and reduce the attack surface by disabling what is unneeded.

SYN-scan all 65535 TCP ports on your own host, identify service versions and the OS, cross-check each open port against ss -tulnp, disable an unnecessary service and re-scan to confirm the port is closed.

YOU'LL BUILD

A port-by-port inventory of your host's attack surface, with an unneeded service disabled and verified closed.

TOOLCHAIN

nmap, ss, systemctl

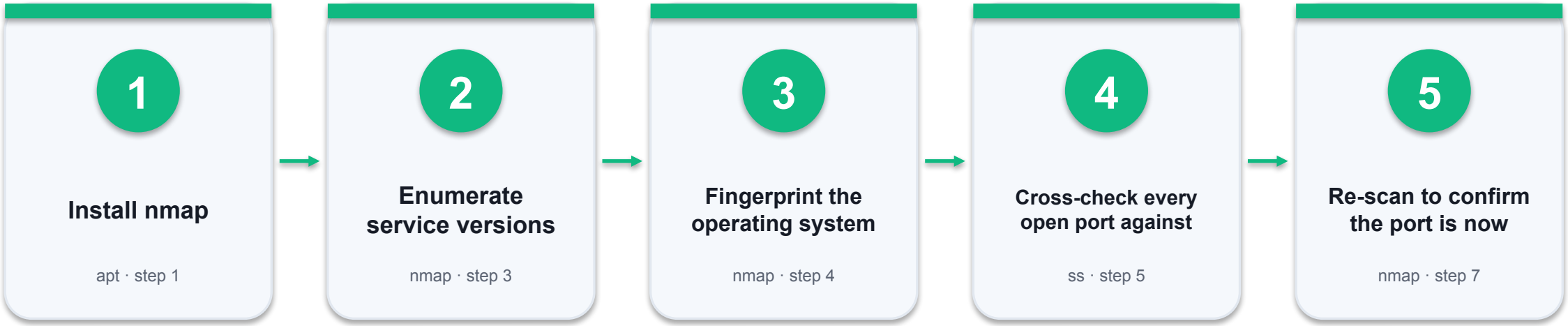
DONE WHEN

The final nmap -sS -p 80 scan reports the port closed, and ss -tulnp no longer lists the disabled service - the attack surface measurably shrank.

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LAB 25 · PROCESS MAP

How Lab 25 Runs



THE POINT

Enumerate exposed services with nmap and reduce the attack surface by disabling what is unneeded.

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LAB 25 · VERIFY

Port Scanning and Service Hardening

✅ Expected result

The final nmap -sS -p 80 scan reports the port closed, and ss -tulnp no longer lists the disabled service - the attack surface measurably shrank.

IF IT DOESN'T WORK

Scan takes very long

-p- probes 65535 ports; in the lab scan a shortlist such as -p 1-1024 instead

nmap -O gives no result

OS fingerprinting needs root privileges and at least one open and one closed port

Port still open after disable

Another process owns it - find the PID with ss -tulnp and stop that unit too

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Attack Catalogue: Vector, Layer, Defence

Attack	Layer	How it works	Primary defence
DoS / DDoS	3-7	Botnet floods links or servers	Scrubbing, rate limits
MAC flooding	2	Overflow CAM table; switch floods	Port security limits
ARP poisoning	2	Forged replies put attacker on-path	Dynamic ARP Inspection
VLAN hopping	2	Double tags / fake trunk cross VLANs	No native VLAN, no DTP
DNS poisoning	7	Fake records redirect victims	DNSSEC, patched resolvers
Evil twin	1-2	Cloned SSID captures clients	802.1X, WIPS monitoring

WHEN IT MATTERS For every attack, know the layer it lives at - the defence always sits at the same layer.

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DoS and DDoS in Depth

1

Botnet and C2

Compromised zombies take orders from a command-and-control server.

2

Volumetric

Saturate the pipe - UDP floods, amplification via open DNS/NTP servers.

3

Protocol attack

Exhaust state tables - SYN floods hold thousands of half-open connections.

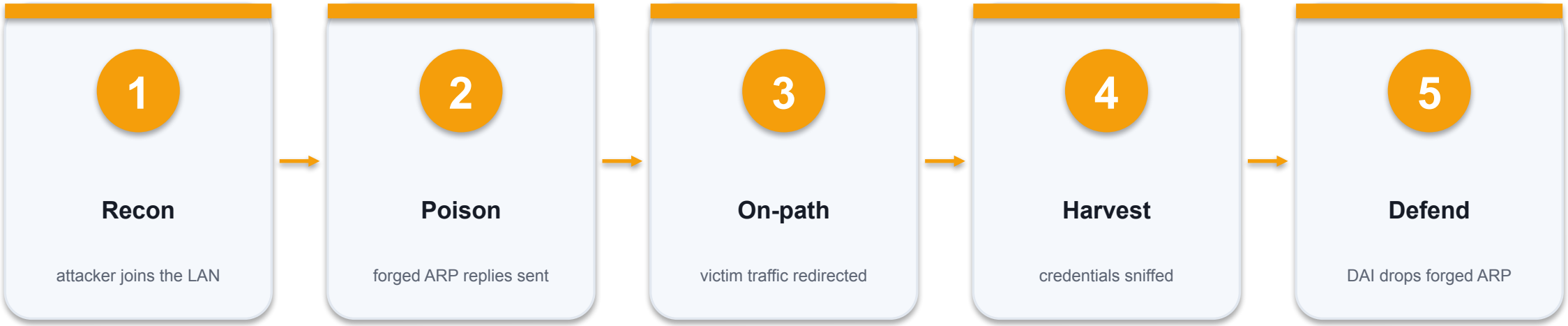
4

Application attack

Cheap requests, expensive responses - HTTP floods and Slowloris.

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Anatomy of an ARP Poisoning Attack



BREAK THE CHAIN

Dynamic ARP Inspection validates every ARP reply against the DHCP snooping binding table - forged replies are dropped at the switch port.

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VLAN Hopping

1

Switch spoofing

The attacker negotiates a trunk via DTP and receives every VLAN - disable DTP, set access mode.

2

Double tagging

Outer tag stripped on the native VLAN, inner tag delivers into the victim VLAN - one-way but real.

3

Native VLAN hygiene

Use a dedicated, unused native VLAN; never carry user traffic on it.

4

Private VLANs

Port isolation stops peer-to-peer traffic even inside the same VLAN.

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Rogue Devices and On-Path Attacks

1

Rogue DHCP server

Hands out attacker-controlled gateway/DNS - a lease outside your network ID is the tell.

2

Rogue AP

An unauthorised access point bridging outsiders straight into the LAN.

3

Evil twin

A cloned corporate SSID; clients auto-join and their traffic is captured.

4

On-path (MITM)

The attacker relays between two parties, reading or altering everything.

5

Defence: snooping + 802.1X

DHCP snooping drops rogue offers; port authentication blocks rogue gear.

6

Defence: WIPS

Wireless IPS detects unknown SSIDs and contains rogue radios.

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Spoofing Attacks

1

MAC spoofing

Clone an allowed MAC address to slip past port filters.

2

IP spoofing

Forge the source address - the engine behind amplification DDoS.

3

DNS spoofing / poisoning

Corrupt resolver caches so good names resolve to bad hosts.

4

Defences

Ingress filtering (BCP 38), DNSSEC validation, Dynamic ARP Inspection.

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Switch and Network Hardening

1

DHCP snooping

Only trusted uplink ports may send DHCP offers - kills rogue DHCP.

2

Dynamic ARP Inspection

Validates ARP against the snooping bindings - kills ARP poisoning.

3

Port security

Limit MACs per port, err-disable on violation - kills MAC flooding.

4

RA guard

Blocks rogue IPv6 router advertisements at access ports.

5

Control plane policing

Rate-limits traffic to the device CPU so it survives a DoS.

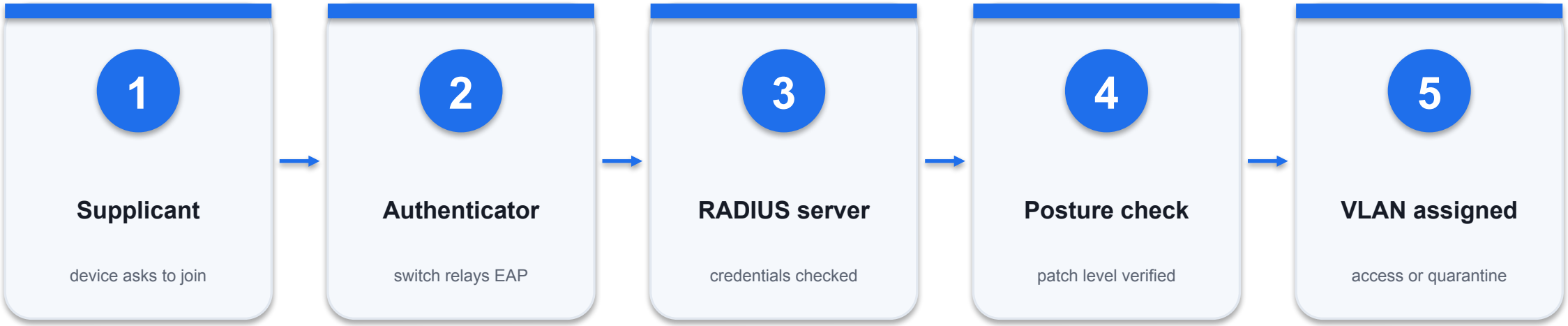
6

Disable unused ports

A patch-panel jack nobody uses should terminate nowhere.

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802.1X Network Access Control



WHY IT MATTERS

Nothing gets a usable port until identity and device health are proven - NAC turns every access port and SSID into a checkpoint.

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Segmentation Enforcement Zones

Zone	What lives there	Why segregate it
IoT	Cameras, sensors, smart devices	Rarely patched; assume compromised
IIoT / OT	Industrial sensors, actuators	Uptime-critical; no direct IT access
SCADA / ICS	Plant supervisory control	Legacy protocols with no auth
Guest	Visitor phones and laptops	Internet only - zero internal reach
BYOD	Staff personal devices	Unmanaged; NAC posture, limited access

WHEN IT MATTERS Segment by trust and blast radius: a hacked camera must never see the finance VLAN.

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Deception Technologies

1

Honeypot

A deliberately vulnerable decoy host - any touch on it is by definition suspicious.

2

Honeynet

A whole decoy network of honeypots modelling your real estate.

3

Why deceive

Attackers burn time on fakes while defenders harvest indicators and early warning.

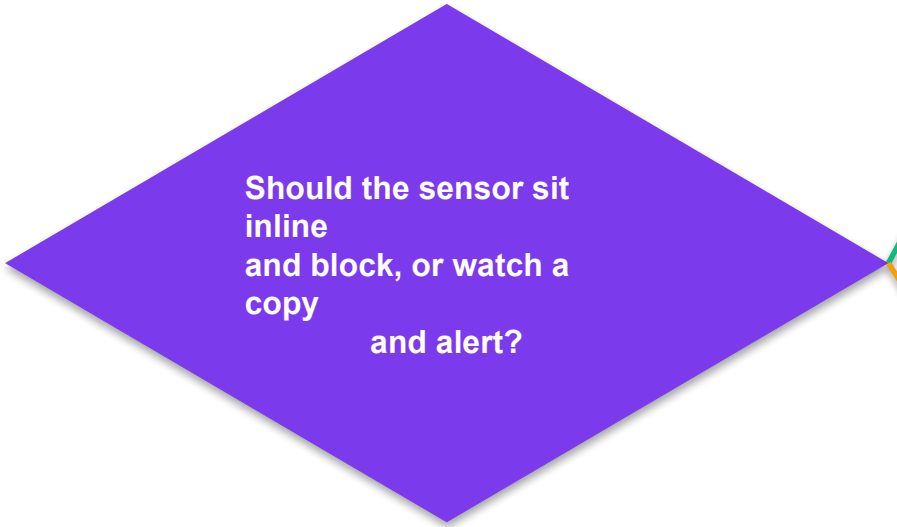
4

Placement

In the screened subnet or a dedicated VLAN - never among production data.

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IDS or IPS - Where Does the Sensor Sit?



INLINE - IPS

Sits in the traffic path and drops malicious packets in real time. Risk: a false positive blocks legitimate traffic; adds latency.

PASSIVE - IDS

Watches a mirror/TAP copy and raises alerts for analysts. No traffic risk, but detection only - response is manual.

Suricata can run either mode - Lab 26 runs it passively as an IDS on eth0.

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How IDS/IPS Detect

1

Signature-based

Match known patterns (ET rules) - precise, but blind to novel attacks.

2

Anomaly-based

Baseline normal, alert on deviation - catches zero-days, more false positives.

3

Rule feeds

Emerging Threats and Snort VRT keep signatures current - update daily.

4

Where alerts land

Suricata writes fast.log for humans and eve.json for SIEM pipelines.

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Reading a Suricata Alert (eve.json)

The alert you trigger in Lab 26, as the JSON event your SIEM ingests.

```
$ cat eve.json | jq 'select(.event_type=="alert")'
{
  "timestamp": "2026-08-19T10:42:07+0800",
  "event_type": "alert",
  "src_ip": "203.0.113.99",
  "dest_ip": "10.0.0.5",
  "alert": {
    "signature": "GPL ATTACK_RESPONSE
        id check returned root",
    "category": "Potentially Bad Traffic",
    "severity": 2
  }
}
```

event_type

One JSON object per event - jq filters alerts out of the flow, DNS and TLS records.

signature

The ET rule that fired: the HTTP response contained uid=0(root) - a compromise tell.

src / dest

Who attacked whom - the fields a SIEM correlates across all your sensors.

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DOMAIN 4.0 · HANDS-ON

Intrusion Detection with Suricata

LAB 26

Run Suricata as an IDS, trigger a known signature and read the alert in fast.log and eve.json.

Install Suricata, pull the Emerging Threats ruleset, start the engine on eth0, generate benign trigger traffic from testmynids.org, then trace the resulting alert through fast.log and the JSON eve.json event stream.

YOU'LL BUILD

A live IDS on eth0 with a current ET ruleset, plus a captured alert you can parse as JSON with jq.

TOOLCHAIN

suricata, suricata-update, jq, curl, TIS Cybersecurity Simulator

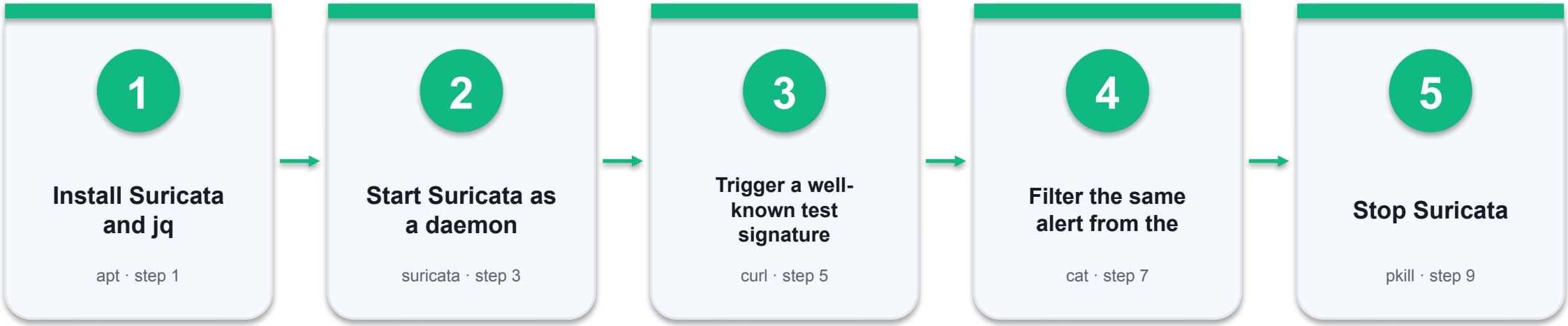
DONE WHEN

fast.log shows an alert like GPL ATTACK_RESPONSE id check returned root; filtering eve.json for event_type alert returns the same event with src/dest IPs and severity.

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LAB 26 · PROCESS MAP

How Lab 26 Runs



THE POINT

Run Suricata as an IDS, trigger a known signature and read the alert in fast.log and eve.json.

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LAB 26 · VERIFY

Intrusion Detection with Suricata

✅ Expected result

fast.log shows an alert like GPL ATTACK_RESPONSE id check returned root; filtering eve.json for event_type alert returns the same event with src/dest IPs and severity.

IF IT DOESN'T WORK

No alerts appear

Confirm Suricata listens on the right interface (check ip a), then re-run the testmynids curl

suricata-update fails

Check Internet access - the rules download from the Emerging Threats feed

eve.json is empty

Give the engine a few seconds after the trigger; check suricata.log for startup errors

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Access Control Lists (ACLs)

1

Top-down evaluation

Rules are checked in order; the FIRST match wins and processing stops.

2

Implicit deny

Whatever no rule matches is dropped - place your permits above the deny.

3

Match criteria

Source and destination IP, protocol, port - the classic 5-tuple.

4

Where ACLs live

Router interfaces, switch VLANs, cloud security groups, iptables chains.

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An iptables ACL, Line by Line

The Lab 27 rule set: permit MySQL from one subnet, drop the rest.

```
$ iptables -I INPUT -p tcp --dport 3306 \
    -s 10.0.0.0/24 -j ACCEPT
$ iptables -A INPUT -p tcp --dport 3306 -j DROP
$ iptables -L INPUT -n -v --line-numbers

num  target  prot  source        dpt
1    ACCEPT  tcp   10.0.0.0/24   3306
2    DROP    tcp   0.0.0.0/0     3306
```

-I vs -A

-I inserts at the top, -A appends at the bottom - placement decides what matches first.

First match wins

10.0.0.0/24 hits rule 1 and is accepted; everyone else falls through to the DROP.

Scope

Only port 3306 is filtered here - pair with a default-deny policy for real hardening.

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URL and Content Filtering

1

URL filtering

Allow or block by domain and category - malware, phishing, gambling lists.

2

Content filtering

Inspect payloads - DLP stops card numbers leaving; AV strips attachments.

3

DNS filtering

The cheapest control: a resolver that refuses to resolve known-bad domains.

4

TLS inspection

Filtering encrypted traffic needs a re-signing proxy - governance required.

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DOMAIN 4.0 · HANDS-ON

Building an ACL with iptables

LAB 27

Build an ordered iptables ACL that permits MySQL from one subnet and drops it from everywhere else.

Insert an ACCEPT rule for TCP 3306 from 10.0.0.0/24 above an appended DROP, verify ordering with --line-numbers, test both paths with nc from allowed and blocked sources, then persist the rules across reboots.

YOU'LL BUILD

A working two-rule ACL on the INPUT chain proving top-down, first-match evaluation, saved to survive reboot.

TOOLCHAIN

iptables, netcat, iptables-persistent

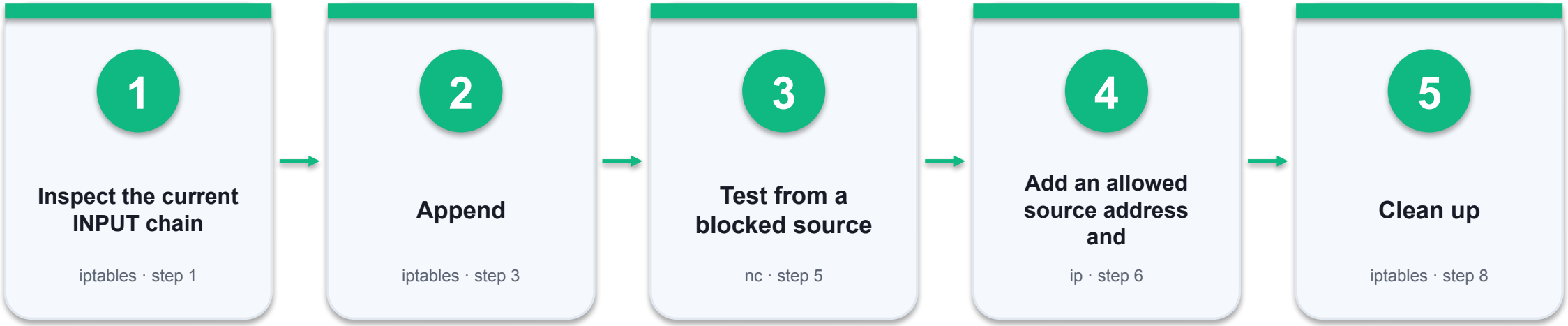
DONE WHEN

The probe from a blocked source hangs (dropped); the probe from 10.0.0.5 gets an instant refused/RST - accepted by the ACL, nothing listening. The filter demonstrably works.

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LAB 27 · PROCESS MAP

How Lab 27 Runs



THE POINT

Build an ordered iptables ACL that permits MySQL from one subnet and drops it from everywhere else.

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LAB 27 · VERIFY

Building an ACL with iptables

✅ Expected result

The probe from a blocked source hangs (dropped); the probe from 10.0.0.5 gets an instant refused/RST - accepted by the ACL, nothing listening. The filter demonstrably works.

IF IT DOESN'T WORK

ACL never blocks anything

The ACCEPT may sit below the DROP - first match wins; verify order with --line-numbers

Rules vanish after reboot

Install iptables-persistent and run netfilter-persistent save to write /etc/iptables/rules.v4

nc test looks inconclusive

With no MySQL listening, allowed sources see instant RST (refused); dropped sources see a timeout

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Physical Security

1

Cameras (CCTV)

Detective control - record and deter; cover entrances, corridors and racks.

2

Locks

Preventive - server rooms, racks and cages; smart locks log every entry.

3

Badge readers

Authenticate entry per person and leave an audit trail.

4

Access control vestibule

A two-door airlock defeats tailgating - one person per badge.

5

Security guards

Flexible deterrent and response - verify visitors and deliveries.

6

Environmental sensors

Door-open, temperature and humidity alarms in the data centre.

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Physical Control Categories

Preventive

- Stop the event: locks, MFA, firewalls
- Access control vestibules
- Strongest when layered together

Detective

- Discover the event: CCTV, IDS, logs
- Audits and SIEM alerting
- Enables fast, informed response

Deterrent / compensating

- Warning signs, visible cameras
- Guards deter casual attempts
- Compensating controls cover gaps

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System Life Cycle and Secure Disposal

1

Asset tags and inventory

You cannot secure what you do not track - tag and record every device.

2

Secure wipe

DoD 5220.22-M-style overwrite or crypto-erase before any disposal.

3

Factory reset

Acceptable only for devices that never held sensitive data.

4

Chain of custody

A documented audit trail from decommission to certified destruction.

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Exam Checkpoint - Numbers to Memorise

- RADIUS: UDP 1812/1813 (legacy 1645/1646) - TACACS+: TCP 49
- LDAP 389, LDAPS 636, Kerberos 88, HTTPS 443, SSH 22
- IDS detects on a mirror copy; IPS sits inline and blocks
- Screened subnet (DMZ) hosts public services between two firewalls
- MFA needs two DIFFERENT factors: know / have / are

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DOMAIN 4 · RECAP

Layers, least privilege, verify everything.

Domain 4 in one line: encrypt in transit and at rest, authenticate with multiple factors, segment aggressively, harden every device, and watch the logs.

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Recap — Network Security

- You can now: Configure a default-deny host firewall with UFW: global allows, source-restricted rules, rule management.
- You can now: Generate a self-signed X.509 certificate, serve HTTPS from nginx and inspect the TLS handshake.
- You can now: Harden SSH: Ed25519 key authentication, password login disabled, config validated before reload.
- You can now: Build a site-to-site WireGuard tunnel between two network namespaces and verify the handshake.
- You can now: Enumerate exposed services with nmap and reduce the attack surface by disabling what is unneeded.
- You can now: Run Suricata as an IDS, trigger a known signature and read the alert in fast.log and eve.json.
- You can now: Build an ordered iptables ACL that permits MySQL from one subnet and drops it from everywhere else.

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5.0

DOMAIN 5.0 · 24% OF THE EXAM

Network Troubleshooting

Methodology · cabling & physical interfaces · network services · performance · tools

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WHAT YOU'LL DO

Hands-On Labs — Network Troubleshooting

Labs 28–30

- Layer 1/2 Diagnostics with ip and ethtool
- Reachability with ping, traceroute and mtr
- DNS Troubleshooting with dig and resolvectl

Labs 31–33

- ARP and the IPv6 Neighbour Table
- Packet Capture with tcpdump and TIS PCAP Analyzer
- MTU and Path-MTU Discovery

Labs 34–36

- Diagnosing Routing Problems
- Simulating Slow and Lossy Links with tc
- Service Reachability Triage with curl, nc and openssl

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DOMAIN 5 · 24% OF THE EXAM

Network Troubleshooting.

Domain 5 is 24% of the N10-009 exam - the biggest single slice. Master one method, then apply it to cables, switches, routes, DNS and services.

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Why a Method Beats Guesswork

1

Symptoms are not causes

A 'slow website' can be DNS, routing, loss, MTU or the app itself - the method finds which before you change anything.

2

Gather before you touch

Information gathered first - symptoms, scope, timing - is evidence; information gathered after a change is contaminated.

3

The OSI model is your map

Every fault lives at a layer; walking the stack top-down, bottom-up or from the middle turns chaos into a search.

4

Change one thing at a time

Isolate variables - if you change three things and it works, you still do not know the cause and cannot prevent it.

5

Prevention is part of the fix

A ticket is not closed at 'it works' - it closes at 'it cannot happen again', or at least 'we will see it coming'.

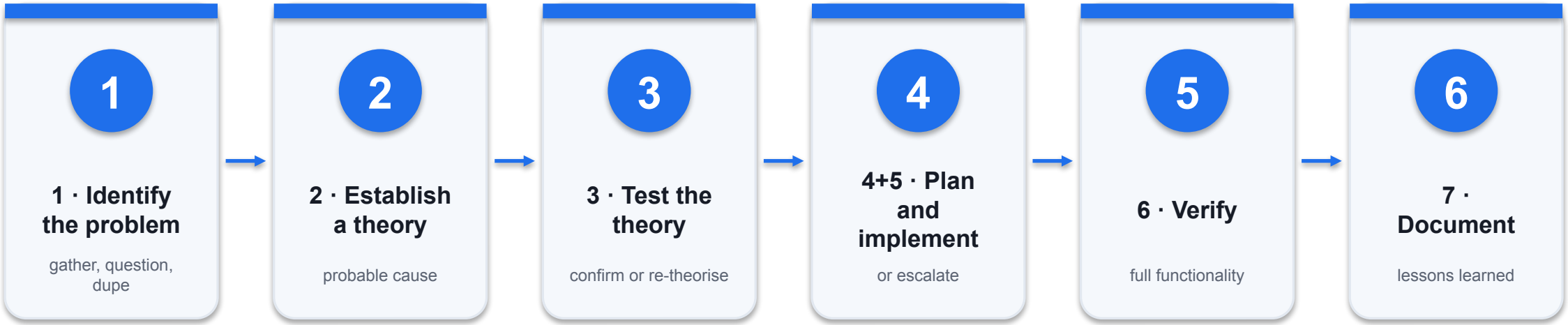
6

Documentation compounds

Yesterday's documented outage is today's 10-minute fix - lessons learned are the cheapest training you will get.

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CompTIA's 7-Step Troubleshooting Method



WHY IT MATTERS

Exam scenario questions state which step you are in and ask for the next action - knowing the order is free marks.

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Step 1 - Identify the Problem

1

Gather information

Ticket text, monitoring alerts, logs, topology diagrams - collect the facts before forming any opinion.

2

Question users

Ask open questions first ('what were you doing?'), then closed ones ('does it fail on Wi-Fi too?').

3

Identify symptoms

Pin down exactly what fails, for whom, since when and how often - 'the internet is down' is never the real symptom.

4

Determine if anything changed

New firmware, a moved desk, a patch window, a new switch - most outages follow a change.

5

Duplicate the problem

If you can reproduce it on demand, you can test theories against it safely and prove the eventual fix.

6

Approach problems individually

Two simultaneous complaints may be two faults - splitting them avoids chasing a cause that explains only half.

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Questioning Users and Finding the Change

Ask the user

- What exactly do you see - error text, spinner, timeout?
- When did it last work, and what changed since?
- Does it affect one app, one site or everything?
- Are colleagues nearby affected too?
- Does it follow you to another desk or network?

Ask the environment

- Check the change log and maintenance windows
- Compare against monitoring baselines and alerts
- Did patching, firmware or config pushes run overnight?
- Look for moved equipment, new cables, new devices
- Scope it: one user, one switch, one VLAN or one site?

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Step 2 - Establish a Theory of Probable Cause

1

Question the obvious

Cable seated? Power on? Right SSID? Caps Lock? The embarrassing causes are the most common - eliminate them first.

2

Top-to-bottom OSI

Start at L7 and walk down - natural when the app clearly fails but the network looks healthy.

3

Bottom-to-top OSI

Start at L1 and walk up - natural for a single dead desk: cable, link light, then IP, then services.

4

Divide and conquer

Start mid-stack: ping the gateway. Success rules out L1-L3 locally; failure rules out everything above.

5

Consider multiple approaches

Hold more than one candidate theory and rank them by likelihood and by how cheap they are to test.

6

Use scope as evidence

One user points down the stack at their gear; a whole site points up at shared infrastructure - WAN, DNS, DHCP, core.

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Divide and Conquer - Where to Start



YES - start in the middle

Shared infrastructure is suspect. Ping the gateway and a known IP: works means look up (DNS, app); fails means look down (L1-L3).

NO - start at the bottom

A single victim is usually local. Check cable and link light, then IP/mask/gateway, then walk up the stack layer by layer.

Divide and conquer halves the search space with every test - the binary search of troubleshooting.

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Step 3 - Test the Theory

Theory confirmed

- Determine the next steps to resolve
- Move straight to a plan of action
- Record what proved the cause

Theory busted

- Re-establish a new theory
- Re-question the obvious
- Widen the scope of what you test

Beyond your access or skill

- Escalate - do not keep digging
- Hand over the evidence so far
- List the tests already performed

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Steps 4 and 5 - Plan, Implement or Escalate

1

Plan before you touch

Write the steps, the order, and who does what - a fix applied ad hoc at 2 a.m. becomes tomorrow's mystery config.

2

Identify potential effects

Will the fix drop users, restart services, or trigger STP recalculation? Say so before it happens.

3

Schedule the change

Non-urgent fixes go in a change window with approval; outage fixes still get a one-line plan first.

4

Prepare the rollback

Know how to undo the change before you make it - save the running config, snapshot, or note the old value.

5

Implement or escalate

Execute your plan - or hand it to the vendor, ISP or senior engineer if it exceeds your authority or access.

6

Verify as you go

Test after each step of the plan, not only at the end, so a failed step is caught immediately.

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Steps 6 and 7 - Verify and Document

1. **Verify FULL functionality**
   Test more than the reported symptom - the user's app, neighbours on the same switch, and anything your change touched.

2. **Have the user confirm**
   The ticket closes when the reporter agrees it works, not when your ping succeeds.

3. **Implement preventive measures**
   Monitoring alerts, config backups, UPS, spare cables, tightened change control - stop the repeat.

4. **Document findings**
   What was broken and what the root cause was - written for the next engineer, not for yourself.

5. **Document actions and outcomes**
   Exactly what was changed, where, and the result - this becomes tomorrow's change log.

6. **Capture lessons learned**
   What slowed the diagnosis? What would have caught it sooner? Feed that into runbooks and training.

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The 7 Steps at a Glance

Step	Goal	Key actions
1 Identify	Know the real problem	Gather info, question users, what changed?
2 Theorise	Rank probable causes	Question the obvious; walk the OSI stack
3 Test	Confirm or discard	Prove the theory, re-theorise or escalate
4+5 Plan + implement	Fix safely	Plan steps and effects; act or escalate
6 Verify	Whole system works	Full functionality + preventive measures
7 Document	Keep the knowledge	Findings, actions, outcomes, lessons

WHEN IT MATTERS Exam questions quote a scenario mid-flow and ask for the NEXT step - learn the order cold.

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Symptom to Layer to First Tool

Layer	Typical symptom	First tool to reach for
L1 Physical	No link light, error counters climb	Cable tester, ethtool, link LEDs
L2 Data Link	Floods, loops, VLAN oddities	ip neigh / arp, switch counters
L3 Network	Some destinations unreachable	ping, traceroute/mtr, ip route
L4 Transport	Connects to some ports only	nc -zv, ss -tlnp, firewall rules
L7 Application	TCP fine but service errors	curl -v, dig, server logs

WHEN IT MATTERS Pick the layer the symptom names, test there, then move up or down on the result.

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The Method in 60 Seconds - a Dry Run

1. Identify: one user, wired, no web since desk move
2. Theory: cable damaged in the move (L1 first)
3. Test: swap patch lead - link light returns
4. Plan+fix: replace lead, label the run
5. Verify + document: user confirms, ticket notes

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OBJ 5.2 · CABLING & INTERFACES

Physical first: most 'network down' calls end at a cable.

Objective 5.2: cabling faults, interface errors and hardware issues - the layer where troubleshooting most often starts and ends.

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Incorrect Cable Type

1

Wrong Cat rating

Cat 5e on a 10G link negotiates down or throws errors - match the category to the speed and the run length.

2

UTP where STP is needed

Unshielded cable beside motors, lifts or fluorescent ballasts picks up EMI - shielded (STP) or a new path fixes it.

3

Single-mode vs multimode

SMF (yellow, laser, long haul) and MMF (aqua/orange, short haul) do not mix - and neither do their transceivers.

4

TX/RX transposed

Fibre pair crossed at one end: both link lights stay dark. Flip the pair at ONE end only.

5

Copper vs fibre reach

UTP tops out at 100 m per horizontal run (EIA/TIA allows 90 m + patch); beyond that use fibre or add a switch.

6

The jacket tells a story

Read the print on the cable jacket - Cat level, shielding and plenum rating are all written on it.

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Signal Degradation

1

Attenuation

Signal weakens with distance - runs past spec show as errors and negotiated-down speeds, worst at higher frequencies.

2

Crosstalk

Signal bleeding between pairs, usually from sloppy termination - pairs untwisted too far at the plug.

3

NEXT and FEXT

Near-end crosstalk is measured at the transmitter end, far-end at the receiver - certifiers report both in dB.

4

EMI / interference

Motors, HVAC, power runs and fluorescent lighting induce noise into copper - reroute the cable or shield it.

5

dB loss on fibre

Dirty connectors, tight bends and bad splices eat the optical budget - measure with an optical power meter.

6

The symptom pattern

Degradation rarely kills a link outright - it shows as CRC errors, retransmissions and 'slow' complaints.

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Improper Termination

1

T568A/B mixed ends

One end A, the other B makes an accidental crossover - modern MDI-X often saves you, but certifiers flag it.

2

Split pairs

Wires landed so a circuit uses conductors from two different twisted pairs - passes continuity, fails at speed.

3

Untwisted too far

More than ~13 mm untwisted at the punch-down or plug invites crosstalk - re-twist and re-terminate.

4

Bad punch-down / loose IDC

Half-seated conductors at the patch panel work intermittently - the classic 'wiggle the cable' fault.

5

Damaged RJ45 latch

A broken retention clip lets the plug back out of the jack - the link drops when anyone touches the desk.

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Cable Faults vs Tester Results

Fault	Wiremap / tester shows	Typical cause
Open	Pin fails continuity	Break or unpunched conductor
Short	Two pins joined	Crushed jacket, stray strand
Split pair	Map OK, NEXT fails	Pairs mis-landed on pins
Reversed pair	Pair order flipped	A/B mix at one end
Crossed pairs	1-2 swapped with 3-6	Crossover lead in the run
Excess length	Run over 90-100 m	TDR shows distance vs spec

WHEN IT MATTERS Wiremap finds opens, shorts and miswires; a TDR adds the distance to the fault.

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Cable Test Instruments

1

Cable tester / wiremap

Confirms each of the 8 pins lands on the right pin at the far end and continuity exists - first tool on any drop.

2

Continuity test

Shows whether the conductor is broken anywhere along the run - a pass means the copper path is complete.

3

TDR

Time domain reflectometer - fires a pulse and times the echo to report cable length and the distance to a break.

4

OTDR

The optical TDR for fibre - locates breaks, bad splices and dirty connectors along a fibre run.

5

Cable certifier

Measures NEXT/FEXT, attenuation and return loss against the Cat spec - proves a run earns its rating.

6

Optical power meter

Measures received light in dBm - the tool for weak-signal and dirty-connector fibre faults.

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Locating and Verifying: Toner, Loopback, Multimeter

1

Tone generator

Injects a signal onto a pair at one end - the 'Fox' of the classic Fox & Hound pair.

2

Tone probe

Held near cables, it sings when it finds the toned pair - traces unlabeled runs through walls and bundles.

3

Loopback plug

Loops TX back to RX so a NIC or switch port tests its own send/receive path in isolation.

4

Loopback address

127.0.0.1 (::1 in IPv6) tests the local IP stack with no cable at all - can this host even talk to itself?

5

Multimeter

Voltage, current, resistance and continuity - the tool for PoE voltage checks and suspect power.

6

Voltage monitor / UPS

Logs sags and drops in supply power - unexplained rack reboots are a power problem until proven otherwise.

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Reading Port Status

Status	Meaning	Your move
up / up	Admin up, carrier present	Healthy - look elsewhere
up / down	Enabled but no carrier	Cable, far end, transceiver
LOWERLAYERDOWN	Linux: no carrier sensed	Check the physical path
admin down	Deliberately disabled	no shutdown / ip link set up
err-disabled	Switch shut it on error	Fix the cause, then re-enable

WHEN IT MATTERS err-disabled common causes: port-security violation, BPDU guard, link flapping.

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Interface Error Counters

1

CRC errors

Frames that fail the checksum - bad cable, noise or duplex mismatch; any steadily climbing count is a fault.

2

Runts

Frames under 64 bytes, usually collision fragments - the classic signature of a duplex mismatch.

3

Giants

Frames over the MTU - typically an MTU or jumbo-frame mismatch between devices.

4

Drops / discards

The device threw frames away - buffers full (congestion) or frames it could not process.

5

Duplex mismatch

One side full, one side half: link stays up but collisions, CRCs and runts soar under load - forced-speed misconfig is the usual cause.

6

Read the DELTA

Absolute counts mean little - clear or note the counters, wait, re-read; the growth rate convicts.

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Reading an ethtool Counter Dump

Lab 28's core skill: convict a bad link from its counters.

```
$ ethtool eth0 | grep -E 'Speed|Duplex|Link'
        Speed: 1000Mb/s
        Duplex: Half
        Link detected: yes
$ ethtool -S eth0 | grep -iE 'crc|err|drop'
     rx_crc_errors: 18201
     rx_errors: 18244
     rx_dropped: 12
     tx_errors: 0
```

HALF DUPLEX

Gigabit should negotiate full duplex - Half means autoneg failed or one side was forced.

RX CRC CLIMBING

18k CRC errors received: corrupt frames arriving - bad cable, noise or that duplex mismatch.

TX CLEAN

Zero TX errors - our side sends fine; the fault is inbound: cable, far end or negotiation.

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Hardware Issues: PoE and Transceivers

1

PoE budget exhausted

Every switch has a total PoE wattage; add one camera too many and the newest device simply gets no power.

2

PoE standard mismatch

802.3af (15.4 W), at/PoE+ (30 W), bt/PoE++ (60-90 W) - a device asking for more than the port class delivers won't boot.

3

Transceiver mismatch

SFP wavelength, speed and SMF/MMF type must match at BOTH ends - and some switches reject non-vendor optics.

4

Transceiver signal strength

Read optical RX power via DDM/DOM or a power meter: too low means dirty or bent fibre, too high can saturate the receiver.

5

Check the power draw

show power inline (or vendor equivalent) lists per-port draw against budget - first stop for a dead PoE device.

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Wired Connection Scenarios

Symptom

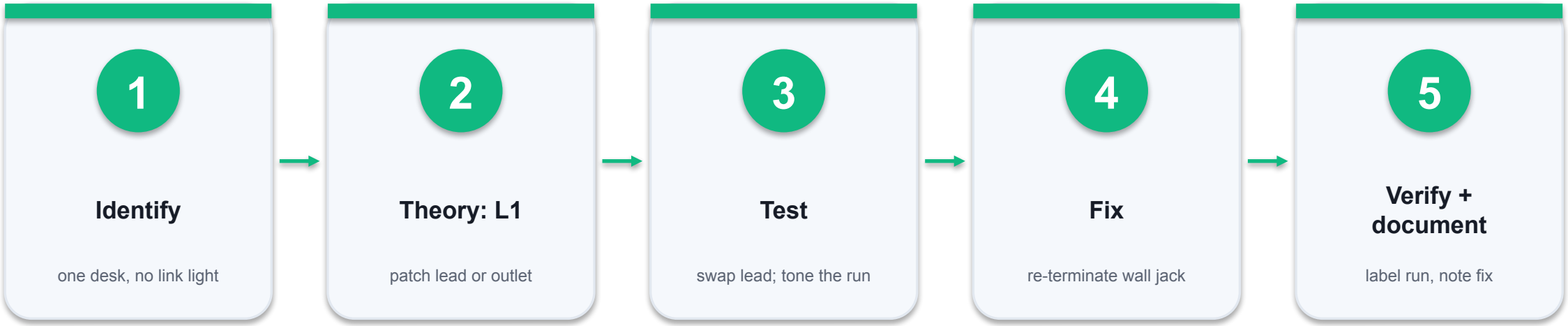
- No link light on NIC or switch
- Link up but slow / errors under load
- VoIP or video is choppy (jitter)
- Works until someone touches the desk
- Switch port LED never blinks

Likely cause and first action

- Cable, port or far end - swap one at a time
- Cable spec below network speed - check Cat rating
- Buffer/QoS or increase speed - measure first
- Re-terminate - loose plug or damaged latch
- Try another port; confirm it is not an uplink port

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Scenario: The Dead Desk Port



PATTERN

Bottom-to-top fits single-user wired faults: cheap L1 tests first, and the toner finds which patch-panel port serves the desk.

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DOMAIN 5.0 · HANDS-ON

Layer 1/2 Diagnostics with ip and ethtool

LAB 28

Diagnose cable and interface faults from link state, speed/duplex and error counters.

Use ip and ethtool on a Killercoda Ubuntu VM to read interface state, verify speed and duplex, hunt CRC and frame error counters, and bounce a NIC to recover the link.

YOU'LL BUILD

A verified eth0: link state decoded, speed/duplex confirmed and every error counter read and interpreted.

TOOLCHAIN

ethtool, iproute2, Killercoda Ubuntu playground

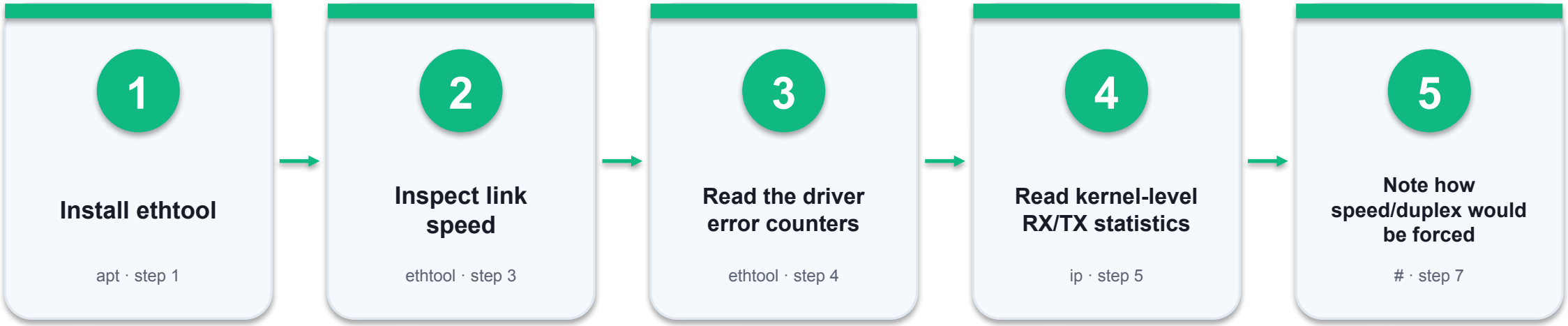
DONE WHEN

ip -brief link shows eth0 UP; ethtool reports Link detected: yes with matching speed/duplex, and you can explain what any non-zero error counter would mean.

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LAB 28 · PROCESS MAP

How Lab 28 Runs



THE POINT

Diagnose cable and interface faults from link state, speed/duplex and error counters.

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LAB 28 · VERIFY

Layer 1/2 Diagnostics with ip and ethtool

✅ Expected result

ip -brief link shows eth0 UP; ethtool reports Link detected: yes with matching speed/duplex, and you can explain what any non-zero error counter would mean.

IF IT DOESN'T WORK

ethtool: command not found

Run apt update && apt install -y ethtool first.

State shows LOWERLAYERDOWN

No carrier - in the real world check the cable/SFP; in the lab bring it up with ip link set eth0 up.

Counters show no errors

Normal on a clean virtual NIC - the skill is knowing where CRC and frame errors would appear.

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The First-Response Command-Line Kit

1

ping

Sends ICMP echoes to measure reachability, RTT and loss - the baseline for every path complaint.

2

traceroute / tracert

Linux traceroute = Windows tracert: reveals every L3 hop to the destination and the RTT to each.

3

pathping

Windows hybrid - traces the path, then pings each hop over time for per-hop loss statistics.

4

mtr

Live traceroute + ping combined; its Loss% column shows exactly where loss begins - best first tool for path issues.

5

nslookup / dig

Query DNS directly, choosing the server, the record type and even the full delegation trace.

6

Speed testers

Browser or CLI bandwidth tests verify the up/down throughput actually delivered to one machine.

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Reading a traceroute

Three RTT samples per hop - and the hops that lie.

```
$ traceroute 8.8.8.8
 1  10.0.0.1      0.4 ms  0.3 ms  0.3 ms
 2  100.64.12.1   2.1 ms  2.4 ms  2.0 ms
 3  * * *
 4  72.14.204.9   8.9 ms  9.2 ms  8.8 ms
 5  dns.google    9.0 ms  8.9 ms  9.1 ms
```

HOP = ROUTER

Each line is one L3 hop; the three columns are three RTT samples to that router.

*** * * AT HOP 3**

That router won't answer probes - filtered, not broken: hops 4 and 5 still respond.

WHERE LATENCY JUMPS

A jump persisting to the last hop marks the slow segment; on one hop only it is rate-limiting.

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Traceroute Gotchas

1

* * * is usually fine

Routers deprioritise or drop probe replies - only worry when every hop after it also fails.

2

ICMP is deprioritised

Routers answer probes on the slow CPU path while real transit is forwarded in hardware - hop RTTs exaggerate.

3

TCP traceroute

traceroute -T -p 443 uses TCP SYN probes that pass firewalls blocking ICMP or UDP probes.

4

Windows vs Linux probes

tracert uses ICMP echoes; Linux traceroute defaults to UDP - a firewall may pass one and block the other.

5

Paths change mid-trace

ECMP load balancing can split probes across paths - mtr gives a steadier per-hop view.

6

The return path is invisible

You see the outbound path only - the reply may come back a different way (asymmetric routing).

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Reading an mtr Report

mtr -rwc 20 sends 20 cycles and prints per-hop loss.

```
$ mtr -rwc 20 8.8.8.8
HOST                Loss%   Snt   Avg  Wrst
1. 10.0.0.1          0.0%    20   0.4   1.1
2. isp-core-rtr     60.0%    20   2.2   3.7
3. 72.14.212.1       0.0%    20   8.7  11.9
4. dns.google        0.0%    20   9.0  11.4
```

60% AT HOP 2

Looks alarming - but hops 3 and 4 show 0% loss, so transit THROUGH hop 2 is fine.

RATE-LIMITING

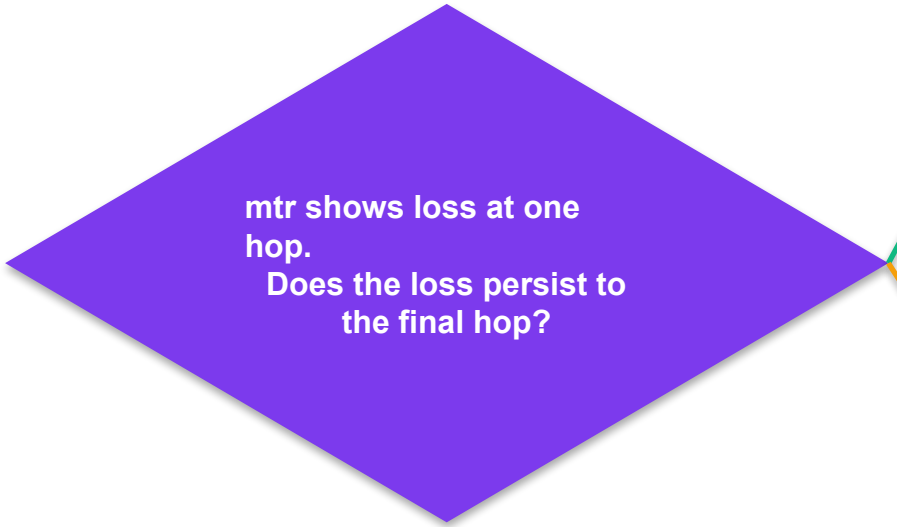
Hop 2 rate-limits its own ICMP replies. Real loss persists on every hop below it.

REAL LOSS

Loss beginning at hop N and continuing to the final hop - report that segment to its owner.

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Is That Packet Loss Real?



YES - real loss

The path drops packets from that hop onward - localise the segment and engage its owner (ISP or WAN provider) with the mtr report.

NO - cosmetic

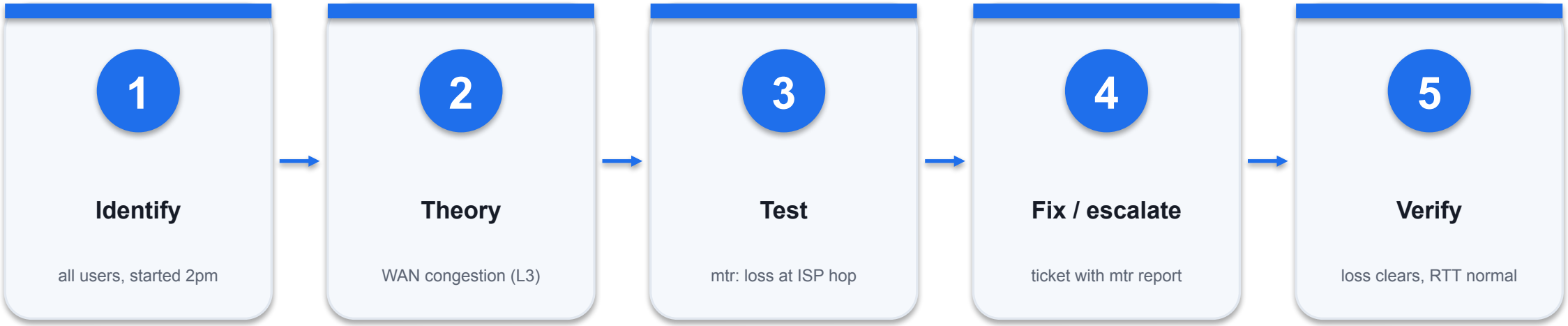
That single router is rate-limiting ICMP replies while forwarding fine - transit is unaffected; do not chase it.

Only loss that continues all the way down the report is service-affecting.

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OBJ 5.1 · METHOD APPLIED

Scenario: The Whole Office Is Slow



PATTERN

Whole-site symptoms point at shared links - measure first, because carriers act on an mtr report, not on 'it feels slow'.

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DOMAIN 5.0 · HANDS-ON

Reachability with ping, traceroute and mtr

LAB 29

Localise latency and packet loss along a network path with ping, traceroute and mtr.

Baseline RTT and loss with ping, expose every L3 hop with traceroute, combine both views in a 20-cycle mtr report, then bypass ICMP filters with a TCP traceroute on port 443.

YOU'LL BUILD

An mtr loss report for 8.8.8.8 and a TCP traceroute that passes firewalls which block ICMP.

TOOLCHAIN

iputils-ping, traceroute, mtr-tiny

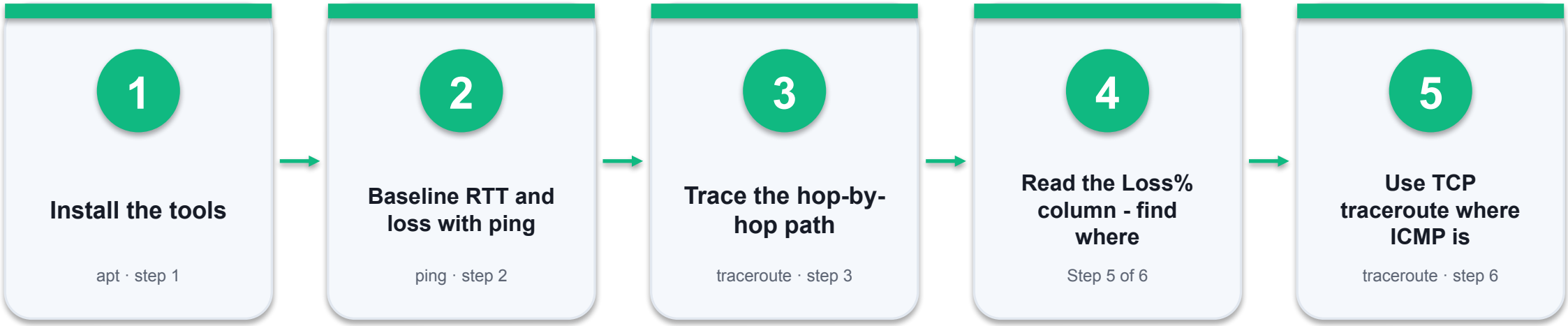
DONE WHEN

ping reports 0% loss, traceroute lists each hop with three RTT samples, and the mtr report shows 0.0% loss on the final hop.

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LAB 29 · PROCESS MAP

How Lab 29 Runs



THE POINT

Localise latency and packet loss along a network path with ping, traceroute and mtr.

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LAB 29 · VERIFY

Reachability with ping, traceroute and mtr

✅ Expected result

ping reports 0% loss, traceroute lists each hop with three RTT samples, and the mtr report shows 0.0% loss on the final hop.

IF IT DOESN'T WORK

traceroute prints * * *

That hop drops ICMP/UDP probes - usually filtered, not broken; try traceroute -T -p 443 instead.

mtr: command not found

Install the report-mode build with apt install -y mtr-tiny.

Loss shown on one middle hop only

ICMP rate-limiting on that router - only loss persisting to the final hop is real.

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DNS Troubleshooting Basics

1

Test with an IP, not a name

If the site loads by IP address while the name fails, connectivity is fine - the fault is DNS.

2

Find the configured resolver

Linux: cat /etc/resolv.conf or resolvectl status; Windows: ipconfig /all - know WHO you are asking.

3

Flush the cache

ipconfig /flushdns (Windows) or resolvectl flush-caches (Linux) clears poisoned or stale answers.

4

Query a server directly

dig @1.1.1.1 name or nslookup name 8.8.8.8 bypasses the local resolver to isolate which side is broken.

5

Compare two resolvers

Same name, two servers: one NXDOMAIN and one answer convicts the failing resolver, not the domain.

6

Check the record, not just the name

Wrong or stale A/AAAA/CNAME records resolve 'successfully' to the wrong place - read the answer.

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Decoding DNS Response Codes

Status	Meaning	Likely cause
NOERROR	Query answered	Healthy (still read the answer!)
NXDOMAIN	Name does not exist	Typo, expired or missing record
SERVFAIL	Server-side failure	DNSSEC failure, broken zone
REFUSED	Server will not answer	ACL - you are not its client
Timeout	No response at all	Wrong server IP, port 53 blocked

WHEN IT MATTERS dig prints the status in its header line: ->>HEADER<<- opcode: QUERY, status: NXDOMAIN

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Walking the Hierarchy with dig +trace

+trace resolves from the root down, showing each delegation.

```
$ dig +trace example.com
.             NS   a.root-servers.net.
com.          NS   a.gtld-servers.net.
example.com.  NS   a.iana-servers.net.
example.com.  A    96.7.128.175
```

ROOT HANDS OFF

The root servers do not know example.com - they delegate to the .com TLD servers.

TLD DELEGATES

The .com servers point at the domain's authoritative servers - the zone's real home.

WHERE IT BREAKS

The last successful delegation shows which level failed - registrar, zone or record.

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Client-Side vs Server-Side DNS Faults

Client side

- Wrong DNS server configured (DHCP or static)
- Stale cache - flush and retry
- Search suffix appending the wrong zone
- hosts file override hijacking the name
- VPN swapping the resolver silently

Server side

- Zone expired or misloaded (SERVFAIL)
- Missing or wrong A, AAAA or CNAME record
- DNSSEC validation failure upstream
- Forwarder or recursion broken
- ACL refusing your source (REFUSED)

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Scenario: 'The Site Is Down' - But Is It?



PATTERN

The IP-vs-name test splits the universe in one move: if the IP works, stop blaming the network and interrogate DNS.

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DOMAIN 5.0 · HANDS-ON

DNS Troubleshooting with dig and resolvectl

LAB 30

Isolate DNS faults by comparing resolvers, tracing delegation and reading response codes.

Identify which resolver the system really uses, compare answers from 1.1.1.1 and 8.8.8.8, walk the delegation chain with dig +trace, time a lookup and flush the systemd-resolved cache.

YOU'LL BUILD

A decoded dig +trace delegation chain plus a flushed resolver cache and the NXDOMAIN/SERVFAIL/REFUSED decode.

TOOLCHAIN

dnsutils (dig), systemd-resolved (resolvectl)

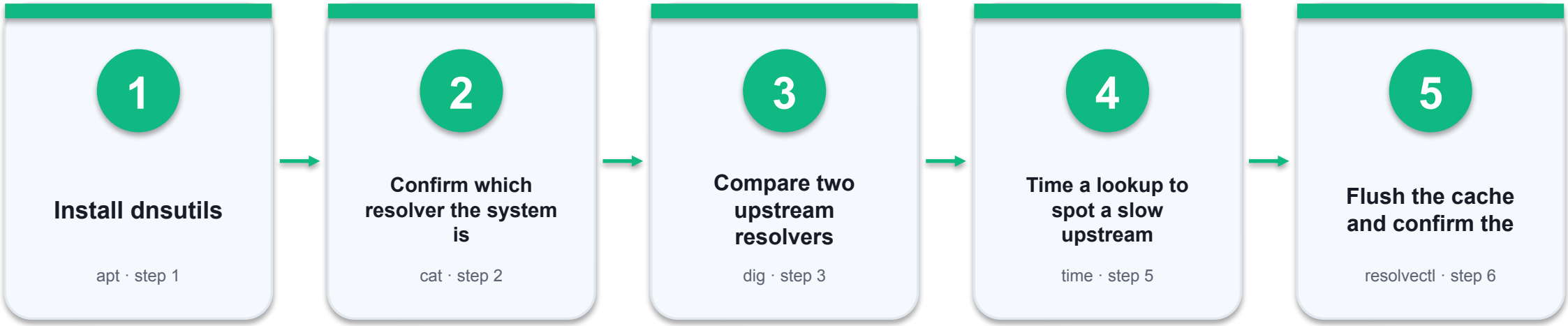
DONE WHEN

dig +short returns the same A record from both resolvers, +trace shows root -> .com -> authoritative delegation, and resolvectl statistics resets after the flush.

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LAB 30 · PROCESS MAP

How Lab 30 Runs



THE POINT

Isolate DNS faults by comparing resolvers, tracing delegation and reading response codes.

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LAB 30 · VERIFY

DNS Troubleshooting with dig and resolvectl

✅ Expected result

dig +short returns the same A record from both resolvers, +trace shows root -> .com -> authoritative delegation, and resolvectl statistics resets after the flush.

IF IT DOESN'T WORK

resolv.conf points to 127.0.0.53

Expected - systemd-resolved proxies DNS; use resolvectl status to see the real upstream.

SERVFAIL from one resolver

Upstream or DNSSEC failure - compare against a second resolver to confirm it is server-side.

dig +trace hangs

Some networks block outbound DNS to the root servers; rerun on the Killercoda VM.

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ARP and Neighbour Discovery

1

What ARP does

Maps an IPv4 address to a MAC so frames can be delivered on the local segment - no ARP entry, no delivery.

2

IPv6 uses NDP

IPv6 replaced ARP with ICMPv6 Neighbour Solicitation/Advertisement - same job, viewed with ip -6 neigh.

3

REACHABLE / STALE / FAILED

REACHABLE: recently confirmed. STALE: known, unconfirmed. FAILED: no answer - the neighbour is gone or blocked.

4

arping

Sends real ARP requests to one host - proves L2 adjacency even when ICMP ping is firewalled.

5

Gratuitous ARP

Unsolicited 'this IP is mine' broadcasts - normal after failover, but also the tool of ARP spoofers.

6

Flush to reset

ip neigh flush all (or arp -d) clears stale entries - the IP-layer equivalent of bouncing the NIC.

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Duplicate IPs and ARP Poisoning

1

The duplicate-IP symptom

Connectivity that flip-flops - works, breaks, works - as two hosts fight over one address in peers' ARP caches.

2

Where duplicates come from

Static IPs inside a DHCP scope, a cloned VM, or a second DHCP server handing out the same range.

3

Detect with DAD

arping -D probes your own address - 'address already in use' convicts a duplicate on the segment.

4

Find the culprit by MAC

The ARP reply carries the offender's MAC - trace it to a switch port via the MAC address table.

5

ARP poisoning

An attacker answers ARP for the gateway to sit in the middle - defend with dynamic ARP inspection (DAI).

6

Pin critical neighbours

A static ARP entry (ip neigh add ...) locks the gateway or server MAC where poisoning is a real risk.

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Switching Issues: Loops, VLANs, ACLs

1

STP loop symptoms

LED storm on every port, CPU spikes, MAC table churn, whole-segment meltdown - seconds after someone loops a cable.

2

Why loops melt networks

Ethernet frames have no TTL - a broadcast circles forever and multiplies at every switch: broadcast storm.

3

STP blocking surprises

A port stuck in blocking after a topology change - or a cheap unmanaged switch that silently runs no STP at all.

4

Incorrect VLAN assignment

Port in the wrong VLAN: link up, DHCP answers from the wrong subnet, gateway unreachable - check switchport config.

5

Trunk / native VLAN mismatch

Mismatched allowed-lists or native VLANs silently drop exactly one VLAN's traffic across a link.

6

ACLs eating traffic

'Everything works except X between two subnets' is an ACL or firewall rule until proven otherwise - check hit counters.

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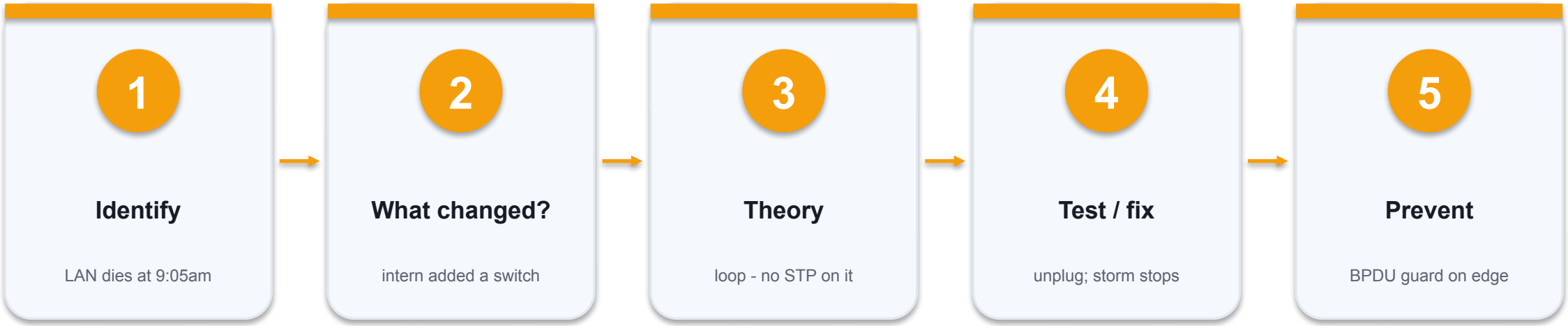
Layer 2 Faults: Symptom to Check

Fault	Classic symptom	First check
Switching loop	Storm: all LEDs solid, LAN dead	STP status, recent cabling
Blocked port	Uplink idle, path suboptimal	spanning-tree state per port
Wrong VLAN	Link up, DHCP from wrong subnet	switchport access vlan
Trunk mismatch	One VLAN dead across a link	allowed / native VLAN lists
Duplicate IP	Flip-flop reachability	arping -D, ARP tables
ACL block	One app/port pair fails, rest OK	ACL hit counters

WHEN IT MATTERS Link light on + wrong behaviour = think L2: VLANs, STP, ARP - before re-crimping anything.

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Scenario: New Switch, Network Meltdown



PATTERN

'What changed?' solved this in one question - and the preventive step (BPDU guard) is what makes step 6 more than a checkbox.

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DOMAIN 5.0 · HANDS-ON

ARP and the IPv6 Neighbour Table

LAB 31

Diagnose L2 adjacency problems with ip neigh and arping, including duplicate IP detection.

View and decode neighbour states, probe the default gateway with arping, run duplicate-address detection on your own IP, pin and flush ARP entries, then inspect IPv6 neighbour discovery.

YOU'LL BUILD

A decoded neighbour table with a static entry added and flushed, plus a duplicate-IP probe of your own address.

TOOLCHAIN

iputils-arping, iproute2, ndisc6

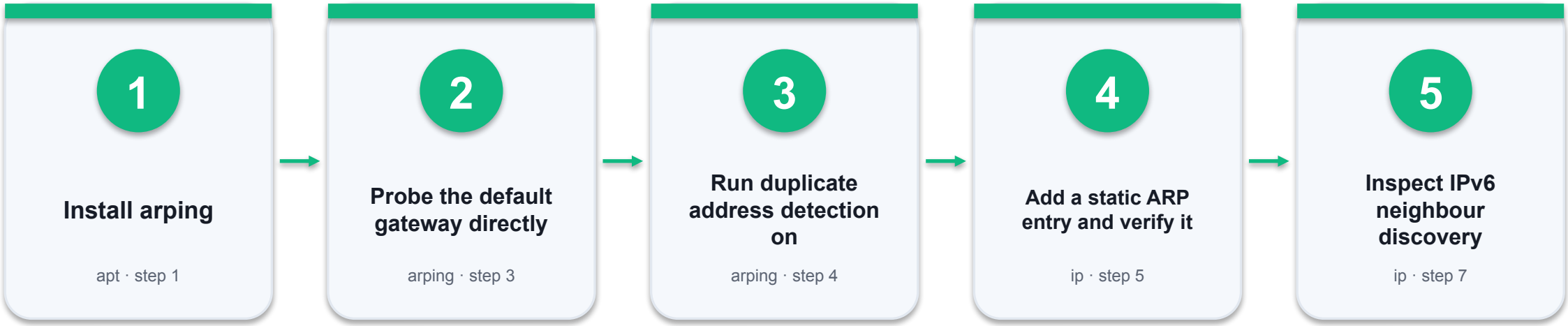
DONE WHEN

ip neigh shows the gateway REACHABLE after arping, DAD reports no duplicate for your IP, and the static 10.0.0.99 entry appears until the flush removes it.

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LAB 31 · PROCESS MAP

How Lab 31 Runs



THE POINT

Diagnose L2 adjacency problems with ip neigh and arping, including duplicate IP detection.

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LAB 31 · VERIFY

ARP and the IPv6 Neighbour Table

✅ Expected result

ip neigh shows the gateway REACHABLE after arping, DAD reports no duplicate for your IP, and the static 10.0.0.99 entry appears until the flush removes it.

IF IT DOESN'T WORK

arping: Operation not permitted

ARP probes need raw sockets - run as root (Killercoda terminals already are).

All entries show STALE

Normal - STALE means known but unconfirmed; traffic or arping promotes it to REACHABLE.

DAD says address already in use

Another host claims your IP - a real duplicate-IP incident; hunt it down by its MAC address.

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Protocol Analyzers

1

What they do

Capture frames off the wire and decode every layer - the ground truth when ping and traceroute run out of answers.

2

Wireshark

The GUI analyzer: live capture, colour rules, stream following and per-protocol statistics.

3

tcpdump

The CLI capture workhorse - runs on servers and routers where no GUI exists; writes .pcap files.

4

tshark

Wireshark's terminal twin - the same dissectors and display filters, with scriptable field output.

5

Capture once, analyse anywhere

Write to .pcap with tcpdump -w, then open the same file in Wireshark, tshark or a web analyzer.

6

Port mirroring / TAP

To see OTHER hosts' traffic you need a SPAN/mirror port or a hardware TAP - switches don't flood unicast to you.

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Capture Filter vs Display Filter

Capture filter (BPF)

- Applied WHILE capturing - packets never stored
- Syntax: 'udp port 53 and host 10.0.0.5'
- Keeps the .pcap small on busy links
- Cannot be undone - discarded is gone
- Same syntax in tcpdump and Wireshark

Display filter

- Applied AFTER capture - hides, never deletes
- Syntax: dns.flags.response==1
- Change it freely while analysing
- Wireshark GUI and tshark -Y
- The web analyzer's search box works this way

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Reading a tcpdump Line

One DNS query and its reply, decoded field by field.

```
$ tcpdump -nn -r /tmp/dns.pcap | head -4
12:04:01.123 IP 10.0.0.5.53622 > 1.1.1.1.53:
    51291+ A? example.com. (29)
12:04:01.131 IP 1.1.1.1.53 > 10.0.0.5.53622:
    51291 1/0/0 A 96.7.128.175 (45)
```

SRC > DST.PORT

10.0.0.5 from ephemeral port 53622 to 1.1.1.1 port 53 - client to resolver.

51291+ A?

Transaction ID 51291, recursion desired (+), question: the A record for example.com.

1/0/0 REPLY

Same ID back: 1 answer, 0 authority, 0 additional - the A record, 8 ms later.

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netstat and ss: Who Is Listening?

1

What they show

Every open port and connection on THIS host - the tool for 'is the service even listening?'.

2

ss replaces netstat

Modern Linux ships ss; the letters carry over: ss -tlnp = TCP listeners, numeric, with processes.

3

-n numeric

Skip DNS and service-name lookups - faster output and no misleading resolved names.

4

-a all / -l listening

-a shows every socket including established connections; -l restricts to listening sockets.

5

-p / -o / -b processes

Which executable and PID own each socket (netstat -b and -o on Windows, ss -p on Linux).

6

-r routing table

netstat -r prints the local routing table - the same data as ip route or route print.

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Reading ss -tlnp

Three listeners - and one classic bind mistake.

```
$ ss -tlnp
State   Local Address:Port    Process
LISTEN  0.0.0.0:22           sshd
LISTEN  127.0.0.1:3306       mysqld
LISTEN  [::]:443             nginx
```

0.0.0.0:22

sshd listens on every IPv4 address - reachable from the network, firewall permitting.

127.0.0.1:3306

MySQL bound to loopback ONLY - remote clients get 'connection refused'; that is the bug.

[::]:443

nginx on the IPv6 wildcard - on dual-stack hosts this usually accepts IPv4 too.

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The Capture Workflow + TIS PCAP Analyzer

1

1 · Filter tight, capture small

Capture with a BPF filter and -c count so the .pcap holds only the conversation you care about.

2

2 · Write to .pcap

tcpdump -w capture.pcap keeps the binary packets - analysis can happen later, elsewhere, repeatedly.

3

3 · First pass with tshark

Pull just the fields you need: -T fields -e dns.qry.name turns packets into a readable table.

4

4 · Go graphical in the browser

The TIS PCAP Analyzer (alfredang.github.io/pcapanalyzer) opens .pcap files in-browser - no install needed.

5

5 · Drill into the layers

Expand any packet to walk Ethernet -> IP -> UDP -> DNS - the OSI model made visible.

6

6 · Search like a display filter

The analyzer's filter box narrows by name or protocol - the same idea as Wireshark's -Y.

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DOMAIN 5.0 · HANDS-ON

Packet Capture with tcpdump and TIS PCAP Analyzer

LAB 32

Capture DNS traffic to a .pcap and analyse it with tshark and the TIS PCAP Analyzer.

Capture ten DNS packets with a BPF capture filter, dissect them with tshark display filters, then move the .pcap to your laptop and explore it graphically in the browser-based TIS PCAP Analyzer.

YOU'LL BUILD

A dns.pcap captured with tcpdump and read three ways: tcpdump summary, tshark fields and the TIS PCAP Analyzer UI.

TOOLCHAIN

tcpdump, tshark, TIS PCAP Analyzer (alfredang.github.io/pcapanalyzer)

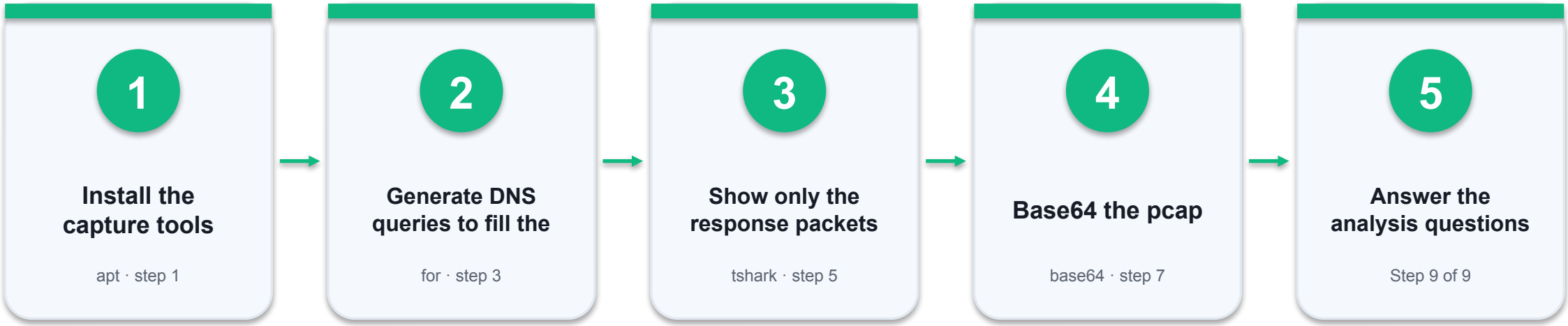
DONE WHEN

The TIS PCAP Analyzer shows your three queries with matching responses: each query leaves a high ephemeral source port for destination port 53, and query/response transaction IDs match.

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LAB 32 · PROCESS MAP

How Lab 32 Runs



THE POINT

Capture DNS traffic to a .pcap and analyse it with tshark and the TIS PCAP Analyzer.

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LAB 32 · VERIFY

Packet Capture with tcpdump and TIS PCAP Analyzer

✅ Expected result

The TIS PCAP Analyzer shows your three queries with matching responses: each query leaves a high ephemeral source port for destination port 53, and query/response transaction IDs match.

IF IT DOESN'T WORK

Capture never reaches 10 packets

Your queries may not match the BPF filter - re-run the dig loop or loosen the filter.

tshark asks about non-root capture

Answer No for the lab - you are reading files, not capturing as a normal user.

Analyzer shows no packets

Re-decode the base64 on your laptop (base64 -d dns.pcap.b64 > dns.pcap) and upload again.

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MTU: When Big Packets Vanish

1

MTU = biggest frame payload

Standard Ethernet carries 1500 bytes of IP packet; tunnels (VPN, GRE, PPPoE) reduce the usable size.

2

The header maths

Payload = MTU - 20 (IP) - 8 (ICMP/UDP) or - 20 (TCP): hence the magic ping size 1472 at MTU 1500.

3

DF: Don't Fragment

With DF set, an oversize packet is dropped and the router returns ICMP Type 3 Code 4: Fragmentation Needed.

4

Path MTU Discovery

Senders probe with DF packets and shrink until they fit - PMTUD depends entirely on those ICMP replies arriving.

5

Jumbo frames

9000-byte MTUs help storage and backup LANs - but EVERY device on the path must agree, or you manufacture giants.

6

Mismatch symptoms

Small packets sail through; anything full-size stalls - a signature no other fault quite copies.

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Proving the MTU with DF-Bit Pings

Lab 33's probe: -M do forbids fragmentation.

```
$ ping -M do -s 1472 -c2 8.8.8.8
1480 bytes from 8.8.8.8: icmp_seq=1 ttl=118
1480 bytes from 8.8.8.8: icmp_seq=2 ttl=118
$ ping -M do -s 1500 -c2 8.8.8.8
ping: local error: message too long, mtu=1500
```

1472 PASSES

1472 payload + 8 ICMP + 20 IP = exactly 1500 - the largest ping that fits standard Ethernet.

1500 REFUSED

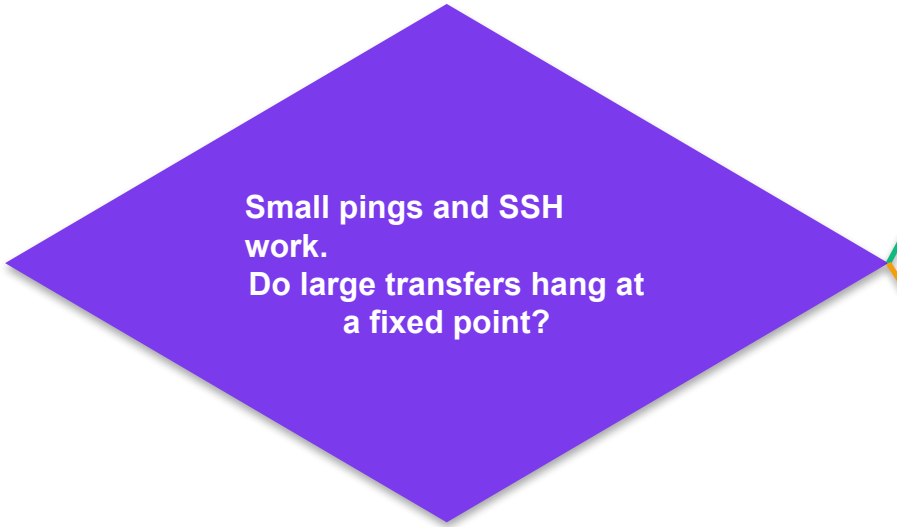
With DF set the kernel won't fragment - 1500 + 28 exceeds the MTU, so the send fails locally.

ON A REAL PATH

A mid-path bottleneck would return ICMP Type 3 Code 4 - unless a firewall eats it.

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The PMTUD Black Hole



YES - suspect MTU

A firewall is eating the Fragmentation Needed ICMP, so senders never shrink. Test with DF pings; clamp TCP MSS on the router.

NO - not MTU

A size-independent fault is congestion, loss or the application - go measure with ping and mtr instead.

MSS clamping: iptables -t mangle ... --clamp-mss-to-pmtu rescues TCP through the hole.

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Living With Broken PMTUD

1

The classic report

'Web page starts loading then hangs' or 'SSH works, SCP stalls' - the first full-size packet is the one that dies.

2

Why ICMP gets blocked

Overzealous firewall policies drop ALL ICMP - killing the Type 3 Code 4 messages PMTUD depends on.

3

MSS clamping

The router rewrites TCP's advertised MSS during the handshake so segments never exceed the path MTU.

4

Lower the tunnel MTU

Setting VPN and tunnel interfaces to a matching MTU (e.g. 1400) prevents oversize packets at the source.

5

Allow the right ICMP

Permit ICMP Type 3 Code 4 - and IPv6 Packet Too Big - through firewalls; PMTUD is mandatory in IPv6.

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DOMAIN 5.0 · HANDS-ON

MTU and Path-MTU Discovery

LAB 33

Demonstrate MTU limits with DF-bit pings and explain PMTUD black holes and MSS clamping.

Compute the ICMP payload maths (MTU minus 28), send Don't-Fragment pings at and over the limit, lower the interface MTU to 1400 and retest both sides of the new limit, then restore it.

YOU'LL BUILD

A demonstrated 1472-byte pass / 1500-byte 'message too long' fail, repeated at MTU 1400, and a restored interface.

TOOLCHAIN

iputils-ping, iproute2

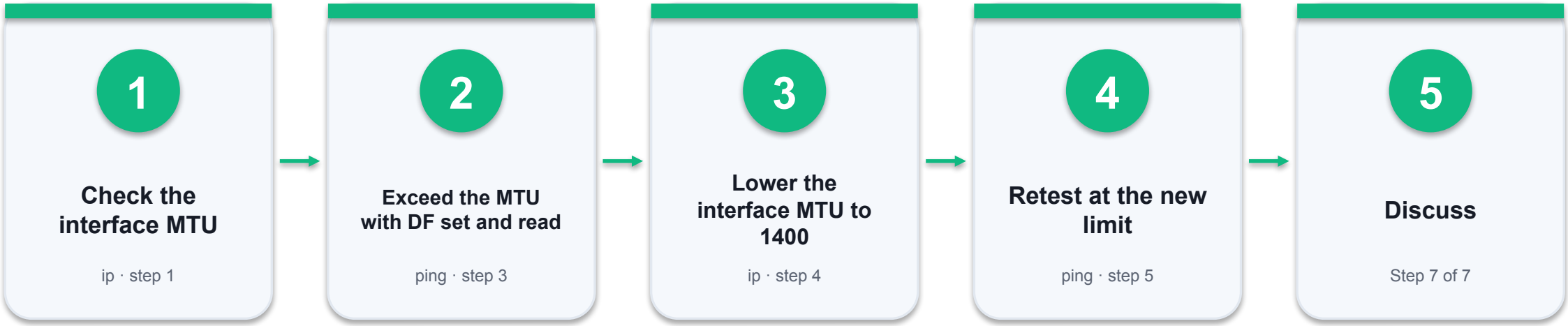
DONE WHEN

1472-byte DF pings succeed at MTU 1500 and 1372-byte pings at MTU 1400, while oversize pings fail with 'message too long' showing the current mtu value.

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LAB 33 · PROCESS MAP

How Lab 33 Runs



THE POINT

Demonstrate MTU limits with DF-bit pings and explain PMTUD black holes and MSS clamping.

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LAB 33 · VERIFY

MTU and Path-MTU Discovery

✅ Expected result

1472-byte DF pings succeed at MTU 1500 and 1372-byte pings at MTU 1400, while oversize pings fail with 'message too long' showing the current mtu value.

IF IT DOESN'T WORK

Oversize ping shows no error

Without -M do the kernel silently fragments - the DF bit is what makes the test honest.

1472 fails too

A tunnel/VPN lowers the effective path MTU - step the size down (1452, 1420...) until it passes.

Forgot to restore the MTU

ip link set eth0 mtu 1500 - a lowered MTU otherwise slows every later lab.

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Routing Table Issues

1

Missing route

No entry covers the destination and no default exists - traffic dies with 'network unreachable'.

2

Wrong or missing default route

The gateway of last resort is wrong or absent: local subnets fine, everything beyond them dead.

3

Wrong next hop

The route exists but points at the wrong router - packets leave, replies never come, or redirects fly.

4

Longest prefix wins

10.99.0.0/24 beats 10.0.0.0/8 regardless of metric - the most specific route always forwards the packet.

5

Metric breaks ties

Between equal prefixes the lower metric wins - how floating static and backup routes stay dormant.

6

ip route get

Ask the kernel directly which route, interface and source IP a destination would use - no guessing.

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Interrogating the Routing Table

Lab 34's key move: make the kernel show its choice.

```
$ ip route
default via 172.30.1.1 dev eth0
10.99.0.0/24 via 172.30.1.1 dev eth0 metric 100
172.30.1.0/24 dev eth0 src 172.30.1.2
$ ip route get 10.99.0.5
10.99.0.5 via 172.30.1.1 dev eth0 src 172.30.1.2
```

THREE ROUTE TYPES

A default, a static /24, and a connected route the kernel added for eth0's own subnet.

ROUTE GET = PROOF

The kernel names the winning route, egress interface and source IP for that destination.

LONGEST PREFIX

10.99.0.5 matches both the /24 and the default - the /24 wins because it is more specific.

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Asymmetric Routing

1

What it is

Traffic leaves by one path and returns by another - legal in IP, but stateful middleboxes hate it.

2

Why firewalls break

A stateful firewall that never saw the outbound SYN drops the returning SYN-ACK as unsolicited.

3

How it happens

Two exits (dual WAN, two routers, ECMP) with inconsistent metrics - or NAT on only one of the paths.

4

How to spot it

traceroute from BOTH ends and compare; mtr -z shows the AS path, where asymmetry jumps out.

5

The usual fixes

Pin flows to one path with policy routing, sync firewall state, or make the metrics consistent.

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Addressing Faults: Symptom to Test

Fault	Classic symptom	Fastest test
Wrong default gateway	LAN fine, nothing beyond it	ping the gw; ip route / ipconfig
Wrong subnet mask	SOME hosts unreachable	compare mask with a peer
DHCP pool exhausted	New hosts get 169.254.x.x	scope stats; lease table
Duplicate IP	Flip-flop connectivity	arping -D your own address
Static/DHCP overlap	Random conflicts appear	audit static assignments

WHEN IT MATTERS APIPA 169.254.x.x means 'I asked DHCP and nobody answered' - the client self-assigned.

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DHCP Pool Exhaustion

1

The symptom

New and returning devices fail to join; victims show APIPA 169.254.x.x self-assigned addresses.

2

Why pools run dry

A scope sized for desks meets BYOD phones and IoT - every device holds a lease whether active or not.

3

Long leases make it worse

8-day leases on guest Wi-Fi hold addresses for one-time visitors a week after they left.

4

Check scope statistics

The DHCP server shows leases used vs available - exhaustion is visible at a glance.

5

The fixes

Shorten lease times, widen the scope, add VLANs with their own scopes, evict statics from the pool.

6

Rogue DHCP servers

A second, often accidental, DHCP server hands out wrong addresses - DHCP snooping on switches blocks it.

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Gateway and Mask Misconfigurations

1

Wrong gateway signature

Everything on the local subnet works; every remote destination fails - off-net traffic has nowhere to go.

2

Test the gateway itself

ping the configured gateway IP: if that fails, check ARP for it; if it works, suspect what lies beyond.

3

Wrong mask signature

The strangest fault: SOME remote hosts work, some local ones don't - the boundary is drawn in the wrong place.

4

Why the mask does that

A too-wide mask claims remote hosts as 'local' (ARPed, never routed); too narrow sends local peers via the gateway.

5

Compare with a healthy peer

ip addr against a working neighbour finds mask and gateway typos in seconds - baselines beat theory.

6

Scope tells you where to look

DHCP misconfig hits everyone on the scope; a static typo hits one machine.

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Scenario: Ping This, But Not That



PATTERN

Partial reachability is routing until proven otherwise - and ip route get turns 'I think' into 'the kernel says'.

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DOMAIN 5.0 · HANDS-ON

Diagnosing Routing Problems

LAB 34

Read and manipulate the routing table; prove which route a destination uses with ip route get.

Print the routing table, query the kernel's route choice for two destinations, add a static route with a metric, inspect precedence and longest-prefix match, then remove the test route.

YOU'LL BUILD

A routing table with a temporary 10.99.0.0/24 static route, proven with ip route get and cleanly removed.

TOOLCHAIN

iproute2, traceroute

DONE WHEN

ip route get 10.99.0.5 switches from the default route to the new static route while it exists, and reverts to the default after ip route del.

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LAB 34 · PROCESS MAP

How Lab 34 Runs



THE POINT

Read and manipulate the routing table; prove which route a destination uses with ip route get.

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LAB 34 · VERIFY

Diagnosing Routing Problems

✅ Expected result

ip route get 10.99.0.5 switches from the default route to the new static route while it exists, and reverts to the default after ip route del.

IF IT DOESN'T WORK

RTNETLINK: file exists

The route is already present - delete it first with ip route del 10.99.0.0/24.

Route added but not used

A longer prefix elsewhere wins - longest-prefix match beats metric every time.

No default route at all

Re-add it: ip route add default via <gateway IP> dev eth0.

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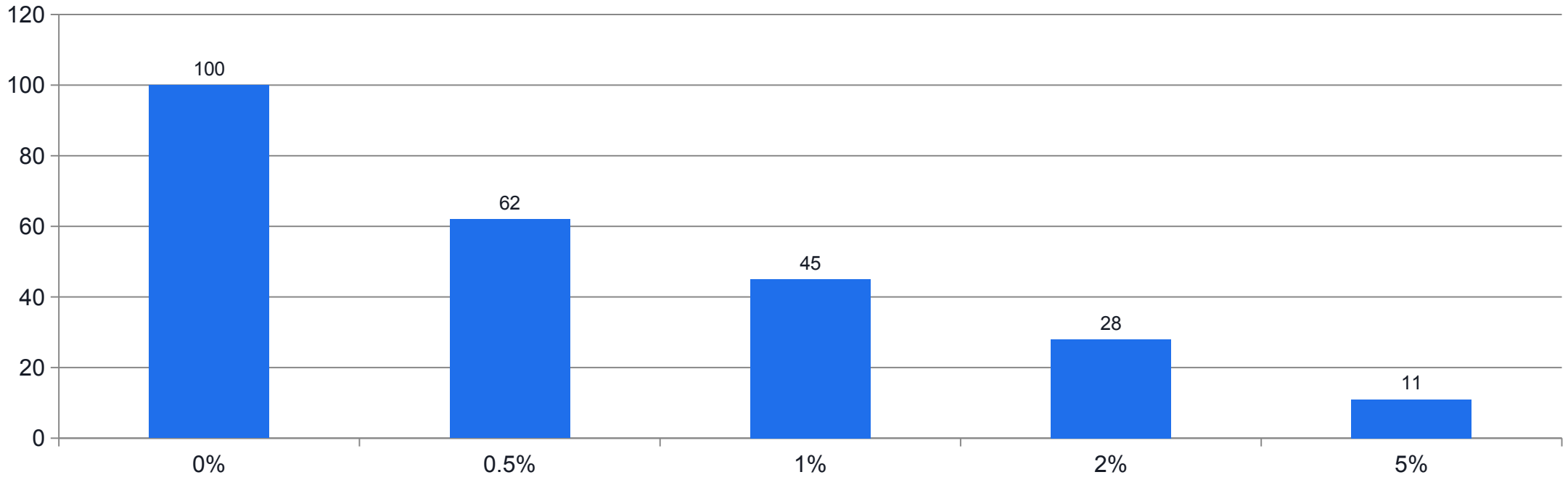
Latency vs Loss vs Jitter

Metric	What it is	Hurts most	Measure with
Latency	Round-trip delay	Interactive apps, gaming	ping RTT, mtr Avg
Packet loss	Frames never arrive	TCP throughput collapse	ping / mtr Loss%
Jitter	Variation in delay	VoIP and live video	mtr Wrst-Avg, iperf3 -u
Bandwidth	Path capacity	Bulk transfer, backup	iperf3, speed testers

WHEN IT MATTERS VoIP wants jitter under ~30 ms and loss under 1% - buffers trade jitter for latency.

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A Little Loss Destroys a Lot of Throughput



WHAT THE DATA SHOWS

TCP treats every loss as congestion and backs off - 1% loss roughly halves throughput long before a link is 'full'. This is why Lab 35's netem loss test feels so dramatic.

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Congestion and Bottlenecks

1

Oversubscribed uplinks

48 access ports funnelling into one 1G uplink - fine on average, saturated at 9 a.m. sharp.

2

The queueing signature

Latency climbs FIRST as buffers fill; loss follows when they overflow - watch the RTT trend before loss.

3

Duplex mismatch again

Still the classic false 'congestion': collisions and CRC errors under load on one specific link.

4

Find the top talkers

Interface counters, NetFlow/sFlow or a capture answer WHO is eating the link before you upgrade it.

5

Broadcast storms

A loop or a chatty protocol can consume a LAN with traffic nobody asked for - see the L2 slides.

6

QoS as triage

Prioritise voice and video over bulk data - it does not add capacity, it decides who suffers.

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Wireless Interference and Channel Overlap

1

2.4 GHz: three clean channels

Only 1, 6 and 11 avoid overlap - neighbouring APs on 3 and 8 interfere with BOTH sides.

2

Co-channel vs adjacent

Same-channel APs politely take turns (slow); OVERLAPPING channels corrupt each other (worse).

3

Non-Wi-Fi offenders

Microwave ovens, wireless mice and keyboards, Bluetooth, baby monitors - all camp on 2.4 GHz.

4

Escape to 5 / 6 GHz

Far more non-overlapping channels and less junk - the standing fix for a crowded 2.4 GHz office.

5

Survey before blaming

A Wi-Fi analyzer shows every SSID, channel and RSSI around you - pick the channel the survey supports.

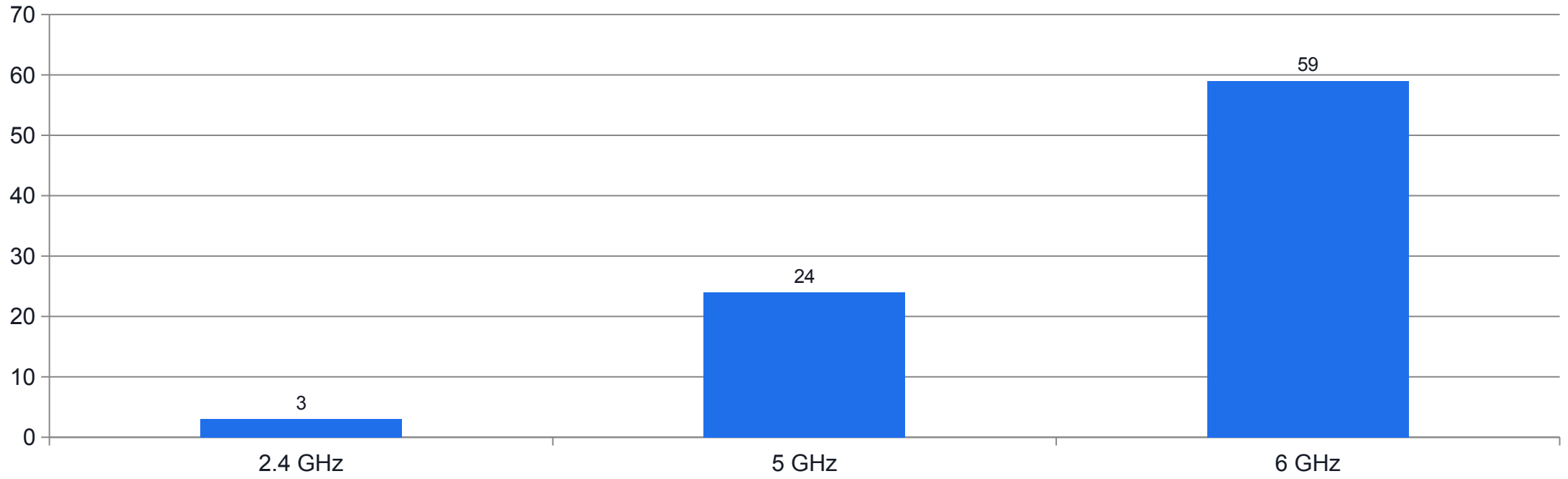
6

Interference symptoms

Connected with strong signal yet slow and jittery - retries eat airtime before anything is 'down'.

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Room to Breathe: Channels by Band



WHAT THE DATA SHOWS

2.4 GHz offers exactly three clean channels, shared with microwaves and Bluetooth; 5 and 6 GHz offer dozens - moving bands beats fighting for channel 6.

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More Wireless Scenarios

1

Over capacity

Too many clients per AP: strong signal, terrible throughput - add APs or steer clients to 5/6 GHz.

2

Incorrect antenna type

Omni antennas in a long corridor, or patch antennas facing the wrong way, waste the cell where users are.

3

Antenna placement

Placement is an art: test and retest - central, high, away from metal and water; re-survey after moving.

4

Cable gain loss

Every metre of antenna coax costs dB - keep antenna cables short or the gain never reaches the air.

5

Old standards drag the cell

Legacy 802.11b/g clients force protection modes that slow everyone - upgrade or isolate them.

6

Roaming problems

Sticky clients cling to a distant AP - tune min-RSSI and roaming thresholds so handoff happens.

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Measuring Performance

1

iperf3

Client/server throughput tester - proves what the path can actually carry, TCP or UDP with loss and jitter stats.

2

Speed testers

Quick browser or CLI checks of real up/down rates to the internet - verify the circuit, not the marketing.

3

Wi-Fi analyzers

Show channels, signal strength and neighbouring networks - the survey tool behind every interference fix.

4

mtr over time

A long-running mtr catches the intermittent loss and jitter that a single ping snapshot misses.

5

tc netem (lab)

Injects delay, jitter and loss on purpose - reproduce the customer's WAN on one VM and test the app against it.

6

Baseline everything

A measurement means little without the 'normal' to compare against - collect baselines before trouble starts.

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Simulating Slow and Lossy Links with tc

LAB 35

Reproduce WAN latency, jitter and loss with tc netem and cap bandwidth with tbf.

Baseline RTT, then use tc netem to inject 200 ms delay, +/-50 ms jitter and 5% loss, verifying each with ping; cap throughput at 1 Mbit/s with tbf, prove it with iperf3, then remove all shaping.

YOU'LL BUILD

A NIC that demonstrably ran at +200 ms delay, 5% loss and a 1 Mbit/s cap - and a clean qdisc afterwards.

TOOLCHAIN

iproute2 (tc/netem/tbf), iperf3

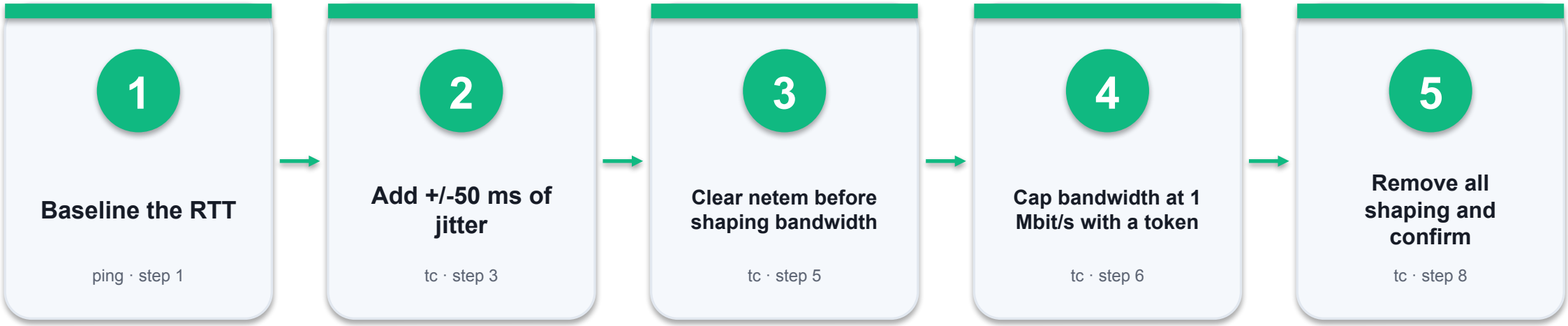
DONE WHEN

ping RTT rises by about 200 ms under netem, roughly 1 in 20 pings times out at 5% loss, and iperf3 reports about 1 Mbit/s under the tbf cap.

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LAB 35 · PROCESS MAP

How Lab 35 Runs



THE POINT

Reproduce WAN latency, jitter and loss with tc netem and cap bandwidth with tbf.

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LAB 35 · VERIFY

Simulating Slow and Lossy Links with tc

✅ Expected result

ping RTT rises by about 200 ms under netem, roughly 1 in 20 pings times out at 5% loss, and iperf3 reports about 1 Mbit/s under the tbf cap.

IF IT DOESN'T WORK

RTNETLINK answers: file exists

A qdisc is already installed - use tc qdisc change, or delete the root qdisc first.

RTT did not change

You shaped the wrong interface - confirm the active NIC with ip -brief link.

iperf3 connection refused

Start the server first (iperf3 -s -D), then run the client against 127.0.0.1.

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WAN Troubleshooting Scenarios

1

Start at your own edge

Most WAN faults are customer-side - cables, the router, config - not the ISP; check yours before calling.

2

Walk to the demarc

LAN interface -> router/modem LAN side -> WAN side -> provider: find the last healthy point.

3

Modem/ONT lights first

Sync and LOS LEDs on the modem or ONT tell you in seconds whether the carrier signal is even present.

4

Test around DNS

ping 8.8.8.8 vs ping google.com separates 'the WAN is down' from 'DNS is down' immediately.

5

Interference and environment

On DSL, coax and wireless WANs, interference is usually consumer-side - unless weather or disaster says otherwise.

6

Evidence for the carrier

mtr reports, timestamps, loss figures - carriers move faster when the ticket contains measurements.

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Layered Service Triage

1. nc -zv host 443 - does TCP connect?
2. openssl s_client - does TLS validate?
3. curl -v - what HTTP status returns?
4. curl -w timings - which phase is slow?
5. Escalate with the failing layer named

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Breaking One Request Into Phases

curl -w prints a stopwatch for every stage of the request.

```
$ curl -so /dev/null https://example.com -w \
  'DNS:%{time_namelookup} TCP:%{time_connect} \
   TLS:%{time_appconnect} TTFB:%{time_starttransfer}'
DNS:2.104 TCP:2.121 TLS:2.189 TTFB:2.239
```

DNS: 2.1 SECONDS

Nearly the whole wait happened before any packet reached the server - the resolver is to blame.

GAPS, NOT VALUES

Read the DIFFERENCES: TCP-DNS = handshake, TLS-TCP = crypto, TTFB-TLS = server think time.

WHO TO CALL

Slow DNS: resolver team (Lab 30). Slow TTFB: app team. Slow TCP: network path (Lab 29).

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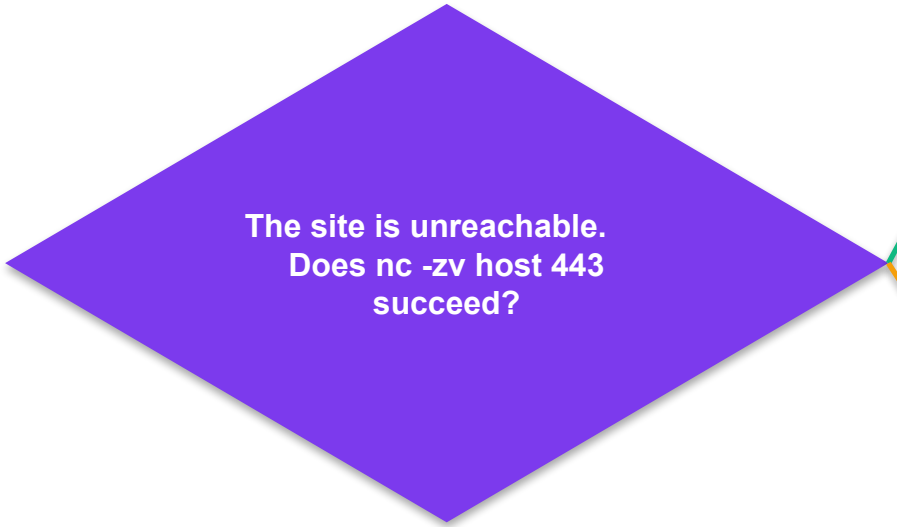
HTTP Status Codes for Troubleshooters

Code	Meaning	Prime suspect
200	OK	Nothing - move on
301 / 302	Redirect	Follow it - check the target
401 / 403	Auth required / forbidden	Credentials, ACLs, permissions
404	Not found	URL path, deployment
500	Application error	The app itself - read its logs
502/503/504	Bad gateway/overload/timeout	Upstream server or proxy

WHEN IT MATTERS Any HTTP answer at all proves DNS, routing, TCP and TLS all worked - you are at Layer 7.

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The nc -zv Fork



YES - go up the stack

Routing, firewall and listener are all fine. Test TLS (openssl s_client), then HTTP (curl -v) - the fault is above L4.

NO - stay low

No TCP handshake. Check DNS resolution, then the path (mtr), then firewalls, then whether anything listens (ss -tlnp).

One command splits the whole stack in half - divide and conquer as a habit.

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Symptom Sprint: Name the Likely Cause

1

Small pings OK, downloads stall

MTU / PMTUD black hole - run the DF-bit ping test, then clamp MSS (Lab 33).

2

169.254.x.x on new laptops

DHCP exhausted or unreachable - check scope statistics and the lease table (Lab 34).

3

Works by IP, not by name

DNS - compare resolvers with dig, flush the cache (Lab 30).

4

Every LED flashing, LAN dead

Switching loop / broadcast storm - find the new cable, enable BPDU guard (Lab 31).

5

One user flip-flops on/off

Duplicate IP - arping -D, then hunt the MAC (Lab 31).

6

Local fine, remote all dead

Default gateway wrong or down - ping the gateway, check ip route (Lab 34).

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OBJ 5.1 · METHOD APPLIED

Capstone: 'The Website Is Down'



WHY IT MATTERS

Lab 36 runs exactly this triage - each tool eliminates a layer, so the escalation names the failing layer instead of guessing.

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DOMAIN 5.0 · HANDS-ON

Service Reachability Triage with curl, nc and openssl

LAB 36

Isolate a 'site is down' report to TCP, TLS, HTTP or the app using nc, openssl and curl.

Walk the layered triage: prove the TCP handshake with nc -zv, validate the certificate chain with openssl s_client, read HTTP status codes with curl -v, then break one request into DNS/TCP/TLS/TTFB timings.

YOU'LL BUILD

A complete layer-by-layer triage of example.com:443 with a timing breakdown that names the slow phase.

TOOLCHAIN

curl, netcat-openbsd, openssl

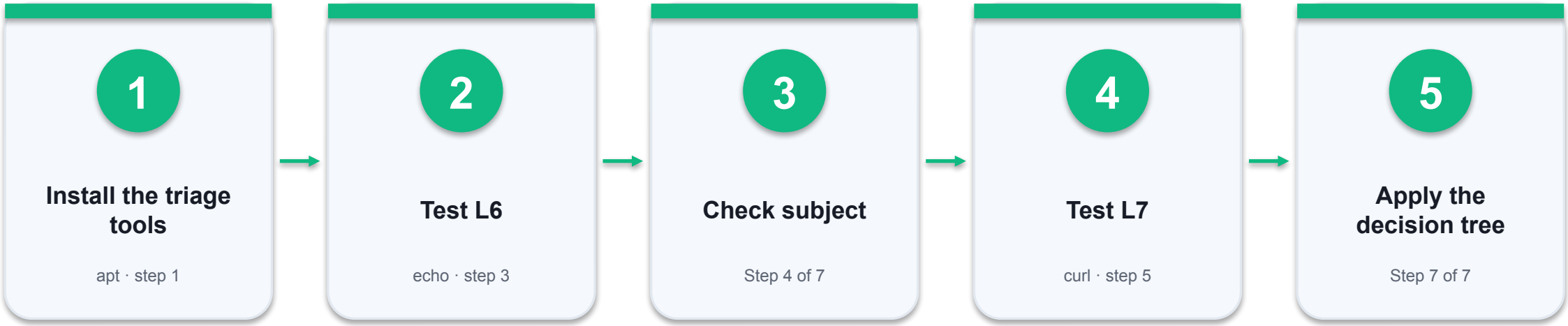
DONE WHEN

nc reports succeeded, openssl shows Verify return code: 0 (ok), curl returns HTTP 200, and the -w timings identify the slowest phase of the request.

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LAB 36 · PROCESS MAP

How Lab 36 Runs



THE POINT

Isolate a 'site is down' report to TCP, TLS, HTTP or the app using nc, openssl and curl.

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LAB 36 · VERIFY

Service Reachability Triage with curl, nc and openssl

✅ Expected result

nc reports succeeded, openssl shows Verify return code: 0 (ok), curl returns HTTP 200, and the -w timings identify the slowest phase of the request.

IF IT DOESN'T WORK

nc hangs then times out

No TCP path - fall back to Lab 29 tools: check DNS, then mtr the host, then firewalls.

Verify return code is non-zero

Expired or untrusted certificate, or SNI mismatch - re-run with the right -servername.

HTTP 502/503/504

TCP and TLS are fine - the fault is the app or its upstream; escalate with your evidence.

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The Troubleshooter's CLI, One Page

Tool	Layer	The question it answers
ping / mtr	L3	Is it reachable? Where is the loss?
traceroute	L3	Which hops does the path take?
ip / arp / ip neigh	L2-L3	Addressing, routes, MAC neighbours
dig / nslookup	L7	What does DNS actually answer?
ss / netstat / nmap	L4	What is listening or open?
tcpdump / Wireshark	L1-L7	What is REALLY on the wire?

WHEN IT MATTERS nmap scans REMOTE ports; ss lists LOCAL sockets - know which side each tool sees.

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The Hardware Kit, One Page

1

Cable tester

First tool on any drop: wiremap and continuity catch opens, shorts and miswires before anything else runs.

2

Toner and probe

The Fox & Hound pair traces which patch-panel port or wall run a cable serves - no more label archaeology.

3

TDR / OTDR

Reports the distance to a break or bad splice on copper (TDR) or fibre (OTDR) - open the right ceiling tile.

4

Loopback plug

Loops TX to RX so a NIC or switch port proves itself with no far end involved.

5

Optical power meter

Reads received light in dBm against the link budget - the number that convicts dirty or bent fibre.

6

Wi-Fi analyzer

Surveys SSIDs, channels and signal strength so channel and AP-placement decisions rest on data.

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DOMAIN 5 · 24% OF THE EXAM

Domain 5 Exam Technique

1

Know the 7 steps by NUMBER

Questions quote a scenario and ask 'what is the NEXT step?' - the order itself is tested.

2

Read the symptom's layer

Every scenario hides a layer clue: link light (L1), VLAN (L2), gateway (L3), port (L4), URL (L7).

3

'What changed?' wins often

When a scenario mentions any recent change, the answer almost always flows from it.

4

Eliminate by scope

One user vs many, one app vs all - scope eliminates half the answer options immediately.

5

Match tool to question

Tool-selection questions abound: know what each CLI and hardware tool uniquely proves.

6

Document is a real answer

After 'verify functionality', 'document findings' IS the correct next action - don't skip it for sounding dull.

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DOMAIN 5 · WRAP-UP

Measure. Don't guess.

Troubleshooting is the discipline of shrinking the possibility space with every test - the seven steps make it repeatable, the tools make it fast, the documentation makes it stick.

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Recap — Network Troubleshooting

- You can now: Diagnose cable and interface faults from link state, speed/duplex and error counters.
- You can now: Localise latency and packet loss along a network path with ping, traceroute and mtr.
- You can now: Isolate DNS faults by comparing resolvers, tracing delegation and reading response codes.
- You can now: Diagnose L2 adjacency problems with ip neigh and arping, including duplicate IP detection.
- You can now: Capture DNS traffic to a .pcap and analyse it with tshark and the TIS PCAP Analyzer.
- You can now: Demonstrate MTU limits with DF-bit pings and explain PMTUD black holes and MSS clamping.
- You can now: Read and manipulate the routing table; prove which route a destination uses with ip route get.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Explained networking concepts

OSI layers, appliances, cloud models, ports & protocols, media, topologies and IPv4/IPv6 subnetting.

2

LO2 · Implemented networks

Static & dynamic routing, NAT/PAT, 802.1Q VLANs, STP, bonding, wireless design and installations.

3

LO3 · Operated networks

SNMP, NetFlow, syslog, NTP, DHCP/DNS services, configuration backup and disaster recovery.

4

LO4 · Secured networks

Firewalls, ACLs, IDS, TLS/PKI, SSH hardening, VPNs, segmentation and attack recognition.

5

LO5 · Troubleshot networks

The seven-step CompTIA methodology with the right diagnostic tool at every layer.

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NEXT STEPS

Preparing for the N10-009 Exam

1

Redo the labs

Repeat every lab from memory until the commands and workflow are automatic.

2

Master subnetting

Drill CIDR/VLSM until a /26 split takes under a minute — use the TIS IP Calculator to self-check.

3

Memorise the ports

FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, SNMP 161/162, LDAP 389, RDP 3389, SIP 5060/5061.

4

Know the blueprint

Revise by the five domains and their weightings — troubleshooting scenarios are the largest share.

5

Practice exams

Attempt timed practice exams and review every wrong answer against the objectives.

6

Book the exam

Schedule N10-009 at a Pearson VUE centre or online when practice scores pass 80%.

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TEST YOURSELF BEFORE THE REAL EXAM

Practice Exam — exams.tertiaryinfotech.com

1

1 · Open the portal

Go to https://exams.tertiaryinfotech.com/practice-exams/comptia/comptia-network-plus

2

2 · Choose Network+

Select the CompTIA Network+ practice exam that matches this course.

3

3 · Sit it timed

Attempt the full paper under exam timing — 90 minutes, no notes.

4

4 · Review and repeat

Review every explanation, revisit the weak domain's labs, then re-take it.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. WSQ CompTIA Certified Security+ Training
2. WSQ CompTIA Certified PenTest+ Training
3. WSQ CompTIA Certified CySA+ Cybersecurity Analyst Training
4. WSQ CompTIA Certified Linux+ Training
5. WSQ CompTIA Certified Cloud+ Training

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WE'RE HERE TO HELP

Support

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- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- LMS / TMS: https://lms-tms.tertiaryinfotech.com

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Assessment

1

Written Assessment (SAQ)

1 hour · open book · short-answer knowledge questions.

2

Practical Performance (PP)

90 minutes · open book · hands-on networking tasks on the lab environment.

3

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM) before you start.

4

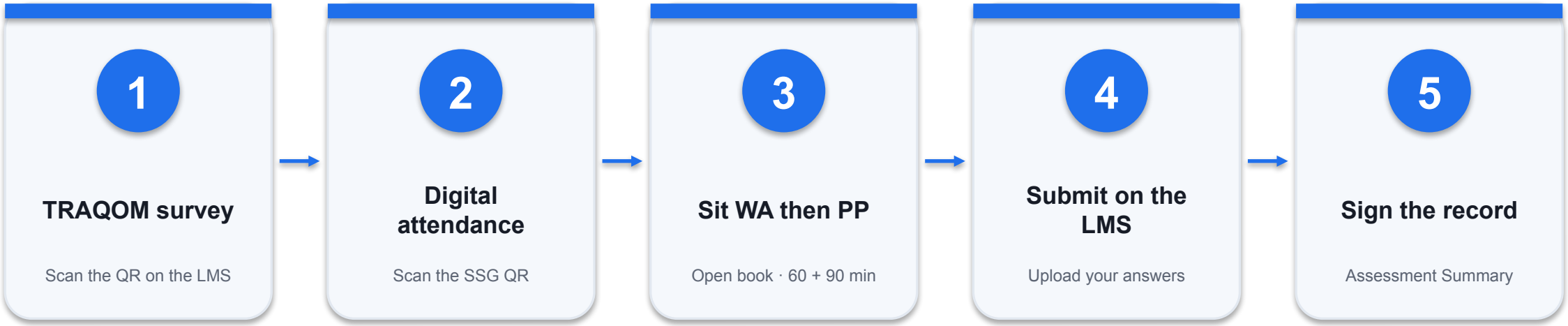
Submit on the LMS

Upload your completed answers at https://lms-tms.tertiaryinfotech.com/

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ON ASSESSMENT DAY

Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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HAPPY NETWORKING

Thank You!

You can now design, implement, operate, secure and troubleshoot modern networks — and you are ready for the CompTIA Network+ (N10-009) exam.

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